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# Experimental Design Analysis (stat408)

Combined by
M waqas(BS-MATHEMATICS)
0333-8000494

**STAT 408**

# COURSE: STA 408
# DESIGN & ANALYSIS OF EXPERIMENT

LECTURE DELIVERED BY

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# COURSE CONTENTS

- Introduction
- Basic principles of Experimental Design; Randomization, Replication and Local control
- Completely Randomized Design (CRD); Description, Layout, Statistical analysis, advantages and limitations
- The Randomized Complete Block Design (RCBD); Description, Layout, Statistical analysis, advantages and limitations.
- Replication within block, missing observations.
- Efficiency of RCBD and CRD.
- The Latin Square Design; Description, Layout, Statistical analysis, advantages and limitations
- Missing observations in Latin Square Design, efficiency of Latin Square
- Factorial Experiment; Main effects, Interaction effects
- Effect in a $2^2$ and $2^3$ in Factorial Experiment, advantages and disadvantages.

# RECOMMENDED BOOKS


- Boniface, D.R. (1995). "*Experiment Design & Statistical Methods*", Chapman & Hall.
- Chaudhry, S.M. and Kamal, S. (2010). *"Introduction to Statistical Theory"* Part II, Ilmi Kitab Khana, Urdu Bazar, Lahore.
- Clarke, G.M. (1994). "*Statistics & Experimental Design*". Edward Arnold.
- Clarke, G.M., and Kempton, R.E. (1997), "*Introduction to the Design & Analysis of Experiments*", Edward Arnold.
- Das, M.N. and Geri, N.C, (1986). "*Design and Analysis of Experiments*", John Wiley, New York.
- Harold, R. L (1992). "Analysis of Variance in Experimental Design". Springer Verlage
- Mead, R. (1988). "*The Design of Experiments*". Cambridge University Press, Cambridge.
- Montgomery, D. C., Design and analysis of experiments, Edition: 7$^{th}$ ed. Publisher: New Jersey : John Wiley, 2009

# COURSE: STA 408
# DESIGN & ANALYSIS OF EXPERIMENT

LECTURE NO.1

*Planning an experiment to obtain appropriate data and drawing inference out of the data with respect to any problem under investigation is known as* ***design and analysis of experiment***.

# Why all this?

# INTRODUCTION

- **EXPERIMENT**

An experiment deliberately imposes a treatment on a group of objects or subjects in the interest of observing the response.

- **TREATMENTS**

In experiments, a treatment is something that researchers administer to experimental units.

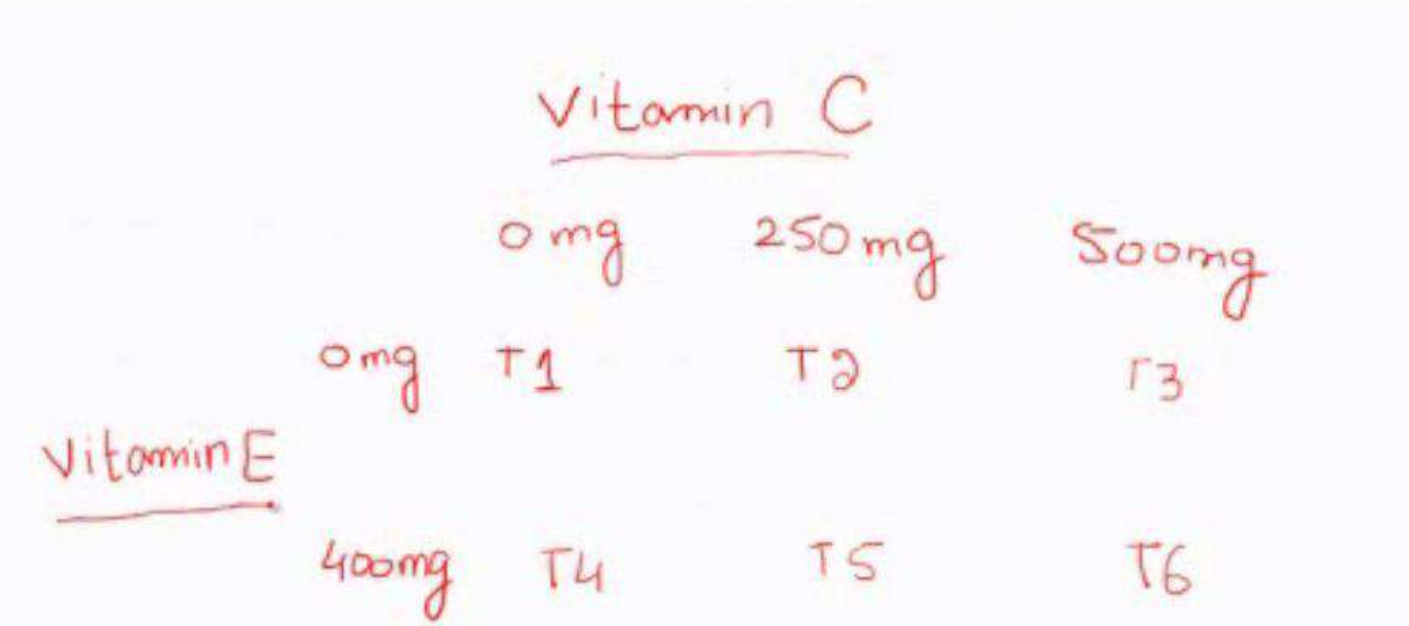
For example, a corn field is divided into four, each part is 'treated' with a different fertilizer to see which produces the most corn.

- **FACTOR**

A factor of an experiment is a controlled independent variable; a variable whose levels are set by the experimenter.

A factor is a general type or category of treatments. Different treatments constitute different levels of a factor.

# EXAMPLE



- **EXPERIMENTAL UNIT**

The experimental unit is the basic entity or unit on which the experiment is performed. i.e. It is an object to which the treatment is applied and in which the variable under investigation is measured and analyzed.

**For example, experimental unit may be:**











# IDENTIFY THE EXPERIMENTAL UNIT ?

- A teacher practices the different teaching methods on different groups in her class to see which yields the best results.

- A doctor treats a patient with a skin condition with different creams to see which is most effective.

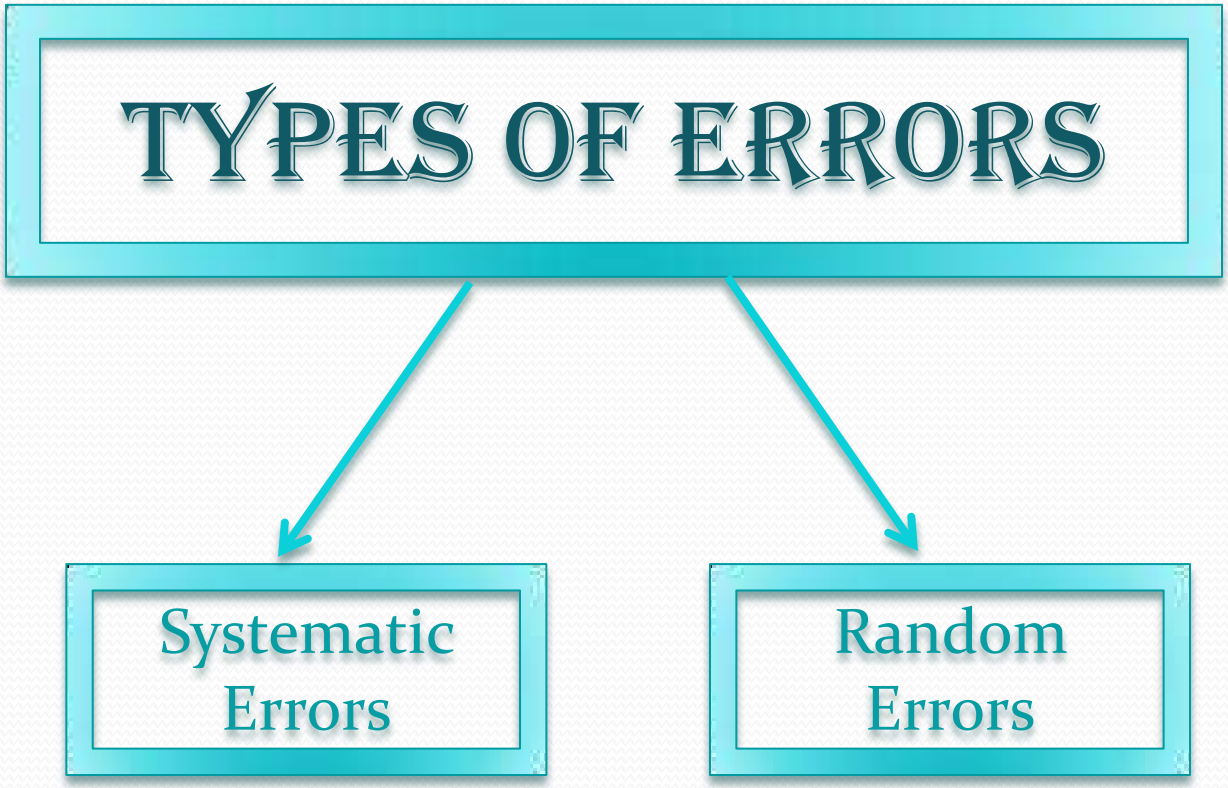
- **EXPERIMENTAL ERROR**

It describes the variation among identically and independently treated experimental units.

The various **origins of experimental error** include:

- ✓ The natural variation among the experimental units.
- ✓ Inability to reproduce the treatment conditions exactly from one unit to another.
- ✓ Interaction of treatments and experimental units.
- ✓ Any other extraneous factors that influence the measured characteristics.

# TYPES OF ERRORS



- **EXPERIMENTAL DESIGN**

An experimental design is a plan to collect the data relevant to the problem under investigation. In such a way as to provide a basis for valid and objective inferences about the stated problem.

The plan usually consists of the selection of the treatments , specifications of the experimental layouts, allocation of the treatments and collection of observation for analysis.

- **PURPOSE OF EXPERIMENTAL DESIGN**

  - ✓ Get maximum information for minimum expenditure in the minimum possible time
  - ✓ Helps to reduce the experimental error
  - ✓ To ignore spurious effects, if any. *(FOR DETAIL, SEE THE LECTURE)*
  - ✓ To evaluate and examine the outcomes critically and logically.

- **STEPS IN DESIGNING AN EXPERIMENT**

A statistically designed experiment may consist of the following steps.

- ✓ Statement of the problem
- ✓ Formulation of the hypothesis
- ✓ Choice of response or dependent variable.
- ✓ Selection of the factors to be varied.
- ✓ Choice of levels of these factors i.e. qualitative or quantitative and fixed or random.
- ✓ How factors are to be combined.
- ✓ Number of observations to be taken.

- ✓ Methods of randomization to be used.
- ✓ Mathematical model to describe the experiment.
- ✓ Data collection and processing.
- ✓ Application of statistical techniques to the experiment results.
- ✓ Drawing conclusions with measures of reliability of the estimates of any quantities that are evaluated.
- ✓ Evaluation of whole investigations particularly with other investigations on the same or similar problems.

# BASIC PRINCIPLES OF EXPERIMENTAL DESIGN

The basic principles of experimental designs are:

- Randomization
- Replication
- Local Control

These principles make a valid test of significance possible.

# RANDOMIZATION

It is a random process of assigning treatments to the experimental units.

Randomization is usually done by using tables of random numbers, by drawing numbered cards from a well shuffled pack of cards, or by drawing numbered balls from a well-shaken container.

# PURPOSE OF RANDOMIZATION

- ✓ To remove bias and other sources of extraneous variation, which are not controllable .
- ✓ To form the basis of any valid statistical test.

# REPLICATION

Replication is basically a repetition of the basic experiment.

In other words, it is a complete run for all the treatments to be tested in the experiment.

In all experiments, some variation is introduced because of the fact that the experimental units such as individuals or plots of land in agriculture experiments, cannot be physically identical. This type of variation can be removed by using a number of experimental units. We therefore perform the experiment more than once, i.e., we repeat the basic experiment. An individual repetition is called a replicate.

# REPLICATION IS USED ?

- ✓ To secure more accurate estimate of the experimental error.
- ✓ To decrease the experimental error and thereby to increase precision.
- ✓ To obtain more precise estimate of the mean effect of a treatment.

# LOCAL CONTROL

A term referring to the amount of balancing, blocking and grouping of the experimental units.

**Balancing** means that treatments should be assigned to the experimental units in such a way that the result is a balanced arrangements of the treatments.

**Blocking** means that like experimental units should be collected together to form a relatively homogeneous group. A block is also a replicate.

The main purpose of the principle of local control is to increase the efficiency of an experimental design by decreasing the experimental error.

# TYPES OF DESIGNS

1- Systematic designs

2- Randomized designs

# SYSTEMATIC DESIGNS

In systematic designs treatments are applied to the experimental units by some systematic pattern, i.e. by the choice of the experimenter.

## FOR EXAMPLE:

The experimenter wishes to test three treatments and he decides to have four repetitions of each treatment. The arrangements of three treatments over the experimental area could be : *(FOR SOLUTION, LISTEN THE LECTURE)*

# RANDOMIZED DESIGNS

In randomized designs , as the treatments are applied randomly therefore the conclusions drawn are supported by statistical tests.

In this way, inferences are applicable in wider range and the random process minimizes the systematic error.

The analysis of variance techniques are also suitable to randomized designs only.

# CLASSIFICATIONS OF RANDOMIZED DESIGNS

- **Completely Randomized Design (CRD)**
- **Randomized Complete Block Design (RCBD)**
- **Latin Square Design**

*"Detailed discussion will be made in further lectures"*

# EXAMPLE

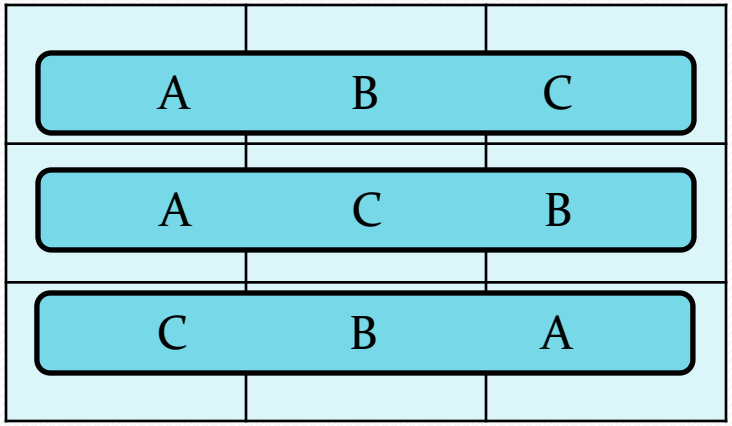
A market gardener wants to test three types of peas, A, B, C, on his land. He has a square plot which he divides into nine equal squares, three to be planted with each type of pea. The problem which he then faces is which square to plant with which type.

- One method is **Completely Randomized Design** which might,

| | | |
|---|---|---|
| C | A | C |
| B | A | A |
| B | B | C |

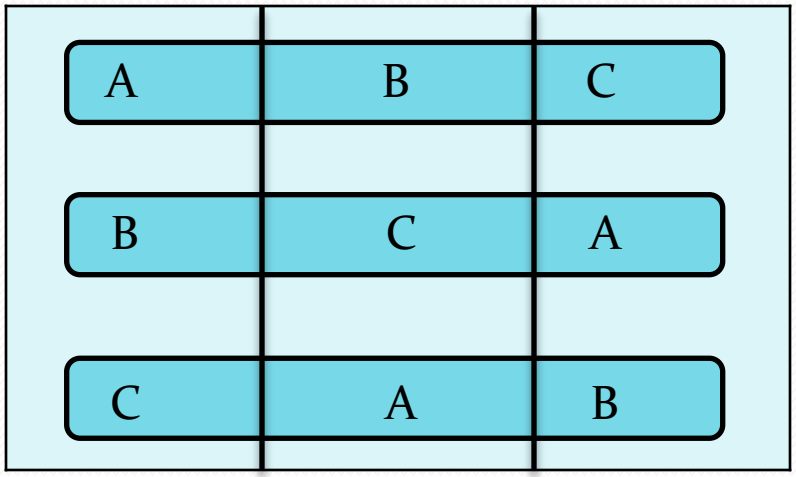
This would be all right if all the plots were equally desirable. If however, there were prevailing north wind so that the northernmost plots were exposed, he might decide to use ,

- **Randomized Complete Block Design**, where each of the types A, B and C is planted once in each west-east block.




If the gardener also felt that the soil to the east was rather better than that to the west, he would use ,

A **Latin Square design**, where each type of pea is planted once in each row (west-east), and once in each column (north-south).



# THANK YOU

# STA 408: DESIGN AND ANALYSIS OF EXPERIEMENTS

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# Contents

1. Basic concepts of ANOVA
2. CR Design

# Analysis of Variance: A Conceptual Overview

- Analysis of Variance (ANOVA) can be used to test for the equality of three or more population means.
- Data obtained from observational or experimental studies can be used for the analysis.
- We want to use the sample results to test the following hypotheses:

$H_0$: $\mu_1 = \mu_2 = \mu_3 = \cdot\ \cdot\ \cdot = \mu_k$

$H_a$: Not all population means are equal

# Analysis of Variance: A Conceptual Overview

$H_0$: $\mu_1 = \mu_2 = \mu_3 = \cdot\ \cdot\ \cdot = \mu_k$

$H_a$: Not all population means are equal

- If $H_0$ is rejected, we cannot conclude that *all* population means are different.
- Rejecting $H_0$ means that at least two population means have different values.

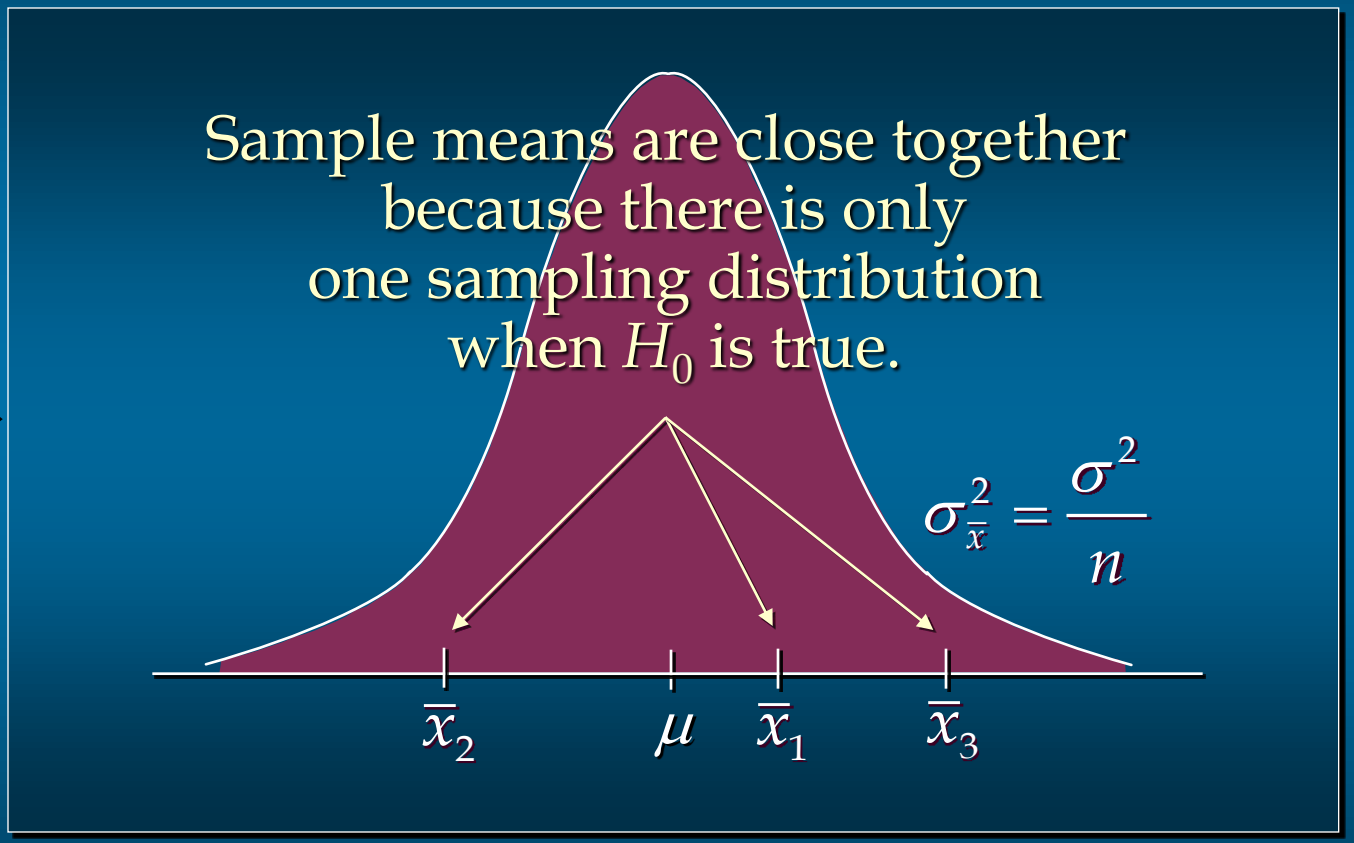
# Analysis of Variance: A Conceptual Overview

- Assumptions for Analysis of Variance

  - For each population, the response (dependent) variable is normally distributed.
  - The variance of the response variable, denoted $\sigma^2$, is the same for all of the populations.
  - The observations must be independent.

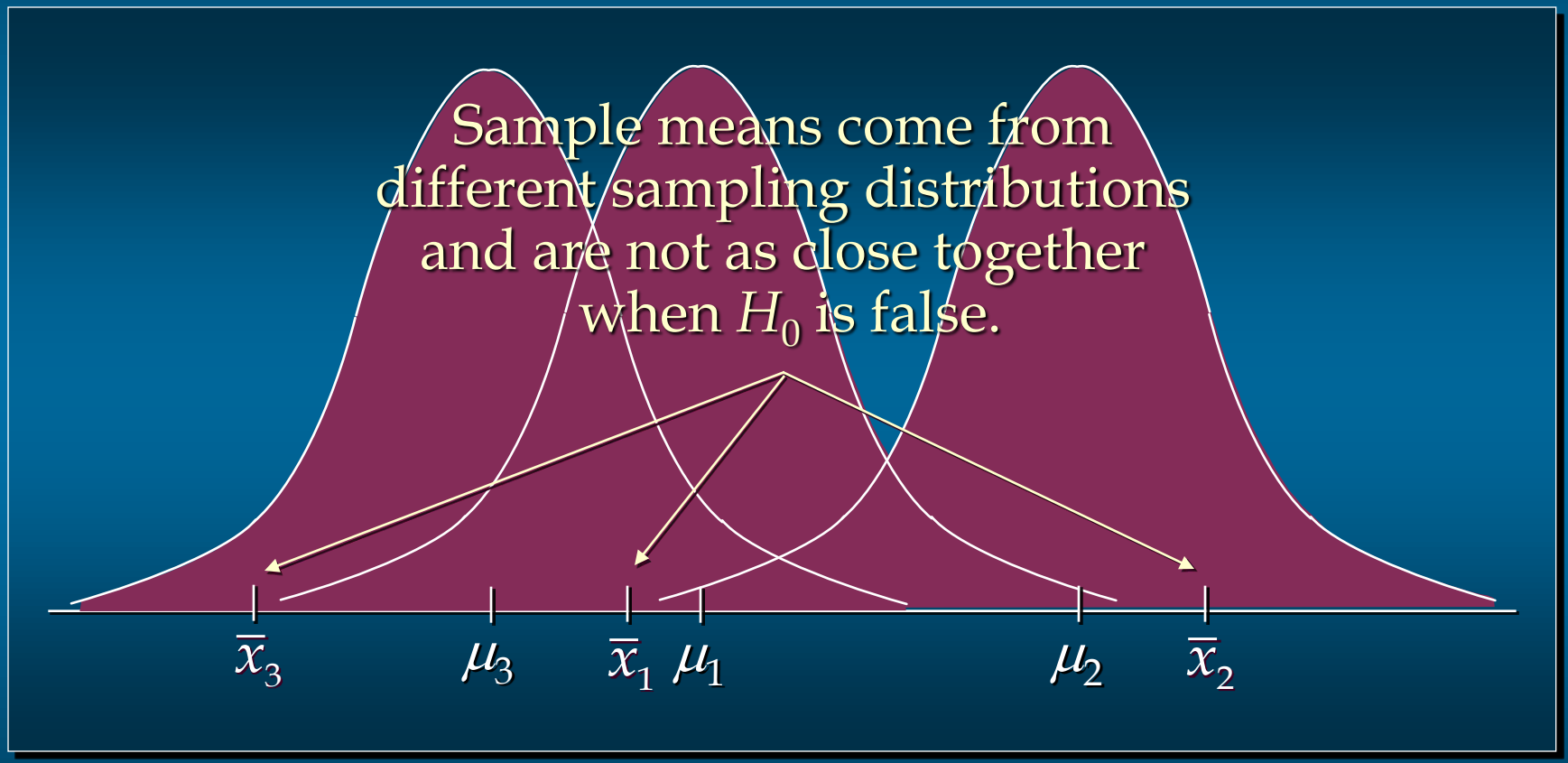
# Analysis of Variance: A Conceptual Overview

- Sampling Distribution of $\bar{x}$ Given $H_0$ is True



# Analysis of Variance: A Conceptual Overview

- Sampling Distribution of $\bar{x}$ Given $H_0$ is False



# Analysis of Variance

- Between-Treatments Estimate of Population Variance
- Within-Treatments Estimate of Population Variance
- Comparing the Variance Estimates: The *F* Test
- ANOVA Table

# Between-Treatments Estimate of Population Variance $\sigma^2$

- The estimate of $\sigma^2$ based on the variation of the sample means is called the mean square due to treatments and is denoted by MSTR.

$$\text{MSTR} = \frac{\sum_{j=1}^{k} n_j (\bar{x}_j - \bar{\bar{x}})^2}{k-1}$$

Denominator is the degrees of freedom associated with SSTR

Numerator is called the sum of squares due to treatments (SSTR)

# Within-Treatments Estimate of Population Variance $\sigma^2$

- The estimate of $\sigma^2$ based on the variation of the sample observations within each sample is called the mean square error and is denoted by MSE.

$$\text{MSE} = \frac{\sum_{j=1}^{k}(n_j - 1)s_j^2}{n_T - k}$$

Denominator is the degrees of freedom associated with SSE

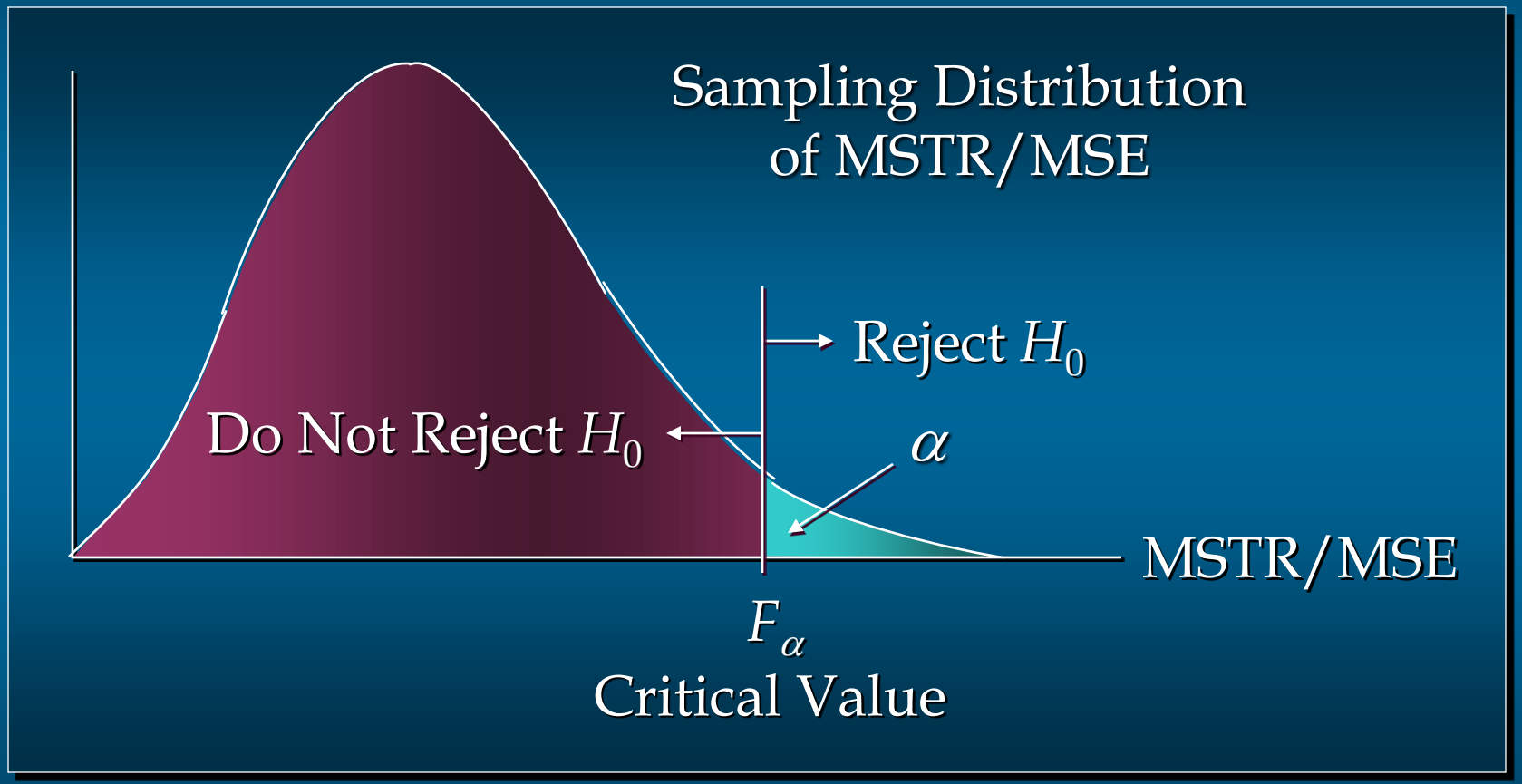
Numerator is called the sum of squares due to error (SSE)

# Comparing the Variance Estimates: The *F* Test

- If the null hypothesis is true and the ANOVA assumptions are valid, the sampling distribution of MSTR/MSE is an *F* distribution with MSTR d.f. equal to $k$ - 1 and MSE d.f. equal to $n_T$ - $k$.
- If the means of the $k$ populations are not equal, the value of MSTR/MSE will be inflated because MSTR overestimates $\sigma^2$.
- Hence, we will reject $H_0$ if the resulting value of MSTR/MSE appears to be too large to have been selected at random from the appropriate *F* distribution.

# Comparing the Variance Estimates: The *F* Test

- Sampling Distribution of MSTR/MSE



# Test for the Equality of $k$ Population Means

- Hypotheses

$H_0$: $\mu_1 = \mu_2 = \mu_3 = \cdot\ \cdot\ \cdot = \mu_k$

$H_a$: Not all population means are equal

- Test Statistic

$F$ = MSTR/MSE

# Test for the Equality of $k$ Population Means

- Rejection Rule
  - $p$-value Approach: Reject $H_0$ if $p$-value $\leq \alpha$
  - Critical Value Approach: Reject $H_0$ if $F \geq F_\alpha$

where the value of $F_\alpha$ is based on an $F$ distribution with $k$ - 1 numerator d.f. and $n_T$ - $k$ denominator d.f.

# ANOVA Table

- The ANOVA table decomposes the variance into two components (sources): a between-group component (treatments) and a within-group component (error). The F-ratio, which in this case equals 15.6234, is a ratio of the between-group estimate to the within-group estimate. Since the P-value of the F-test is less than 0.05, there is a statistically significant difference between the means from one level to another at the 95.0% confidence level.

**ANOVA Table for hours by brand**

| *Source* | *Sum of Squares* | *Df* | *Mean Square* | *F-Ratio* | *P-Value* |
|---|---|---|---|---|---|
| Between groups | 1.08917 | 3 | 0.363057 | 15.62 | 0.0000 |
| Within groups | 0.557714 | 24 | 0.0232381 | | |
| Total (Corr.) | 1.64689 | 27 | | | |

# Completely randomized design (CRD)

- A factor is a variable that the experimenter has selected for investigation (the independent variable).
- A treatment is a level of a factor.
- Experimental units are the objects of interest in the experiment.
- A completely randomized design is an experimental design in which the treatments are randomly assigned to the experimental units.

# COMPLETELY RANDOMIZED DESIGN

A completely randomized (CR) design, which is the simplest type of the basic designs, may be defined as a design in which the treatments are assigned to experimental units completely at random, that is the randomization is done without any restrictions. The design is completely flexible, i.e. any number of treatments and any number of units per treatment may be used. Moreover, the number of units per treatment need not be equal. A completely randomized design is considered to be most useful in situations where (i) the experimental units are homogeneous, (ii) the experiments are small such as laboratory experiments and (iii) some experimental units are likely to be destroyed or to fail to respond.

# CRD

- When the samples of experimental units for each treatment are random and independent of each other
- Design is used to compare the treatment means:

$$H_0 : \mu_1 = \mu_2 = \ldots \mu_k$$

$H_a$ : *At least two of the treatment means differ*

- The hypotheses are tested by comparing the differences between the treatment means.

- Test statistic is calculated using measures of variability within treatment groups and measures of variability between treatment groups

## Steps for Conducting an Analysis of Variance (ANOVA) for a Completely Randomized Design:

- 1- Assure randomness of design, and independence, randomness of samples
- 2- Check normality, equal variance assumptions
- 3- Create ANOVA summary table
- 4- Conduct multiple comparisons for pairs of means as necessary/desired

# Assumptions

**1- Normality:**

You can check on normality using

1- Normal probability plot

2- Kolmogorve test

**2- Constant variance**:

You can check on homogeneity of variances using

1- Histogram Plot

2- leven's test.

# CRD

## ANOVA Summary Table for a Completely Randomized Design

| *Source* | *df* | *SS* | *MS* | *F* |
|---|---|---|---|---|
| *Treatments* | $k-1$ | $SST$ | $MST = \frac{SST}{k-1}$ | $\frac{MST}{MSE}$ |
| *Error* | $n-k$ | $SSE$ | $MSE = \frac{SSE}{n-k}$ | |
| *Total* | $n-1$ | $SS(Total)$ | | |

# Advantages of a CRD

- Very flexible design (i.e. number of treatments and replicates is only limited by the available number of experimental units).
- Statistical analysis is simple compared to other designs.
- Loss of information due to missing data is small compared to other designs due to the larger number of degrees of freedom for the error source of variation.

# Disadvantages

- If experimental units are not homogeneous and you fail to minimize this variation using blocking, there may be a loss of precision.
- Usually the least efficient design unless experimental units are homogeneous.
- Not suited for a large number of treatments.

# Experimental Layout:

An example of the experimental layout for a completely randomized design (CR) using four treatments A, B, C and D, each repeated 3 times, is given below:

# Example:

An experiment was conducted to compare the yields of three varieties of potatoes. Each variety was assigned at random to equal-size plots, four times. The yields were as follows:

| Variety | | |
|---|---|---|
| A | B | C |
| 23 | 18 | 16 |
| 26 | 28 | 25 |
| 20 | 17 | 12 |
| 17 | 21 | 14 |

Test the hypothesis that the three varieties of potatoes are not different in the yielding capabilities.

Example:

Three fertilizer treatments A, B, C each applied to seven plots of strawberry plants, resulted in the following weights of crops (Ib/plot):

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **A** | 24 | 18 | 18 | 29 | 22 | 17 | 15 |
| **B** | 46 | 39 | 37 | 50 | 44 | 45 | 30 |
| **C** | 32 | 30 | 26 | 41 | 36 | 28 | 27 |

Perform the analysis of variance to test the hypothesis of no difference in the treatment effects. Use a 0.01 level of significance.

Solution:

Formulation of Hypothesis:

$H_0 : \mu_1 = \mu_2 = \mu_3$

$H_1 : At\ least\ two\ means\ are\ not\ equal.$

Level of Significance:

$\alpha = 0.05.$

Test Statistics:

$$F = \frac{MS\ Treatment}{MS\ Error}$$

Calculation:

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **A** | 24 | 18 | 18 | 29 | 22 | 17 | 15 | **143** |
| **B** | 46 | 39 | 37 | 50 | 44 | 45 | 30 | **291** |
| **C** | 32 | 30 | 26 | 41 | 36 | 28 | 27 | **220** |
| **Total** | | | | | | | | **654** |

Here $T_1 = 143, T_2 = 291, T_3 = 220$

$T_{..} = 654$

Total SS$= \sum\sum X_{ij}^2 - CF$

CF $= T_{..}^2 / N$ , where N= nk= 7(3) = 21.

CF$= (654)^2 / 21 = 20367.43$

Total SS$= \sum\sum X_{ij}^2 - CF$

Total SS$= 24^2 + 18^2 + 18^2 + \ldots + 28^2 - 20367.43$

$= 22520 - 20367.43 = 2152.57$

Treat SS$= \dfrac{\sum_{j=1}^{k} T_{.j}^2}{n} - CF$

Treat SS= $\frac{\sum_{j=1} T_{.j}^2}{n} - CF$

$$= \frac{1}{7}\left[(143)^2 + (291)^2 + (220)^2\right] - 20367.43$$

$$= 21932.86 - 20367.43 = 1564.43$$

Error SS= Total SS - Treat SS

$$= 2152.57 - 1564.43 = 587.14$$

| Source of Variation | Sum of squares | Degree of freedom | Mean squares | F ratio |
|---|---|---|---|---|
| Treatment | 1565.43 | k-1 = 2 | =1564.43/2 =782.72 | F=782.72/32.62 =24.00 |
| Error | 587.14 | N – k = 18 | =587.14/18 =32.62 | |
| Total | 2152.57 | N – 1 = 20 | | |

Critical Region:

Reject Ho, if

$$F_{cal} \geq F_{\alpha(k-1,N-k)}$$

$$24.00 \geq F_{0.05(2,18)}$$

$$24.00 > 3.55$$

We reject Ho.

Conclusion:

Since the value $24.00 > 3.55$ fall in the critical region, so we reject Ho or also we conclude that the there is a difference between the treatment effects.

# THANK YOU

# STA408
# Design and Analysis of Experiments
# Lesson 3

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# Randomized Complete Block Design

Testing the effects of two independent variables or factors on one *dependent variable.*

# Hypothesis Testing – General Procedure

**Procedure Table**

## Solving Hypothesis-Testing Problems (Traditional Method)

**Step 1** State the hypotheses and identify the claim.

**Step 2** Find the critical value(s) from the appropriate table

**Step 3** Compute the test value.

**Step 4** Make the decision to reject or not reject the null hypothesis.

**Step 5** Summarize the results.

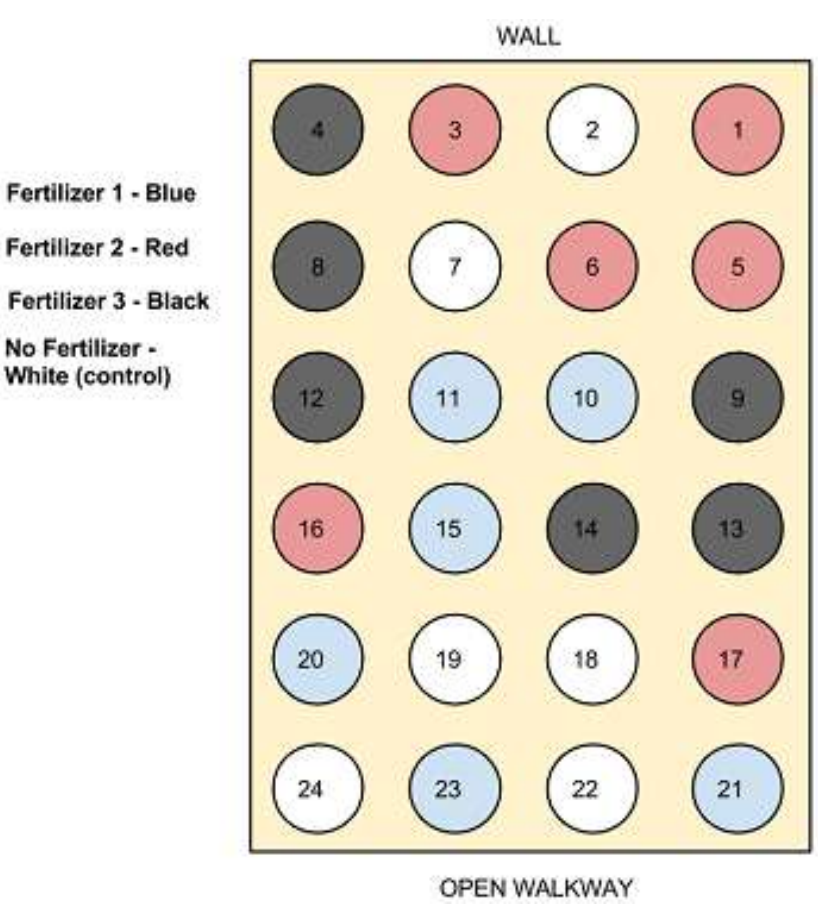
# Main Points

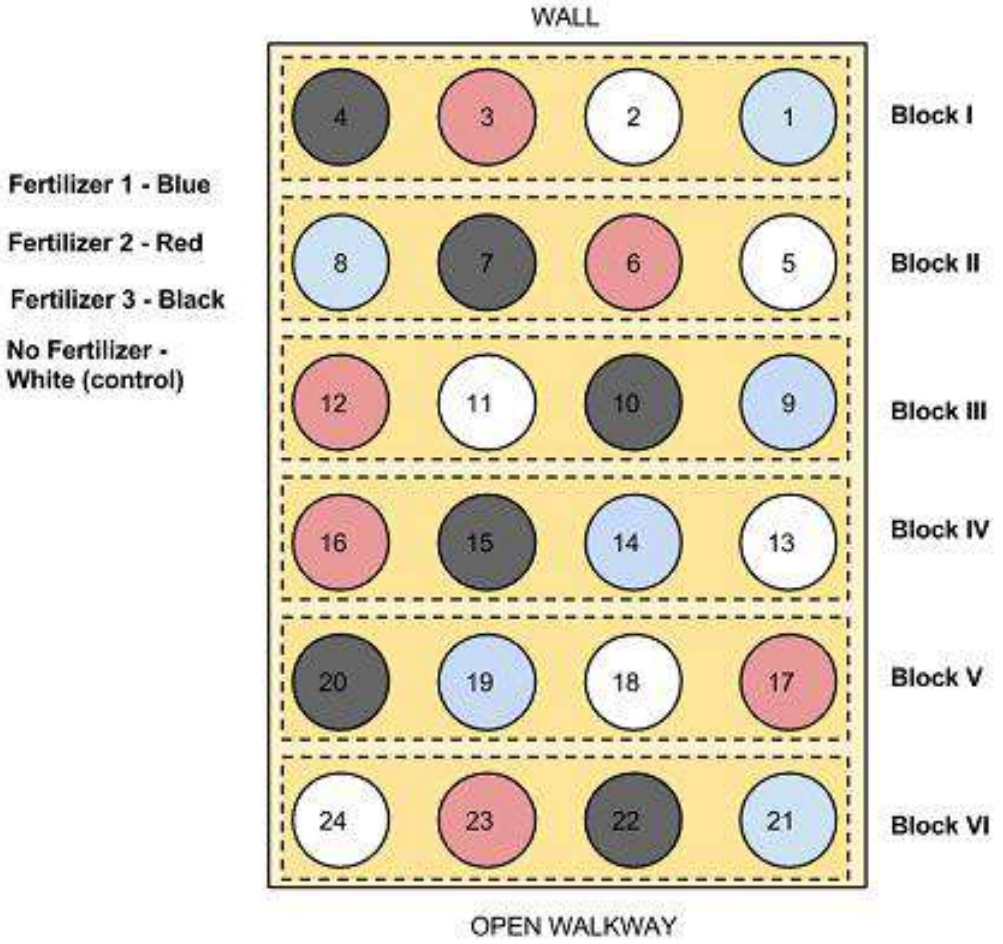
- Experimental material is divided into groups or blocks
  - Why
- Each block contains a complete set of treatments
- Treatments are assigned randomly
  - Randomizations is restricted within block

# Main Points

- A new randomization is done for each block/group
- It has different layouts: 2X3, 3X5, etc
- The RCBD is perhaps the most frequently used experimental design

# CRD vs RCBD


WALL
Fertilizer 1 - Blue
Fertilizer 2 - Red
Fertilizer 3 - Black
No Fertilizer - White (control)
4 3 2 1
8 7 6 5
12 11 10 9
16 15 14 13
20 19 18 17
24 23 22 21
OPEN WALKWAY


WALL
Fertilizer 1 - Blue
Fertilizer 2 - Red
Fertilizer 3 - Black
No Fertilizer - White (control)
4 3 2 1 Block I
8 7 6 5 Block II
12 11 10 9 Block III
16 15 14 13 Block IV
20 19 18 17 Block V
24 23 22 21 Block VI
OPEN WALKWAY

# May look like



# The Agricultural Research Field



# The Agricultural Research Field



# Experimental Layout

Allot treatment randomly




| | | | | |
|---|---|---|---|---|
| BLOCK - I | | | | |
| BLOCK - II | | | | |
| BLOCK - III | | | | |

# Statistical Model

- The liner model for this design is:

$$Y_{ij} = \mu + B_i + T_j + e_{ij}$$

- Where:

$\mu$ = True mean effect

T = Effect of treatment

B = Effect of block/groups

e = effect of random error

# Data Computations Table

| **Block (Groups)** | **Treatments** | | | | | | **Block** | |
|---|---|---|---|---|---|---|---|---|
| | *1* | *2* | … | *j* | … | *k* | Total | Mean |
| *1* | $Y_{11}$ | $Y_{12}$ | | | | | $B_{1\bullet}$ | $\bar{Y}_{1\bullet}$ |
| *2* | | | | | | | | |
| | | | | | | | | |
| *i* | | | | | | | | |
| | | | | | | | | |
| *r* | | | | | | | | |
| Total | $B_{\bullet 1}$ | | | | | | | |
| Means | $\bar{Y}_{\bullet 1}$ | | | | | | | |

# Partitioning of Variations

$$\begin{bmatrix}\text{Total}\\ \text{Variation}\end{bmatrix}=\begin{bmatrix}\text{Variation due to}\\ \text{Treatment}\end{bmatrix}+\begin{bmatrix}\text{Variation due to}\\ \text{Block/Grouping}\end{bmatrix}+\begin{bmatrix}\text{Variation due to}\\ \text{Unknown factors}\end{bmatrix}$$

Total SS = Treatment SS + Block SS + Error SS

$$\sum_i \sum_j (Y_{ij} - \bar{Y})^2 = r \sum_j (\bar{Y}_j - \bar{Y})^2 + k \sum_i (\bar{Y}_i - \bar{Y})^2 + \sum_i \sum_j (Y_{ij} - \bar{Y}_j - \bar{Y}_i - \bar{Y})$$

# AVOVA Table for RCB

| Source of Variation s.o.v | Degree of Freedom d.f | SS = Sum of Squares | MS = Mean Squares | F -test |
|---|---|---|---|---|
| (Variations due to) **Treatment** | K-1 | $SSTr = r\sum_j(\bar{Y}_{\square j} - \bar{Y}_{\square\square})^2$ | $S_t^2 = \frac{SSTr}{k-1}$ | $F_t = \frac{S_t^2}{S_e^2} = \frac{MST}{MSE}$ |
| (Variations due to) **Blocks** | r- 1 | $SSB = k\sum_i(\bar{Y}_{i\square} - \bar{Y}_{\square\square})^2$ | $S_b^2 = \frac{SSB}{r-1}$ | $F_b = \frac{S_b^2}{S_e^2} = \frac{MSB}{MSE}$ |
| (Variations due to) **Error** | (k-1) (r-1) | SSE = By Subtraction | $S_e^2 = \frac{SSTr}{(r-1)(k-1)}$ | |
| **Total** | rk-1 | $SST = \sum_i\sum_j(Y_{ij} - \bar{Y}_{\square\square})^2$ | --- | |

# Example

- Four verities of wheat ere tested in randomized complete clock design in four replications/blocks. Yield in kg/plot is shown in the table below.
- Test the hypothesis that there is no difference in the means of four varieties.

| Blocks | Varieties of wheat | | | |
|---|---|---|---|---|
| | V1 | V2 | V3 | V4 |
| I | 2 | 5 | 4 | 1 |
| II | 2 | 3 | 3 | 1 |
| III | 4 | 6 | 6 | 2 |
| IV | 1 | 4 | 2 | 3 |

# Testing Hypothesis – Example

**Step 1**

Stat the hypothesis.

$H_0 : Tj = 0$ (There is NO difference in the means of four varieties of wheat)
$H_1 : Tj \neq 0$ (There is difference in the means of four varieties)
( Not ALL the four means are equal)

$H_0 : B_i = 0$ (There is NO difference in the means of four varieties of wheat)
$H_1 : B_i \neq 0$ (There is difference in the means of four varieties)
( Not ALL the four means are equal)

**Step 2**

We set the **level of significance (alpha)** $\boldsymbol{\alpha = 0.05}$ **(5%)**

# Testing Hypothesis – Example

**Step 3**

The test-statistic (formula) to be used is:

$$F = \frac{S_t^2}{S_e^2} = \frac{MS(Treatment)}{MS(Error)}$$

*It follows F-distribution with degree of freedom*

***df = $v_1$ = from treatment = k-1***

***df = $v_2$ = from error = (k-1)(r-1)***

# Computations

| Blocks | Varieties (Treatment) | | | | | | |
|---|---|---|---|---|---|---|---|
| | V1 | V2 | V3 | V4 | $Block\ Total$ | $\left(Block\ Total\right)^2$ | Sum of Square of each value |
| I | 2 | 5 | 4 | 1 | 2+5+4+1 = 12 | 12*12 = 144 | $2^2+5^2+4^2+1^2 = 46$ |
| II | 2 | 3 | 3 | 1 | 9 | 81 | 23 |
| III | 4 | 6 | 6 | 2 | 18 | 324 | 92 |
| IV | 1 | 4 | 2 | 3 | 10 | 100 | 30 |
| $\left(\text{Varities Total}\right)$ | | | | | | | |
| $\left(\text{Varities Total}\right)^2$ | | | | | | | |
| Sum of Square of each value | | | | | | | |

# Computations

$$\text{SST} = \text{ Sum of Squares of Total } = \Sigma\Sigma Y^2 - \frac{T_{\square\square}}{n}$$

$$= \left(2^2 + 5^2 + 4^2 + \ldots + 3^2\right) - \frac{\left(2 + 5 + 4 + \ldots + 3\right)^2}{16}$$

$$= 191 - \frac{(49)^2}{16} = 40.94$$

$$\text{SSTr} = \text{ Sum of Squares of Treatment(varities)} = -\Sigma\Sigma Y^2 - \frac{T_{\square\square}}{n}$$

$$= \left(2^2 + 5^2 + 4^2 + \ldots + 3^2\right) - \frac{\left(2 + 5 + 4 + \ldots + 3\right)^2}{16}$$

$$= 191 - \frac{(49)^2}{16} = 40.94$$

# Computations

$$\text{SSTr} = \text{ Sum of Squares of Treatment(varieties)} = \frac{T^2}{r} - \frac{T_{\square\square}}{n}$$

$$= \frac{679}{4} - \frac{(49)^2}{16} = 169.75 - 150.06 = 19.69$$

$$\text{SSB} = \text{ Sum of Squares of Block} = \frac{B^2}{r} - \frac{T_{\square\square}}{n}$$

$$= \frac{649}{4} - \frac{(49)^2}{16} = 162.25 - 150.06 = 12.19$$

$$\text{SSE} = \text{ Sum of Squares of Error} = SST - SSTr - SSB$$

$$= 40.96 - 19.69 - 12.19 = 9.06$$

# ANOVA Table

| Source of Variation s.o.v | Degree of Freedom d.f | SS = Sum of Squares | MS = Mean Squares | Computed F -test |
|---|---|---|---|---|
| (Variations due to) **Treatment** | 4-1 = 3 | 19.69 | 19.69 / 3 = 6.56 | $F_t = \frac{6.56}{1.01} = 6.50$ |
| (Variations due to) **Blocks** | 4 – 1 = 3 | 12.19 | 12.19 / 3 = 4.06 | $F_b = \frac{4.06}{1.01} = 4.02$ |
| (Variations due to) **Error** | | 9.06 | 9.06 / 9 = 1.01 | |
| **Total** | (4*4) – 1 =15 | 40.96 | --- | |

# F – Table Value

**Table H** *(continued)*

$\alpha = 0.05$

| d.f.D.: degrees of freedom, denominator | d.f.N.: degrees of freedom, numerator | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 12 | 15 | 20 |
| 1 | 161.4 | 199.5 | 215.7 | 224.6 | 230.2 | 234.0 | 236.8 | 238.9 | 240.5 | 241.9 | 243.9 | 245.9 | 248.0 |
| 2 | 18.51 | 19.00 | 19.16 | 19.25 | 19.30 | 19.33 | 19.35 | 19.37 | 19.38 | 19.40 | 19.41 | 19.43 | 19.4 |
| 3 | 10.13 | 9.55 | 9.28 | 9.12 | 9.01 | 8.94 | 8.89 | 8.85 | 8.81 | 8.79 | 8.74 | 8.70 | 8.6 |
| 4 | 7.71 | 6.94 | 6.59 | 6.39 | 6.26 | 6.16 | 6.09 | 6.04 | 6.00 | 5.96 | 5.91 | 5.86 | 5.8 |
| 5 | 6.61 | 5.79 | 5.41 | 5.19 | 5.05 | 4.95 | 4.88 | 4.82 | 4.77 | 4.74 | 4.68 | 4.62 | 4.5 |
| 6 | 5.99 | 5.14 | 4.76 | 4.53 | 4.39 | 4.28 | 4.21 | 4.15 | 4.10 | 4.06 | 4.00 | 3.94 | 3.8 |
| 7 | 5.59 | 4.74 | 4.35 | 4.12 | 3.97 | 3.87 | 3.79 | 3.73 | 3.68 | 3.64 | 3.57 | 3.51 | 3.4 |
| 8 | 5.32 | 4.46 | 4.07 | 3.84 | 3.69 | 3.58 | 3.50 | 3.44 | 3.39 | 3.35 | 3.28 | 3.22 | 3.1 |
| 9 | 5.12 | 4.26 | 3.86 | 3.63 | 3.48 | 3.37 | 3.29 | 3.23 | 3.18 | 3.14 | 3.07 | 3.01 | 2.9 |
| 10 | 4.96 | 4.10 | 3.71 | 3.48 | 3.33 | 3.22 | 3.14 | 3.07 | 3.02 | 2.98 | 2.91 | 2.85 | 2.7 |
| 11 | 4.84 | 3.98 | 3.59 | 3.36 | 3.20 | 3.09 | 3.01 | 2.95 | 2.90 | 2.85 | 2.79 | 2.72 | 2.6 |
| 12 | 4.75 | 3.89 | 3.49 | 3.26 | 3.11 | 3.00 | 2.91 | 2.85 | 2.80 | 2.75 | 2.69 | 2.62 | 2.5 |
| 13 | 4.67 | 3.81 | 3.41 | 3.18 | 3.03 | 2.92 | 2.83 | 2.77 | 2.71 | 2.67 | 2.60 | 2.53 | 2.4 |
| 14 | 4.60 | 3.74 | 3.34 | 3.11 | 2.96 | 2.85 | 2.76 | 2.70 | 2.65 | 2.60 | 2.53 | 2.46 | 2.3 |
| 15 | 4.54 | 3.68 | 3.29 | 3.06 | 2.90 | 2.79 | 2.71 | 2.64 | 2.59 | 2.54 | 2.48 | 2.40 | 2.3 |
| 16 | 4.49 | 3.63 | 3.24 | 3.01 | 2.85 | 2.74 | 2.66 | 2.59 | 2.54 | 2.49 | 2.42 | 2.35 | 2.2 |
| 17 | 4.45 | 3.59 | 3.20 | 2.96 | 2.81 | 2.70 | 2.61 | 2.55 | 2.49 | 2.45 | 2.38 | 2.31 | 2.2 |
| 18 | 4.41 | 3.55 | 3.16 | 2.93 | 2.77 | 2.66 | 2.58 | 2.51 | 2.46 | 2.41 | 2.34 | 2.27 | 2.1 |
| 19 | 4.38 | 3.52 | 3.13 | 2.90 | 2.74 | 2.63 | 2.54 | 2.48 | 2.42 | 2.38 | 2.31 | 2.23 | 2.1 |
| 20 | 4.35 | 3.49 | 3.10 | 2.87 | 2.71 | 2.60 | 2.51 | 2.45 | 2.39 | 2.35 | 2.28 | 2.20 | 2.1 |
| 21 | 4.32 | 3.47 | 3.07 | 2.84 | 2.68 | 2.57 | 2.49 | 2.42 | 2.37 | 2.32 | 2.25 | 2.18 | 2.1 |
| 22 | 4.30 | 3.44 | 3.05 | 2.82 | 2.66 | 2.55 | 2.46 | 2.40 | 2.34 | 2.30 | 2.23 | 2.15 | 2.0 |
| 23 | 4.28 | 3.42 | 3.03 | 2.80 | 2.64 | 2.53 | 2.44 | 2.37 | 2.32 | 2.27 | 2.20 | 2.13 | 2.0 |
| 24 | 4.26 | 3.40 | 3.01 | 2.78 | 2.62 | 2.51 | 2.42 | 2.36 | 2.30 | 2.25 | 2.18 | 2.11 | 2.0 |
| 25 | 4.24 | 3.39 | 2.99 | 2.76 | 2.60 | 2.49 | 2.40 | 2.34 | 2.28 | 2.24 | 2.16 | 2.09 | 2.0 |
| 26 | 4.23 | 3.37 | 2.98 | 2.74 | 2.59 | 2.47 | 2.39 | 2.32 | 2.27 | 2.22 | 2.15 | 2.07 | 1.9 |
| 27 | 4.21 | 3.35 | 2.96 | 2.73 | 2.57 | 2.46 | 2.37 | 2.31 | 2.25 | 2.20 | 2.13 | 2.06 | 1.9 |

# Comparison and Conclusion

**Step 5**

$$\textit{Calculated Value of F (treatment)} = 6.56$$

$$\text{Critical/Table Value of F} = 3.86$$

$$\Rightarrow 6.56 > 3.86$$

$$\Rightarrow F_t > F(Table/CriticalValue)$$

**Conclusion and Summarizing:**

Since the computed value of **F (treatment) is GREATER THAN the F Table value, So it** falls in the critical region. We reject our null hypothesis and may conclude that the means of four varieties of wheat are significantly different.

# Comparison and Conclusion

**Step 5**

$$Calculated\ \ Value \text{ of F (Block/Group)} = 4.06$$

$$\text{Critical/Table Value of F} = 3.86$$

$$\Rightarrow 4.06 > 3.86$$

$$\Rightarrow F_t > F(Table / CriticalValue)$$

**Conclusion and Summarizing:**

Since the computed value of **F (Block/group) is GREATER THAN the F Table value, So it** falls in the critical region. We reject our null hypothesis and may conclude that the **BLOCKING or GROUPING was effective.**

# Another example

- The following is the plan of a filed layout testing four varieties A, B, C and D of wheat in each of % blocks. The plot yields in Kgs are also indicated.

| Block I | |
|---|---|
| D | 29.3 |
| B | 33.3 |
| C | 30.8 |
| A | 32.3 |

| Block II | |
|---|---|
| B | 33.0 |
| A | 34.0 |
| C | 34.3 |
| D | 26.0 |

| Block III | |
|---|---|
| D | 29.8 |
| A | 34.3 |
| B | 36.3 |
| C | 35.3 |

| Block IV | |
|---|---|
| B | 36.8 |
| A | 35.00 |
| D | 28.0 |
| C | 32.3 |

| Block V | |
|---|---|
| D | 28.8 |
| C | 35.8 |
| B | 34.5 |
| A | 36.5 |

# Computations

| Blocks | Varieties (Treatment) | | | | | | |
|---|---|---|---|---|---|---|---|
| | V1 | V2 | V3 | V4 | *Block Total* | $\left(\text{Block Total}\right)^2$ | Sum of Square of each value |
| I | | | | | | | |
| II | | | | | | | |
| III | | | | | | | |
| IV | | | | | | | |
| $\left(\text{Varities Total}\right)$ | | | | | | | |
| $\left(\text{Varities Total}\right)^2$ | | | | | | | |
| Sum of Square of each value | | | | | | | |

# ANOVA Table

| Source of Variation s.o.v | Degree of Freedom d.f | SS = Sum of Squares | MS = Mean Squares | Computed F -test |
|---|---|---|---|---|
| (Variations due to) **Treatment** | | | | |
| (Variations due to) **Blocks** | | | | |
| (Variations due to) **Error** | | | | |
| **Total** | | | | |

# Advantages

- The source of **extraneous** variation is controlled by grouping the experimental material and hence the estimate of the experimental error is decreased.
- The design is flexible. Any number of replications may be run and any number of treatments may be tested.
- The experiment can be set up easily
- The statistical analysis is simple and straightforward.
- It is easy to adjust the missing observations

# Disadvantages

- It control the variability only in one direction
- It is not suitable design when the number of treatments is very large or when the blocks are not homogeneous.

- Extraneous Variables are **undesirable** variables that influence the relationship between the variables that an experimenter is examining.
- Another way to think of this, is that these are variables that influence the outcome of an experiment, though they are not the variables that are actually of interest. These variables are undesirable because they add error to an experiment. A major goal in research design is to decrease or control the influence of extraneous variables as much as possible.

# Summary

- Experimental material is divided into groups or blocks
- Each block contains a complete set of treatments
- Treatments are assigned randomly for each block
- Data is organized in ANOVA table
- F-test is used for testing
- Rest of the testing procedure is same.

The END

# COURSE: STA 408
# DESIGN & ANALYSIS OF EXPERIMENT

LECTURE DELIVERED BY

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# COURSE: STA 408
# DESIGN & ANALYSIS OF EXPERIMENT

LECTURE NO.4

# RCBD WITH REPLICATIONS WITHIN BLOCKS

Let us assume that each treatment is replicated *n* times within each block, so that each block contains *nk* plots. The experiment becomes analogous to two way classifications with each cell containing *n* observations. This leads to new statistical model in the form

$$Y_{ijl} = \mu + B_i + \tau_j + (B\tau)_{ij} + e_{ijl}$$

# MISSING OBSERVATIONS IN RCBD

Sometimes it so happens that some experimental units or observations are missing.

FOR EXAMPLE



UPROOTED PLANTS



RECORDS LOST



DIED ANIMALS



DESTROYED PLANTS



BROKEN FLASK



UNCOOPERATIVE PEOPLE

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Suppose,

| | | TREATMENTS | | | | Block Total |
|---|---|---|---|---|---|---|
| | | 1 | 2 | ..... | $k$ | |
| BLOCKS | 1 | $x$ | $y_{12}$ | ..... | $y_{1k}$ | $B'_{1.} + x$ |
| | 2 | $y_{21}$ | $y_{22}$ | ..... | $y_{2k}$ | $B_{2.}$ |
| | ⋮ | ⋮ | ⋮ | | ⋮ | ⋮ |
| | $r$ | $y_{r1}$ | $y_{r2}$ | ..... | $y_{rk}$ | $B_{r.}$ |
| Treatment Total | | $T'_{.1} + x$ | $T_{.2}$ | ..... | $T_{.k}$ | $T'_{..} + x$ |

Performing the analysis of variance, we get the following sums of squares:

| Source of Variation | Sum of Squares |
|---|---|
| Treatments | $\frac{(T_{.1}+x)^2}{r} + \frac{T_{.2}^2}{r} + \cdots + \frac{T_{.t}^2}{r} - \frac{(T_{..}'+x)^2}{rK}$ |
| BLOCKS | $\frac{(B_{1.}'+x)^2}{K} + \frac{B_{2.}^2}{K} + \cdots + \frac{B_{b.}^2}{K} - \frac{(T_{..}'+x)^2}{rK}$ |
| ERROR | By Subtraction |
| Total | $\sum_{ij} y_{ij}^2 + x^2 - \frac{(T_{..}'+x)^2}{rK}$ |

Thus the error sum of squares is obtained as

$$SSE = x^2 + \frac{(T'_{..} + x)^2}{kr} - \frac{(T'_{.1} + x)^2}{r} - \frac{(B'_{1.} + x)^2}{k}$$

+ terms which do not involve $x$.

$$\frac{\partial(SSE)}{\partial x} = 2x + \frac{2(T'_{..} + x)}{kr} - \frac{2(T'_{.1} + x)}{r} - \frac{2(B'_{1.} + x)}{k} \quad \text{---(1)}$$

$$2x + \frac{2(T'_{..} + x)}{kr} - \frac{2(T'_{.1} + x)}{r} - \frac{2(B'_{1.} + x)}{k} = 0$$

$$x + \frac{T'_{..}}{kr} + \frac{x}{kr} - \frac{T'_{.1}}{r} - \frac{x}{r} - \frac{B'_{1.}}{k} - \frac{x}{k} = 0$$

$$\alpha\left[1+\frac{1}{Kn}-\frac{1}{n}-\frac{1}{K}\right]+\frac{\bar{T}'_{..}}{Kr}-\frac{\bar{T}'_{.1}}{n}-\frac{B'_{1.}}{K}=0$$

$$\alpha\left(\frac{Kr-1-K-r}{Kr}\right)=\frac{rB'_{1.}+K\bar{T}'_{.1}-\bar{T}'_{..}}{Kr}$$

$$\alpha=\frac{rB'_{1.}+K\bar{T}'_{.1}-T'_{..}}{(K-1)(r-1)}$$

The formula in general may be written as

$$x = \frac{rB + kT - G}{(k-1)(r-1)}$$

*where*

$r$ = number of replicates

$k$ = number of treatments

$B$ = sum of remaining values in the blocks with the missing observations

$T$ = sum of the remaining values of the treatment with missing observation, and

$G$ = Grand total of all observed values.

# EXAMPLE

The following data, obtained from a randomized complete block design with 3 treatments A, B, C, and 3 blocks, contain one missing observation represented by $x$.

| | | TREATMENTS | | |
|---|---|---|---|---|
| | | A | B | C |
| BLOCKS | I | 5 | 12 | 15 |
| | II | 7 | 10 | 14 |
| | III | 8 | 16 | $x$ |

Estimate the missing observation and prepare a table for analysis of variance.

$$x = \frac{rB + kT - G}{(k-1)(r-1)}$$

$r = 3$, $k = 3$, $B = 8 + 16 = 24$, $T = 15 + 14 = 29$

$G = 87$

$$x = \frac{3(24) + 3(29) - 87}{(3-1)(3-1)} = 18$$

$$\boxed{x = 18}$$

| | | TREATMENTS | | |
|---|---|---|---|---|
| | | A | B | C |
| BLOCKS | I | 5 | 12 | 15 |
| | II | 7 | 10 | 14 |
| | III | 8 | 16 | 18 |

(1) Hypothesis

$H_0: \mu_{.1} = \mu_{.2} = \mu_{.3}$

$H_1$: Not all treatments means are equal

$H_0': \mu_{1.} = \mu_{2.} = \mu_{3.}$

$H_1'$: Not all block means are equal

(2) Level of significance

$\alpha = 0.05$

(3) Test Statistic

$F_1 = \frac{s_t^2}{s_e^2}$ → ms for treatments / ms error

$F_2 = \frac{s_b^2}{s_e^2}$ → ms for blocks / ms error

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(4) Calculation

$$C.F = \frac{(\bar{T}..)^2}{n}$$

$$TSS = \sum_i \sum_j y_{ij}^2 - C.F$$

$$n = rK = 9$$

$$TSS = 158$$

$$C.F = \frac{(105)^2}{9} = 1225$$

$$\bar{T}reatSS = \frac{\sum_j \bar{T}_{\cdot j}^2}{r} - C.F = 126$$

$$BSS = \frac{\sum_i B_{i\cdot}^2}{K} - C.F = 24.7$$

$$ESS = TSS - \bar{T}reatSS - BSS$$

$$ESS = 7.3$$

## ANOVA-TABLE

| Source of Variation | Degrees of freedom | Sum of Squares | Mean Square | F-Test |
|---|---|---|---|---|
| Treatments | $\nu_1 = K-1 = 3-1 = 2$ | $126 = \text{Treat SS}$ | $s_t^2 = \frac{126}{2} = 63$ | $F_1 = \frac{s_t^2}{s_e^2} = \frac{63}{2.43} = 25.9$ |
| Blocks | $\nu_2 = 2$ | $24.7 = \text{BSS}$ | $s_b^2 = \frac{\text{BSS}}{\nu_2} = 12.35$ | $F_2 = \frac{s_b^2}{s_e^2} = \frac{12.35}{2.43} = 5.08$ |
| Error | $\nu_3 = (K-1)(K-1) - 1$ $\nu_3 = 3$ | $7.3 = \text{ESS}$ | $s_e^2 = 2.43$ | |
| Total | 7 | $158 = \text{TSS}$ | — | |

5. Critical Region

Reject Null hypothesis if $\begin{cases} F_1 \geq F_{0.05(2,3)} \\ F_2 \geq F_{0.05(2,3)} \end{cases}$

$25.9 > 9.55 \rightarrow$ Reject $H_0$

$5.08 < 9.55 \rightarrow$ Do not Reject $H_0$'

6. Conclusion.

As our cal value $F_1$ fall in critical Region so, all the Treatment means are not equal.

But the $F_2$ does not lie in reject region. we conclude that all the block means are equal.

# EFFICIENCY OF RCBD RELATIVE TO A CR DESIGN

- ✓ For experimental design, efficiency relates to the ability of a design to achieve the objective with minimal expenditure of resource such as time and money.
- ✓ Its relative efficiency is usually expressed in terms of a ratio of error variances.

$$RE(RCB,CR) = \frac{\text{Estimated MSE for CR design}}{\text{MSE for the RCB design}}$$

$$RE(RCB,CR) = \frac{(r-1)s_b^2 + r(k-1)s_e^2}{(rk-1)s_e^2}$$

MS for blocks

MS for error

# EXAMPLE

Given the following analysis of variance for RCBD:

| Source of Variation | Degrees of freedom | Sum of Squares | Mean Square |
|---|---|---|---|
| Blocks | 9 | 0.4074 | 0.0453 |
| Treatments | 3 | 1.1986 | 0.3995 |
| Error | 27 | 0.6247 | 0.0231 |
| Total | 39 | 2.2309 | - |

Estimate the efficiency of this design relative to completely randomized design.

| Degrees of freedom | Mean Square |
|---|---|
| = 9<br>v − 1 | 0.0453<br>$S_b^2$ = |
| = 3<br>k − 1 | 0.3995<br>$S_t^2$ = |
| 27 | 0.0231<br>$S_e^2$ = |
| = 39<br>vk − 1 | - |

$$RE = \frac{(9)(0.0453) + (10)(3)(0.0231)}{(39)(0.0231)}$$

$$RE = \frac{1.1007}{0.9009} = 1.22 = 122\%$$

# PRACTICE QUESTIONS

1. Determine the missing observation and compute the analysis of variance.

| BLOCKS | TREATMENTS | | | | |
|---|---|---|---|---|---|
| | A | B | C | D | E |
| I | 5 | 6 | 6 | 7 | 4 |
| II | $Y_{2A}$ | 7 | 7 | 8 | 5 |
| III | 7 | 7 | 7 | 9 | 5 |

Yates method

$Y_{2A} = ?$

(1) Hypothesis

$H_0 : \mu_{\cdot A} = \mu_{\cdot B} = \mu_{\cdot C} = \mu_{\cdot D} = \mu_{\cdot E}$

$H_1$: Not all Treatment means are equal

$H_0' : \mu_{1\cdot} = \mu_{2\cdot} = \mu_{3\cdot}$

$H_1'$ : Not all block means are equal

(2) level of significance $\alpha = 0.05$

(3) Test Statistic

$F_1 =$

$F_2 =$

(4) Calculation

$y_{2A} = ?$

ANOVA Table

$[(K-1)(r-1)] - 1$

Sum of Squares

Treatments = 17.6

Blocks = 5.2

Error = 0.8

Total = 23.6

(5) Critical Region

(6) Conclusion

2 Given the following analysis of variance for Randomized Complete Block Design. Fill in the mean squares. Estimate the efficiency of this design relative to Completely Randomized design.

| Source of Variation | Degrees of freedom | Sum of Squares | Mean Square |
|---|---|---|---|
| Treatments | 3 | 134.45 | $S_t^2 = 134.45/3$ |
| Blocks | 4 | 21.46 | |
| Error | 12 | 26.26 | |
| Total | 19 | 182.17 | - |

$$RE(RCBD, CRD) = \frac{\text{Estimated MSE for CRD}}{\text{MSE for RCBD}} = 1.30 \text{ or } 130\%$$

2 Given the following analysis of variance for Randomized Complete Block Design. Fill in the mean squares. Estimate the efficiency of this design relative to Completely Randomized design.

| Source of Variation | Degrees of freedom | Sum of Squares | Mean Square |
|---|---|---|---|
| Treatments | 3 | 134.45 | $S_t^2 = 134.45/3 =$ 44.82 |
| Blocks | 4 | 21.46 | $S_b^2 = 21.46/4 =$ 5.36 |
| Error | 12 | 26.26 | $S_e^2 = 26.26/12 =$ 2.19 |
| Total | 19 | 182.17 | - |

$$RE(RCBD, CRD) = \frac{(r-1)\sigma_b^2 + r(K-1)\sigma_e^2}{(rK-1)\sigma_e^2}$$

$$= \frac{(5-1)(5.36) + (5)(4-1)(2.19)}{[(5\times4)-1](2.19)}$$

$$= \frac{54.29}{41.61} = 1.30 \text{ or } 130\%$$

This indicates that RCBD is 130% as efficient as the CR design.

# THANK YOU

# COURSE: STA 408
# DESIGN & ANALYSIS OF EXPERIMENT

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DESIGN & ANALYSIS OF EXPERIMENT

# THE LATIN SQUARE DESIGN

LECTURE NO.5

# DESCRIPTION

When the variation is found in two directions, it becomes necessary to remove these two sources of variation simultaneously. This is done by simultaneous blocking of experimental units in two mutually perpendicular directions called rows and columns. Since each row and each column is a complete block, the grouping for a balanced arrangement is performed by imposing the restriction that each treatment must appear " once and only once" in each row and in each column.

If there are “k” treatments then the experimental area will be divided into “k” rows and “k” columns in k x k plots or experimental units. As the experiment is laid out in a square pattern and the treatments are then assigned at random to plots or experimental units. Such a double blocking of experimental units and a corresponding double random assignment of treatments is called “**Latin Square Design**”.

# LAYOUTS

Latin squares are always constructed by rotations, e.g. in the case of 4 treatments A, B, C and D, we get

| | | | |
|---|---|---|---|
| A | B | C | D |
| B | C | D | A |
| C | D | A | B |
| D | A | B | C |

A large number of distinct Latin squares can be derived by interchanging rows and columns of a standard Latin square, where a **standard Latin square** is a Latin square in which the treatments in the first row and first column are arranged in alphabetical or numerical order.

A standard Latin square is also called a reduced Latin square.

- There is only one standard square for a 2 x 2 Latin square as,

| | |
|---|---|
| A | B |
| B | A |

- And for 3 x 3 Latin square we have also one standard square such as,

| | | |
|---|---|---|
| A | B | C |
| B | C | A |
| C | A | B |

- But for a 4 x 4 Latin square, there are four possible standard squares as,

| | | | |
|---|---|---|---|
| A | B | C | D |
| B | A | D | C |
| C | D | A | B |
| D | C | B | A |

| | | | |
|---|---|---|---|
| A | B | C | D |
| B | A | D | C |
| C | D | B | A |
| D | C | A | B |

| | | | |
|---|---|---|---|
| A | B | C | D |
| B | C | D | A |
| C | D | A | B |
| D | A | B | C |

| | | | |
|---|---|---|---|
| A | B | C | D |
| B | D | A | C |
| C | A | D | B |
| D | C | B | A |

Each Latin square of k x k yields k! (k-1)! Different squares by permuting the rows except first and all the k columns. According we may obtain from a square of design.

**2 x 2** → 2! (2-1)! = 2

**3 x 3** → 3! (3-1)! = 12

**4 x 4** → 4! (4-1)! = 576

**5 x 5** → 5! (5-1)! = 161280

# MATHEMATICAL MODEL

In Latin square design, the observation $Y_{ij(h)}$,

$$Y_{ij(h)} = \mu + R_i + C_j + \tau_h + e_{ij(h)}$$

where $i, j, h = 1,2,3,\ldots,k.$

The $\mu$ denote the general mean

$R_i$ is the effect of the $i$th row

$C_j$ the effect of the $j$th column

$\tau_h$ the effect of the $h$th treatment

$e_{ij(h)}$ is the experimental error, that is assumed to be normally and independently distributed with mean zero and common variance $\sigma^2$.

1. formulation of Hypothesis
2. level of significance
3. Test Statistic
4. Calculation
5. Critical Region
6. Conclusion

# STATISTICAL ANALYSIS

## 1-Hypothesis

1

$H_0$ : $\mu_{\cdot 1} = \mu_{\cdot 2} = \mu_{\cdot 3} \ldots$

$H_1$ : Not all treatment means are equal

3

$H_0'$ : $\mu_{\cdot 1} - \mu_{\cdot 2} = \mu_{\cdot 3} \ldots$

$H_1'$ : Not all column means are equal

2

$H_0''$ : $\mu_{1\cdot} = \mu_{2\cdot} = \ldots$

$H_1''$ : Not all row means are equal

## 2-Level of Significance

$\alpha = ?$

## 3-Test Statistics

$$F_1 = \frac{s_t^2}{s_e^2}$$

$$F_2 = \frac{s_r^2}{s_e^2}$$

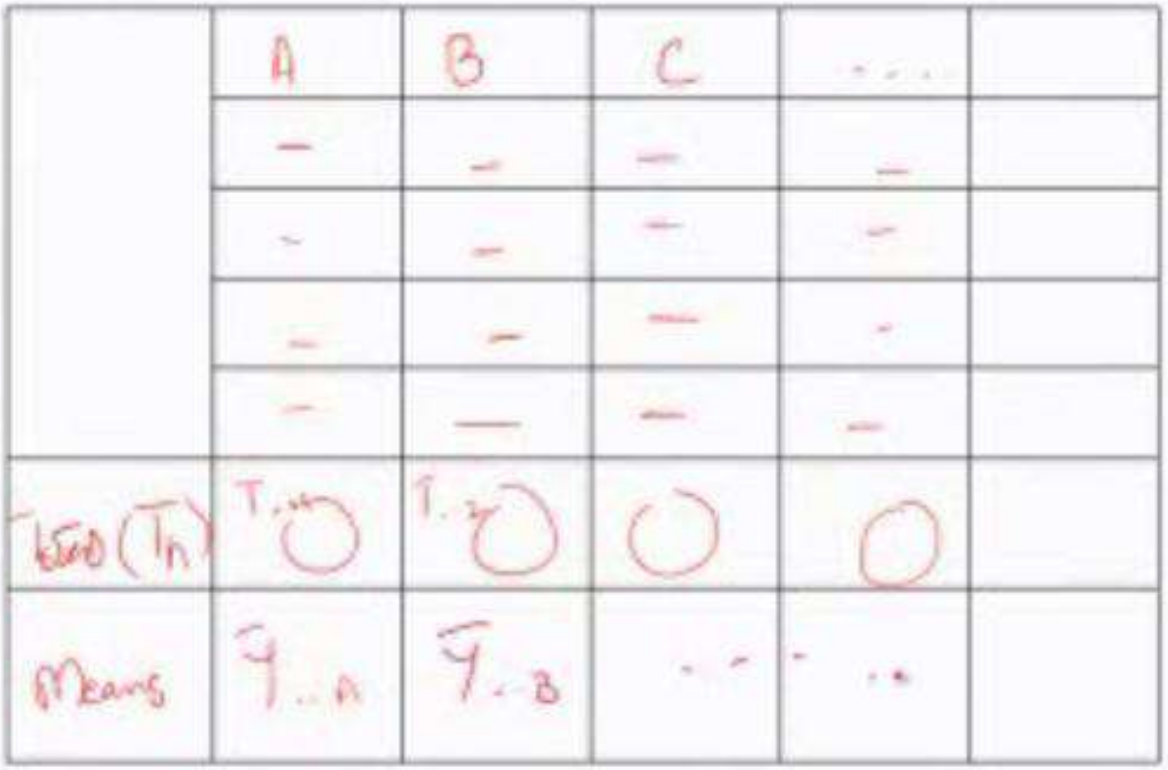
$$F_3 = \frac{s_c^2}{s_e^2}$$

## 4-Calculation

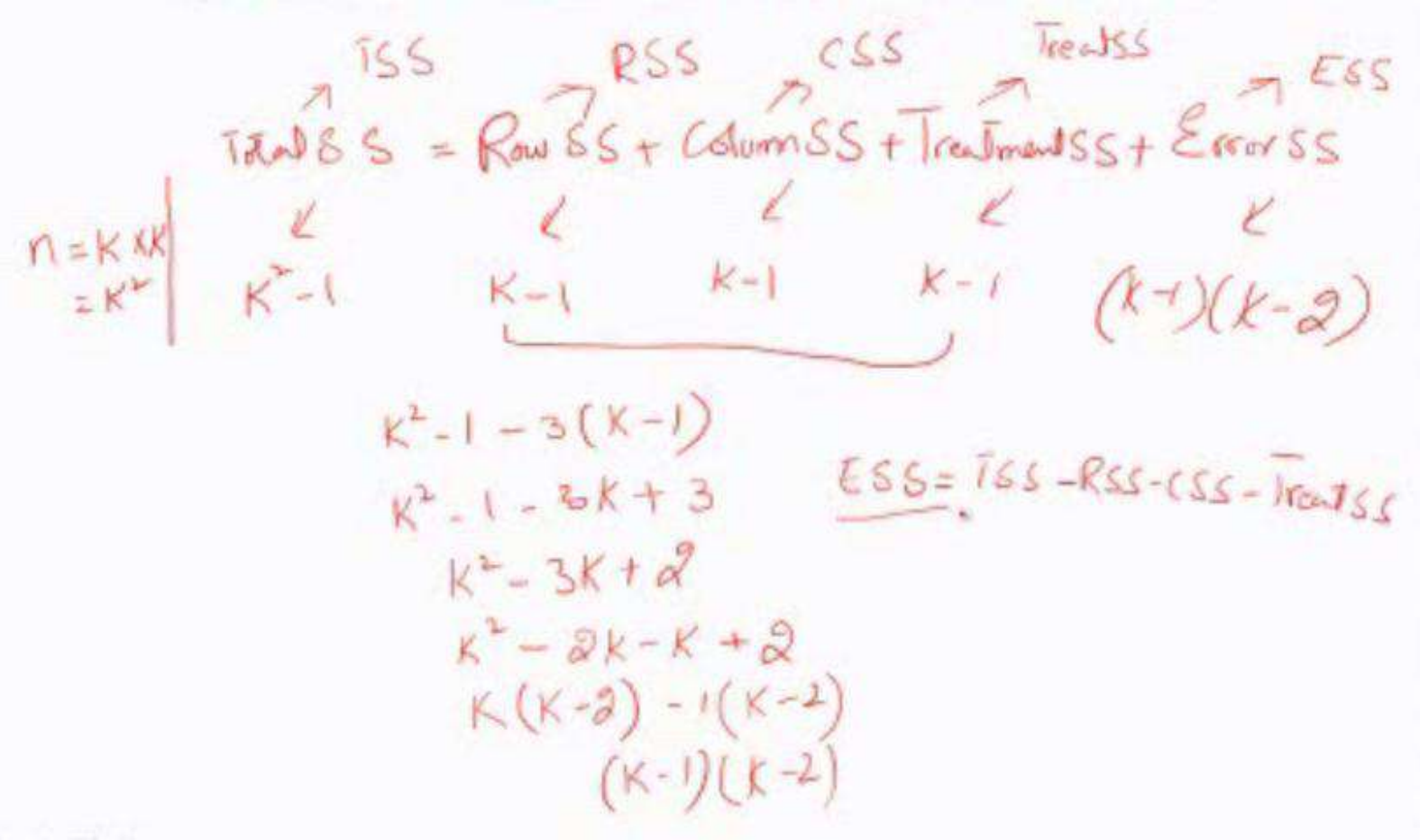
The data obtained from a k x k Latin square design may be arranged in tabular form as,

| | | COLUMNS | | | | | | $R_i$ Total | Means |
|---|---|---|---|---|---|---|---|---|---|
| | | 1 | 2 | 3 | 4 | .... | K | | |
| ROWS | 1 | $Y_{11A}$ | $Y_{12B}$ | $Y_{13C}$ | $Y_{14D}$ | .... | | $R_{1..}$ | $\bar{Y}_{1..}$ |
| | 2 | | | | | | | $R_{2..}$ | $\bar{Y}_{2..}$ |
| | 3 | | | | | | | $R_{3..}$ | $\bar{Y}_{3..}$ |
| | 4 | $Y_{41D}$ | $Y_{42A}$ | $Y_{43B}$ | $Y_{44C}$ | | | | |
| | | | | | | | | | |
| | K | | | | | | | | |
| Total $C_j$ | | $C_{.1.}$ | $C_{.2.}$ | | | | | $G = T_{...}$ | |
| Means | | $\bar{Y}_{.1.}$ | $\bar{Y}_{.2.}$ | $\bar{Y}_{.3.}$ | | | | | $\bar{Y}_{...}$ |

# COMPUTATIONS FOR THE TREATMENT MEANS



# PARTITIONING OF TOTAL SUM OF SQUARES



$$C.F = \frac{(T_{...})^2}{n} = \frac{G^2}{k^2}$$

$$TSS = \sum_i \sum_j \left( Y^2_{ij(h)} \right) - C.F$$

$$RSS = \frac{\sum_i R_i^2}{k} - C.F$$

$$CSS = \frac{\sum_j C_j^2}{k} - C.F$$

$$\text{Treat SS} = \frac{\sum_h T_h^2}{k} - C.F.$$

# ANOVA-TABLE for k x k Latin Square Design

| Source of Variation | Degrees of freedom | Sum of Squares | Mean Square | F-Test |
|---|---|---|---|---|
| Between Treatment | $\nu_1 = k-1$ | $\text{Treat SS} = \frac{\sum T_h^2}{k} - C.F$ | $s_t^2 = \frac{\text{Treat SS}}{\nu_1}$ | $F_1 = s_t^2/s_e^2$ |
| B/W Rows | $\nu_2 = k-1$ | $RSS = \frac{\sum R_i^2}{k} - C.F$ | $s_R^2 = \frac{RSS}{\nu_2}$ | $F_2 = s_R^2/s_e^2$ |
| B/W Columns | $\nu_3 = k-1$ | $CSS = \frac{\sum C_j^2}{k} - C.F$ | $s_C^2 = \frac{CSS}{\nu_3}$ | $F_3 = s_C^2/s_e^2$ |
| Within Errors | $\nu_4 = (k-1)(k-2)$ | $TSS - \text{Treat SS} - CSS - RSS = ESS$ | $s_e^2 = \frac{ESS}{\nu_4}$ | |
| Total | $k^2-1$ | $\sum_i \sum_j (Y_{ij(h)}^2) - CF$ | — | |

## 5-Crirtical region

Reject the null hypotheses when

$$F_1 \geq F_{\alpha,(\nu_1,\nu_4)}$$

$$F_2 \geq F_{\alpha,(\nu_2,\nu_4)}$$

$$F_3 \geq F_{\alpha,(\nu_3,\nu_4)}$$

## 6-Conclusion

# EXAMPLE

The atmosphere in 4 different districts of a large town was sampled, the samples being taken at 4 different heights. Four different tests for the presence of a certain chemical were made on the samples. The arrangement is shown in the following table with the percentage by weight of the chemical as determined by the tests. Letter denote the difference tests.

| | | DISTRICTS | | | |
|---|---|---|---|---|---|
| | | **1** | **2** | **3** | **4** |
| **HEIGHTS** | **1** | **A** 8 | **B** 5.3 | **C** 4.1 | **D** 5 |
| | **2** | **D** 6.8 | **A** 4.9 | **B** 4.1 | **C** 3.2 |
| | **3** | **B** 6.3 | **C** 4.7 | **D** 4.0 | **A** 5 |
| | **4** | **C** 5.7 | **D** 3.3 | **A** 4.0 | **B** 4.2 |

Is there any evidence of significant variation from district to district and between heights in the percentage of the chemical present in the atmosphere..

## 1-Hypothesis

$H_0: \mu_{\cdot 1} = \mu_{\cdot 2} = \mu_{\cdot 3} = \mu_{\cdot 4}$

$H_1:$ Not all treatments means are equal.

$H_0': \mu_{1\cdot} = \mu_{2\cdot} = \mu_{3\cdot} = \mu_{4\cdot}$

$H_1':$ Not all row means are equal

$H_0'': \mu_{\cdot 1} = \mu_{\cdot 2} = \mu_{\cdot 3} = \mu_{\cdot 4}$

$H_1'':$ Not all column means are equal.

## 2-Level of Significance

$\alpha = 0.05$

## 3-Test Statistic

$$F_1 = \frac{s_t^2}{s_e^2}$$ — Tests

$$F_2 = \frac{s_r^2}{s_e^2}$$ — Weights

$$F_3 = \frac{s_c^2}{s_e^2}$$ — Districts

## 4-Calculations

| HEIGHTS | DISTRICTS | | | | $R_i$ | $R_i^2$ |
|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | | |
| 1 | A 8 | B 5.3 | C 4.1 | D 5 | 22.4 | 501.76 |
| 2 | D 6.8 | A 4.9 | B 4.1 | C 3.2 | 19.0 | 361.00 |
| 3 | B 6.3 | C 4.7 | D 4.0 | A 5 | 20.0 | 400 |
| 4 | C 5.7 | D 3.3 | A 4.0 | B 4.2 | 17.2 | 295.84 |
| $C_j$ | 26.8 | 18.2 | 16.2 | 17.4 | 78.6 | 1558.60 |
| $C_j^2$ | 718.4 | 331.24 | 262.44 | 302.76 | 1614.68 | — |
| $\sum \sum r_{ij(ki)}$ | 182.42 | 85.48 | 65.62 | 77.88 | 411.00 | |

| | TESTS | | | | |
|---|---|---|---|---|---|
| | A | B | C | D | |
| | 8 | 5.3 | 4.1 | 5 | |
| | 4.9 | 4.1 | 3.2 | 6.8 | |
| | 5 | 6.3 | 4.7 | 4.0 | |
| | 4.0 | 4.2 | 5.7 | 3.3 | Total |
| $T_n$ | 81.9 | 19.9 | 17.7 | 19.1 | 78.6 |
| $T_n^2$ | 479.61 | 396.01 | 313.29 | 364.81 | 1553.72 |

$$TSS = \sum_i \sum_j y_{ij(h)}^2 - C.F \qquad C.F = \frac{G^2}{K^2} = \frac{(78.6)^2}{16}$$

$$= 411.00 - 386.1225 \qquad C.F = 386.1225$$

$$= 24.8775$$

$$RSS = \frac{\sum R_i^2}{K} - C.F = \frac{1558.60}{4} - 386.1225$$

$$RSS = 3.5275$$

Table

$$CSS = \frac{\sum C_j^2}{K} - C.F = \frac{1614.68}{4} - 386.1225$$

$$CSS = 17.5475$$

$$TreatSS = \frac{\sum T_h^2}{K} - C.F = \frac{1553.72}{4} - 386.1225$$

$$TreatSS = 2.3075$$

$$ESS = TSS - RSS - CSS - TreatSS$$

$$ESS = 1.4950$$

# ANOVA-TABLE for 4 x 4 Latin Square Design

| Source of Variation | Degrees of freedom | Sum of Squares | Mean Square | F-Test |
|---|---|---|---|---|
| B/W Treatments | $v_1 = 4-1=3$ | Treat SS = 2.3075 | $s_t^2 =$ 5.8492 | $F_1 = 23.47$ |
| B/W Rows | $v_2 = 3$ | RSS = 3.5275 | $s_r^2 =$ 1.1758 | $F_2 = 4.72$ |
| B/W Columns | $v_3 = 3$ | CSS = 17.5475 | $s_c^2 =$ 0.7692 | $F_3 = 3.09$ |
| Error | $v_4 = 6$ | 1.4950 | $s_e^2 =$ 0.2490 | |
| Total | 15 | 24.8725 | | |

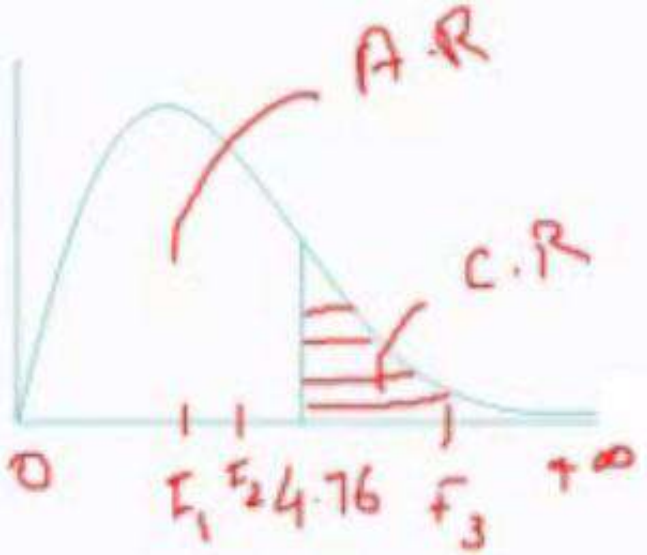
## 5-Crirtical Region

Reject Null hypothesis if

3.09 $\leftarrow F_1 > F_{0.05(3,6)} = 4.76$

4.72 $\leftarrow F_2 > F_{0.05(3,6)} = 4.76$

23.47 $\leftarrow F_3 > F_{0.05(3,4)} = 4.76$




F-table

## 6-Conclusion

By comparing the computed values of *F* with the corresponding critical *F*, we conclude that the variation among the districts is significant and the variation among the heights is insignificant. Also, there is no difference between the sensitivity of the test.

# THANK YOU

# COURSE: STA 408
# DESIGN & ANALYSIS OF EXPERIMENT

LECTURE DELIVERED BY

**SARAH AHMAD**
DEPARTMENT OF STATISTICS
VIRTUAL UNIVERSITY OF PAKISTAN



# COURSE: STA 408
# DESIGN & ANALYSIS OF EXPERIMENT

LECTURE NO.6

# ADVANTAGES & DISADVANTAGES

- ✓ A Latin square design is generally more efficient than RCBD.
- ✓ A Latin square design reduces the error variance by controlling two sources of variation.
- ✓ The analysis of a Latin square design is simple and remains relatively simple with missing observations.

- ✓ A Latin square design is less flexible than RCBD. It is practical only for 5 to 10 treatments. When the number of treatments exceeds 10, the design is seldom used.
- ✓ For a small number of treatments, a Latin Square design does not provide a sufficient number of replicates to give a valid estimate of error.
- ✓ Replication in a Latin square design is costly.
- ✓ In agriculture experimentation, the land requirement is rigid, the actual layout in the field may be laborious and approach to the central most plot becomes difficult.

# MISSING OBSERVATIONS IN LATIN SQUARE DESIGN

STA: 408

Suppose,

| | | COLUMNS | | | | | | Total |
|---|---|---|---|---|---|---|---|---|
| | | 1 | 2 | 3 | 4 | ..... | $k$ | |
| ROWS | 1 | $x$ | $Y_{12B}$ | $Y_{13C}$ | $Y_{14D}$ | ..... | - - - | $R_1' + x$ |
| | 2 | ⋮ | ⋮ | ⋮ | ⋮ | ..... | ⋮ | $R_2$ |
| | 3 | ⋮ | ⋮ | ⋮ | ⋮ | ⋮ | ⋮ | $R_3$ |
| | 4 | $Y_{41D}$ | $Y_{42A}$ | $Y_{43B}$ | $Y_{44C}$ | - - - | - - - | $R_4$ |
| | | ⋮ | ⋮ | ⋮ | ⋮ | ⋮ | ⋮ | ⋮ |
| | $k$ | ⋮ | ⋮ | ⋮ | ⋮ | ⋮ | ⋮ | ⋮ |
| Total | | $C_1' + x$ | $C_2$ | $C_3$ | $C_4$ | - - - | - - - | $G' + x$ |

| | TREATMENTS | | | | |
|---|---|---|---|---|---|
| | A | B | C | ..... | *k* |
| | — | — | — | ..... | — |
| | — | — | — | ..... | — |
| Total | $T_1' + 2$ | $T_2$ | $T_3$ | - - - - | $T_k$ |

Performing the analysis of variance, we get the following sums of squares:

| Source of Variation | Sum of Squares |
|---|---|
| Rows | $\frac{(R_1'+T)^2+\sum_{i=2}^{K} R_i^2}{K} - \frac{(G'+T)^2}{K^2} = RSS$ |
| Columns | $\frac{(C_1'+T)^2+\sum_{j=2}^{K} C_j^2}{K} - \frac{(G'+T)^2}{K^2} = CSS$ |
| Treatments | $\frac{(T_1'+T)^2+\sum_{k=2}^{K} T_k^2}{K} - \frac{(G'+T)^2}{K^2} = TrSS$ |
| Error | By Subtraction |
| Total | $T^2+\sum_i\sum_j Y_{ij}^2 - \frac{(G'+T)^2}{K^2} = TSS$ |

Thus the error sum of squares is obtained as

$$ESS = TSS - RSS - CSS - T_6SS$$

$$ESS = x^2 + \sum_i \sum_j y_{ij}^2 - \frac{(G'+x)^2}{k^2} - \frac{(R_1'+x)^2}{k} - \frac{\sum_{i=2}^{k} R_i^2}{k} + \frac{(G'+x)^2}{k^2}$$

$$- \frac{(C_1'+x)^2}{k} - \frac{\sum_{j=2}^{k} C_j^2}{k} + \frac{(G'+x)^2}{k^2}$$

$$- \frac{(T_1'+x)^2}{k} - \frac{\sum_{h=2}^{k} T_h^2}{k} + \frac{(G'+x)^2}{k^2}$$

$$ESS = x^2 + \frac{2(G_1' + x)^2}{k^2} - \frac{1}{k}\left[(R_1' + x)^2 + (C_1' + x)^2 + (T_1' + x)^2\right]$$

+ Terms in which $x$ is not involved

Differentiate ESS w.r.t $x$,

$$\frac{\partial(ESS)}{\partial x} = 2x + \frac{(2\times 2)(G_1' + x)}{k^2} - \frac{1}{k}\left[2(R_1' + x) + 2(C_1' + x) + 2(T_1' + x)\right]$$

$$2x + \frac{4(G_1' + x)}{k^2} - \frac{2}{k}\left[(R_1' + x) + (C_1' + x) + (T_1' + x)\right] = 0$$

$$2x + \frac{4G'}{K^2} + \frac{4x}{K^2} - \frac{2R_1'}{K} - \frac{2x}{K} - \frac{2C_1'}{K} - \frac{2x}{K} - \frac{2\bar{T}_1'}{K} - \frac{2x}{K} = 0$$

$$2x + \frac{4x}{K^2} - \frac{2x}{K} - \frac{2x}{K} - \frac{2x}{K} - \frac{2R_1'}{K} - \frac{2C_1'}{K} - \frac{2\bar{T}_1'}{K} + \frac{4G'}{K^2} = 0$$

$$2x\left(1 + \frac{2}{K^2} - \frac{1}{K} - \frac{1}{K} - \frac{1}{K}\right) = \frac{2}{K}\left(R_1' + C_1' + \bar{T}_1'\right) - \frac{4G'}{K^2}$$

$$x\left(1 + \frac{2}{K^2} - \frac{3}{K}\right) = \frac{2}{2K}\left(R_1' + C_1' + \bar{T}_1'\right) - \frac{4^2}{2}\frac{G_1'}{K^2}$$

$$\chi\left(\frac{K^2-3K+2}{K^2}\right) = \frac{1}{K}(R_1' + C_1' + T_1') - \frac{2G'}{K^2}$$

$$\chi\left(\frac{K^2-K-2K+2}{K^2}\right) = \frac{K(R_1' + C_1' + T_1') - 2G'}{K^2}$$

$$\chi\left(\frac{K(K-1)-2(K-1)}{K^2}\right) = \text{''}$$

$$\chi\left\{\frac{(K-1)(K-2)}{K^2}\right\} = \text{''}$$

$$\chi = \frac{K(R_1' + C_1' + T_1') - 2G'}{K^2} \times \frac{K^2}{(K-1)(K-2)}$$

$$\chi = \frac{K(R_1' + C_1' + T_1') - 2G'}{(K-1)(K-2)}$$

$$x = \frac{k(R_i' + C_j' + T_h') - 2G'}{(k-1)(k-2)}$$

no. of treatments

Row total (missing obs)

Column total (missing obs)

Grand total

no. of treatments

$$x = \frac{K(R+C+T) - 2G}{(K-1)(K-2)}$$

# EXAMPLE

An agriculture experiment was conducted on the Latin square plan to test the effect on yield due to change of treatment (5 kinds) and also to variation of soil in each of two perpendicular directions. The results are set out in the Latin square below, in which letters correspond to treatments, while rows and columns correspond to the two perpendicular directions.

| | | | | |
|---|---|---|---|---|
| A 7.4 | D $x$ | E 5.8 | B 12.0 | C 14.3 |
| C 11.8 | B 6.5 | A 8.7 | E 7.6 | D 7.9 |
| D 10.1 | C 17.9 | B 9.0 | A 8.5 | E 7.1 |
| E 8.8 | A 10.1 | C 15.7 | D 11.1 | B 7.4 |
| B 11.8 | E 8.8 | D 14.3 | C 18.4 | A 10.1 |

Estimate the missing observation and perform an analysis of variance.

| | | | | | |
|---|---|---|---|---|---|
| A 7.4 | D $x$ | E 5.8 | B 12.0 | C 14.3 | R=39.5 |
| C 11.8 | B 6.5 | A 8.7 | E 7.6 | D 7.9 | |
| D 10.1 | C 17.9 | B 9.0 | A 8.5 | E 7.1 | |
| E 8.8 | A 10.1 | C 15.7 | D 11.1 | B 7.4 | |
| B 11.8 | E 8.8 | D 14.3 | C 18.4 | A 10.1 | |
| | C=43.3 | | | | G=251.1 |

| | | | | | |
|---|---|---|---|---|---|
| **Treatment D :** | 7.9 | 10.1 | 11.1 | 14.3 | 43.4 |

$$x = \frac{k(R + C + T) - 2G}{(k-1)(k-2)}$$

$k = 5$

$$x = \frac{(5)(39.5 + 43.3 + 43.4) - 2(251.1)}{(5-1)(5-2)}$$

$$x = \frac{631 - 502.2}{12} = \frac{128.8}{12} = 10.7333 \quad \boxed{x = 10.7}$$

STA 408

① Hypotheses

$H_0$: All effects on yield due to variation of soil in columns are zero.

$H_1$: - - - - - - - - are not zero.

$H_0'$: - - - - - - - - rows are zero

$H_1'$: - - - - - - - - rows are not equal to zero.

$H_0''$: All effects on yield due to change in treatments are zero.

$H_1''$: - - - - - - - are not equal to zero.

| | COLUMNS | | | | | | |
|---|---|---|---|---|---|---|---|
| ROWS | 1 | 2 | 3 | 4 | 5 | $R_i$ | $R_i^2$ |
| 1 | A 7.4 | D 10.7 | E 5.8 | B 12.0 | C 14.3 | 50.2 | 2520.04 |
| 2 | C 11.8 | B 6.5 | A 8.7 | E 7.6 | D 7.9 | 42.5 | 1806.25 |
| 3 | D 10.1 | C 17.9 | B 9.0 | A 8.5 | E 7.1 | 52.6 | 2766.76 |
| 4 | E 8.8 | A 10.1 | C 15.7 | D 11.1 | B 7.4 | 53.1 | 2819.615 |
| 5 | B 11.8 | E 8.8 | D 14.3 | C 18.4 | A 10.1 | 63.4 | 4019.56 |
| $C_j$ | 49.9 | 54 | 53.5 | 57.6 | 46.8 | T.. 261.8 | 13932.22 |
| $C_j^2$ | 2490.01 | 2916 | 2862.25 | 3317.76 | 2190.24 | 13776.26 | |
| $\sum\sum y_{ij}^2$ | 512.69 | 656.6 | 641.31 | 735.78 | 474.08 | 3020.46 | |

| | TREATMENTS | | | | | |
|---|---|---|---|---|---|---|
| | A | B | C | D | E | |
| | 7.4 | 12 | 14.3 | 10.7 | 5.8 | |
| | 8.7 | 6.5 | 11.8 | 7.9 | 7.6 | |
| | 8.5 | 9 | 17.9 | 10.1 | 7.1 | |
| | 10.1 | 7.4 | 15.7 | 11.1 | 8.8 | |
| | 10.1 | 11.8 | 18.4 | 14.3 | 8.8 | |
| $T_h^2$ | 2007.04 | 2180.89 | 6099.61 | 2926.81 | 1451.61 | 14665.96 |

$$C.F = \frac{(T_{..})^2}{k^2} = \frac{(261.8)^2}{25} = 2741.5696$$

$$TSS = \sum_i \sum_j Y_{ij}^2(h) - C.F = 3020.46 - 2741.5696$$

$$TSS = 278.8904$$

$$CSS = \frac{\sum_j C_j^2}{k} - C.F = \frac{13776.26}{5} - 2741.5696$$

$$CSS = 13.6824$$

$$RSS = \frac{\sum_i R_i^2}{k} - C.F = \frac{13932.22}{5} - 2741.5696$$

$$RSS = 44.8744$$

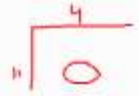
## ANOVA-TABLE

| Source of Variation | Degrees of freedom | Sum of Squares | Mean Square | F-Test |
|---|---|---|---|---|
| Columns | $K-1 = 5-1 = 4$ | 13.6824 | $S_c^2 = 3.4206$ | $F_1 = 1.3105$ |
| Rows | 4 | 44.8744 | $S_r^2 = 11.2186$ | $F_2 = 4.2981$ |
| Treatments | 4 | 191.6224 | $S_t^2 = 47.9056$ | $F_3 = 18.3539$ |
| Error | $(K-1)(K-2)-1 = 11$ | 28.7114 | $S_e^2 = 2.6101$ | |
| Total | 23 | 278.8904 | — | — |

(5) Critical Region

Reject null hypothesis if $F_{cal} > F_{tab}$

$F_{0.05(4,11)} = 3.36$




(6) Conclusion

As $F_1 < F_{tab}$, we do not reject $H_0$

$F_2 > F_{tab}$, $H_0'$ is rejected

$F_3 > F_{tab}$, $H_0''$ reject

(a) The variation of soil in columns is not significant

(b) There are significant differences in soil in rows.

(c) The effects on yield due to change in treatments are significantly different.

# EXAMPLE

The following data represent the yield (in tons per ha) of four cultivars of Khaya arranged in a 4 x 4 Latin square with one missing value. The four cultivars are A (KH4), B (KH2O), C (KH29) and D (KS15).

| | COLUMNS | | | |
|---|---|---|---|---|
| ROWS | 1 | 2 | 3 | 4 |
| I | 6.2 (C) | 5.1 (D) | 3.3 (B) | 5.7 (A) |
| II | 5.1 (B) | 3.8 (A) | *M* (C) | 3.9 (D) |
| III | 5.4 (D) | 4.8 (C) | 6.9 (A) | 4.6 (B) |
| IV | 3.2 (A) | 2.9 (B) | 5.2 (D) | 5.4 (C) |

Estimate the missing observation and perform the analysis of variance to test for significant differences among the cultivars.

① Hypothesis

$H_0$: There is no significant difference in the yield of the cultivars

$H_1$: There is significant .....

② $\alpha = 0.05$

③ $F = \frac{s_t^2}{s_e^2}$

④ Calculation

⑤ Critical Region

⑥ Conclusion

| | COLUMNS | | | | |
|---|---|---|---|---|---|
| ROWS | 1 | 2 | 3 | 4 | Total |
| I | 6.2 (C) | 5.1 (D) | 3.3 (B) | 5.7 (A) | |
| II | 5.1 (B) | 3.8 (A) | M (C) | 3.9 (D) | R = 12.8 |
| III | 5.4 (D) | 4.8 (C) | 6.9 (A) | 4.6 (B) | |
| IV | 3.2 (A) | 2.9 (B) | 5.2 (D) | 5.4 (C) | |
| Total | | | C = 15.4 | | G = 71.4 |

K = 4

Total

| Treatment C : | 6.2 | M | 4.8 | 5.4 | 16.4 |
|---|---|---|---|---|---|

$$\hat{\tau} = \frac{K(R+C+T) - 2G}{(K-1)(K-2)} = \frac{(4)(12.8+15.4+16.4) - 2(71.4)}{(4-1)(4-2)}$$

$$\hat{\tau} = 5.9$$

| ROWS | COLUMNS | | | |
|---|---|---|---|---|
| | 1 | 2 | 3 | 4 |
| I | 6.2 (C) | 5.1 (D) | 3.3 (B) | 5.7 (A) |
| II | 5.1 (B) | 3.8 (A) | 5.9 (C) | 3.9 (D) |
| III | 5.4 (D) | 4.8 (C) | 6.9 (A) | 4.6 (B) |
| IV | 3.2 (A) | 2.9 (B) | 5.2 (D) | 5.4 (C) |

Sum of Squares

RSS = 3.4675

CSS = 2.9325

TrSS = 5.1825

ESS = 8.115

TSS = 19.6975

C.F = 374.4225

# THANK YOU

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# COURSE: STA 408
# DESIGN & ANALYSIS OF EXPERIMENT

LECTURE DELIVERED BY

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# COURSE: STA 408
# DESIGN & ANALYSIS OF EXPERIMENT

LECTURE NO.7

# EFFICIENCY OF LATIN SQUARE DESIGN

- **Relative to RCBD**

When taking rows as blocks

$$\frac{1}{k}\left[\frac{s_c^2}{s_e^2}+(k-1)\right]$$

ms for columns; MS Error; no of rows

$= RE(LS, RCBD_{rows})$

When taking columns as blocks

$$\frac{1}{k}\left[\frac{s_r^2}{s_e^2}+(k-1)\right]$$

MS rows; MS error

$= RE(LS, RCBD_{cols})$

- **Relative to CR Design**

MS rows MS columns MS error

$$RE(LS, CR) = \frac{1}{k}\left[\frac{s_r^2 + s_c^2 + (k-1)s_e^2}{(k+1)s_e^2}\right]$$

# EXAMPLE

Consider an experiment to investigate the effect of four different diets on milk production of cows. There are four cows in the study. During each lactation period the cows receive a different diet. Assume that there is a washout period between diets so that previous diet does not affect future results. Lactation period and cows are used as blocking variables.

A 4 *x* 4 Latin square design is implemented.

| | Cow | | | |
|---|---|---|---|---|
| Period | 1 | 2 | 3 | 4 |
| 1 | A=38 | B=39 | C=45 | D=41 |
| 2 | B=32 | C=37 | D=38 | A=30 |
| 3 | C=35 | D=36 | A=37 | B=32 |
| 4 | D=33 | A=30 | B=35 | C=33 |

=49.06 18.2292 0.8125

The mean square period, mean square cows and mean square error are 49.06, 18.2292 and 0.8125 respectively. Estimate the relative efficiency of Latin square design to RCBD

i) by taking the cows as blocks

ii) by taking lactation periods as blocks

① by taking the columns as blocks

$$R_E(LS, RCBD_{cols}) = \frac{1}{k}\left[\frac{\sigma_r^2}{\sigma_e^2} + (k-1)\right]$$

$$= \frac{1}{4}\left[\frac{49.06}{0.8125} + (4-1)\right] = \frac{1}{4}\left[\frac{49.06}{0.8125} + 3\right]$$

$$= 15.8454$$

$$= 15.85$$

(ii) by Taking rows as blocks

$$RE(LS, RCBD_{rows}) = \frac{1}{K}\left[\frac{\sigma_c^2}{\sigma_e^2} + (K-1)\right]$$

$$= \frac{1}{4}\left[\frac{18.2292}{0.8125} + 3\right] = 6.36$$

# EXAMPLE

Consider the given ANOVA table for 5 brands of air cleaner in 5x5 Latin Square and find the efficiency of this design relative to CR design.

| Source of variation | Degrees of freedom | Sum of squares | Mean Square | $F$ |
|---|---|---|---|---|
| Rows | 4 | 44512.56 | 11128.14 | 3.85 |
| Columns | 4 | 2177.76 | 544.44 | 0.19 |
| Treatments | 4 | 58034.16 | 14508.54 | 5.02 |
| Error | 12 | 34647.52 | 2887.29 | |
| Total | 24 | 139372.00 | | |

$$RE(LS, CR) = \frac{\hat{\sigma}_r^2 + \hat{\sigma}_c^2 + (K-1)\hat{\sigma}_e^2}{(K+1)\hat{\sigma}_e^2}$$

$$= \frac{11128.14 + 544.44 + (5-1)(2887.29)}{(5+1)(2887.29)} = 1.34 = 134\%$$

That is, approximately 34% more observations per treatment would be required in a completely randomized design to obtain the same precision for estimating the treatment means as with this Latin square design.

# MULTIPLE COMPARISON TESTS

If $F$ -test after the analysis of variance rejects our null hypothesis, all we can conclude is that the $k$ population means are not all equal. The conclusion might not be sufficient to satisfy the experimenter, rather he would like to know which means (or set of means) might differ significantly from each other.

For this purpose, several tests based on different view points, have been developed to make comparisons between pair of means. Such tests, used as a follow-up to $F$ - tests are called the **multiple comparison tests**.

**MOST COMMONLY USED TESTS**

- Fisher's Least Significance Difference (or LSD) test
- The Student Newman-keul's Multiple Range test
- The Duncan's Multiple Range test
- The Scheffe's test
- The Tukey's *T*-method

# *LSD*-TEST

When the null hypothesis of equal means is rejected by *F*-test after the ANOVA, we can test the significance of differences between means of $k$ samples (or treatments) by computing the smallest difference that would be judge significant and to compare the absolute values of all differences of means with it. This smallest difference is called the least significant difference or *LSD* and is given by

$$LSD = t_{\alpha/2,(\nu)}\sqrt{\frac{2(\text{MSE})}{r}}$$

Two means are declared to come from populations with significantly different means, when the absolute value of their difference exceeds the *LSD*.

# EXAMPLE

Five fertilizers A, B, C, D, and E were tested by arranging plants in a Latin Square design in a field. The rows and columns in the table are rows and columns in the field. The yields in bushels per plot are shown.

| | | COLUMNS | | | | |
|---|---|---|---|---|---|---|
| | | 1 | 2 | 3 | 4 | 5 |
| ROWS | 1 | B 4.9 | D 6.4 | E 3.3 | A 9.5 | C 11.8 |
| | 2 | C 9.3 | A 4.0 | B 6.2 | E 5.1 | D 5.4 |
| | 3 | D 7.6 | C 15.4 | A 6.5 | B 6.0 | E 4.6 |
| | 4 | E 5.3 | B 7.6 | C 13.2 | D 8.6 | A 4.9 |
| | 5 | A 9.3 | E 6.3 | D 11.8 | C 15.9 | B 7.6 |

$\alpha < 0.05$

Analyze the data for evidence at the 5 percent level that the mean yields are not equal for the five fertilizers. Also pick up the best fertilizer.

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## 1-Hypothesis

$H_0$ : The mean yields for all the five fertilizers are equal.

$H_1$ : Not all mean yields for the fertilizers are equal.

## 2-Level of significance

$\alpha = 0.05$

## 3-Test Statistic

$$F = \frac{MS\ for\ fertilizers}{MS\ for\ error} = \frac{s_t^2}{s_e^2}$$

## 4-Calculations

| ROWS | COLUMNS | | | | | $R_i$ | $R_i^2$ |
|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | | |
| 1 | B 4.9 | D 6.4 | E 3.3 | A 9.5 | C 11.8 | 35.9 | 1288.81 |
| 2 | C 9.3 | A 4.0 | B 6.2 | E 5.1 | D 5.4 | 30 | 900 |
| 3 | D 7.6 | C 15.4 | A 6.5 | B 6.0 | E 4.6 | 40.1 | 1608.10 |
| 4 | E 5.3 | B 7.6 | C 13.2 | D 8.6 | A 4.9 | 39.6 | 1568.16 |
| 5 | A 9.3 | E 6.3 | D 11.8 | C 15.9 | B 7.6 | 50.9 | 2590.81 |
| $C_j$ | 36.4 | 39.7 | 41.0 | 45.1 | 34.3 | 196.5 | 7955.79 |
| $C_j^2$ | 1324.96 | 1576.09 | 1681.00 | 2034.01 | 1176.49 | 7792.55 | |

$$\sum_i \sum_j y_{ij(k)}^2 = 1829.83$$

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| | FERTILIZERS | | | | |
|---|---|---|---|---|---|
| | A | B | C | D | E |
| | 9.5 | 4.9 | 11.8 | 6.4 | 3.3 |
| | 4.0 | 6.2 | 9.3 | 5.4 | 5.1 |
| | 6.5 | 6.0 | 15.4 | 7.6 | 4.6 |
| | 4.9 | 7.6 | 13.2 | 8.6 | 5.3 |
| | 9.3 | 7.6 | 15.9 | 11.8 | 6.3 |
| Total ($T_h$) | 34.2 | 32.3 | 65.6 | 39.8 | 24.6 |
| Means | 6.84 | 6.46 | 13.12 | 7.96 | 4.92 |

$$TSS = \sum_i \sum_j Y_{ij(lh)}^2 - C.F$$

$$C.F = \frac{G^2}{K^2}$$

$$RSS = \frac{\sum_i R_i^2}{K} - C.F$$

$$CSS = \frac{\sum_j C_j^2}{K} - C.F$$

$$ESS = TSS - RSS - CSS - TrtSS$$

$$TrtSS = \frac{\sum_h T_h^2}{K} - C.F$$

## ANOVA-TABLE for 5 x 5 Latin Square Design

| Source of Variation | Degrees of freedom | Sum of Squares | Mean Square | *F*-Test |
|---|---|---|---|---|
| Rows | 4 | 46.67 | 11.67 | |
| Columns | 4 | 14.02 | 3.51 | |
| Treatments (Fertilizers) | 4 | 196.61 | 49.15 | $F = \frac{49.15}{2.34} = 21.00$ |
| Error | 12 | 28.04 | 2.34 | |
| Total | 24 | 285.34 | | |

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## 5-Crirtical region

Reject $H_0$ if $F \geq F_{0.05(4,12)}$

$F \geq 3.26$

$F_{cal} = 21.00$

$21.00 > 3.26$

F-table

## 6-Conclusion

Since the computed value of F = 21.00 falls in the critical region, we therefore reject our null hypothesis. Hence we conclude that the data provide sufficient evidence to indicate at the 5% significance level that the mean yields for the fertilizers are not equal.

# LEAST SIGNIFICANCE DIFFERENCE (*LSD*)

Now, we compute the Least Significance Difference (LSD) to determine which fertilizer is to be considered the best one.

| A | B | C | D | E |
|---|---|---|---|---|
| $\bar{y}_A$ | $\bar{y}_B$ | $\bar{y}_C$ | $\bar{y}_D$ | $\bar{y}_E$ |
| 6.84 | 6.46 | 13.12 | 7.96 | 4.92 |

$$LSD = t_{\alpha/2,\nu}\sqrt{\frac{2MSE}{R}}$$

$$LSD = t_{0.025,12}\sqrt{\frac{2(2.34)}{5}}$$

$$LSD = 2.18(0.96) = 2.09$$

$^{5}C_2 = {}^{5}C_2 = 10$ pairs

| PAIRS | | | LSD-Value | Decision |
|---|---|---|---|---|
| $\lvert\bar{y}_A - \bar{y}_B\rvert$ | $= \lvert 6.84 - 6.46\rvert = 0.38$ | $<$ | 2.09 | Insignificant |
| $\lvert\bar{y}_A - \bar{y}_C\rvert$ | $= \lvert 6.84 - 13.12\rvert = 6.28$ | $>$ | 2.09 | – Significant |
| $\lvert\bar{y}_A - \bar{y}_D\rvert$ | $= \lvert 6.84 - 7.96\rvert = 1.12$ | $<$ | 2.09 | Insignificant |
| $\lvert\bar{y}_A - \bar{y}_E\rvert$ | $= \lvert 6.84 - 4.92\rvert = 1.92$ | $<$ | 2.09 | " |
| $\lvert\bar{y}_B - \bar{y}_C\rvert$ | $= \lvert 6.46 - 13.12\rvert = 6.66$ | $>$ | 2.09 | – Significant |
| $\lvert\bar{y}_B - \bar{y}_D\rvert$ | $= \lvert 6.46 - 7.96\rvert = 1.5$ | $<$ | 2.09 | Insignificant |
| $\lvert\bar{y}_B - \bar{y}_E\rvert$ | $= \lvert 6.46 - 4.92\rvert = 1.54$ | $<$ | 2.09 | " |
| $\lvert\bar{y}_C - \bar{y}_D\rvert$ | $= \lvert 13.12 - 7.96\rvert = 5.16$ | $>$ | 2.09 | – Significant |
| $\lvert\bar{y}_C - \bar{y}_E\rvert$ | $= \lvert 13.12 - 4.92\rvert = 8.2$ | $>$ | 2.09 | – " |
| $\lvert\bar{y}_D - \bar{y}_E\rvert$ | $= \lvert 7.96 - 4.92\rvert = 3.04$ | $>$ | 2.09 | – " |

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# EXAMPLE

Perform the analysis of variance on the following data and analyze the treatment means using the LSD test with a 0.05 level of significance.

| | TREATMENTS | | | | | |
|---|---|---|---|---|---|---|
| **BLOCKS** | **1** | **2** | **3** | **4** | **5** | **6** |
| **1** | 1 | 3 | 6 | 4 | 3 | 2 |
| **2** | 1 | 4 | 4 | 8 | 5 | 1 |
| **3** | 3 | 6 | 7 | 8 | 4 | 3 |
| **4** | 2 | 3 | 2 | 3 | 2 | 1 |

## 1-Hypothesis

$$H_0 : \mu_{.1} = \mu_{.2} = \mu_{.3} = \mu_{.4} = \mu_{.5} = \mu_{.6}$$

$H_1$ : Not all six treatment means are equal.

## 2-Level of significance

$\alpha = 0.05$

## 3-Test Statistic

$$F = \frac{\text{MS for Treatments}}{\text{MS for Error}} = \frac{s_t^2}{s_e^2}$$

## 4-Calculations

| Blocks | Treatments | | | | | $B_{i\cdot}$ | $B_{i\cdot}^2$ |
|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | | |
| 1 | 1 | 3 | 6 | 4 | 3 | | |
| 2 | 1 | 4 | 4 | 8 | 5 | | |
| 3 | 3 | 6 | 7 | 8 | 4 | | |
| 4 | 2 | 3 | 2 | 3 | 2 | | |
| $T_{\cdot j}$ | | | | | | | |
| $T_{\cdot j}^2$ | | | | | | | |
| Means | | | | | | | |
| $\sum_i \sum_j y_{ij}^2$ | | | | | | | |

## ANOVA-TABLE

| Source of Variation | Degrees of freedom | Sum of Squares | Mean Square | *F*-Test |
|---|---|---|---|---|
| Between Treatments | 5 | 51.83 | 10.37 | $F = \frac{10.37}{1.57} = 6.61$ |
| Between blocks | 3 | 28.50 | 9.50 | |
| Error | 15 | 23.50 | 1.57 | |
| Total | 23 | 103.83 | – | |

## 5-Crirtical region

Reject $H_0$ if $F \geq F_{0.05(5,15)}$ (5: d.f for treatment, 15: for error)

$F_{cal} = 6.61$

$6.61 > 2.90$

$F_{0.05(5,15)} = 2.90$

## 6-Conclusion

We reject the null hypothesis of equal treatment means as the computed value of $F = 6.61$ exceeds the table value. As the null hypothesis is rejected, using $F$ - test, we therefore apply the *LSD* test to find out which means differ from each other.

The Least Significant Difference (*LSD*) is given by the relation

$$LSD = t_{\alpha/2,(\nu)}\sqrt{\frac{2(\text{MSE})}{r}}$$

$$LSD = t_{0.025,(15)}\sqrt{\frac{2(1.57)}{4}} = 1.89$$

$$LSD = 1.89$$

| Pairs | | | | *LSD*-Value | Decision |
|---|---|---|---|---|---|
| $\lvert\bar{y}_1 - \bar{y}_2\rvert$ | $\lvert 1.75 - 4.00\rvert$ | $= 2.25$ | $>$ | 1.89 | Significant |
| $\lvert\bar{y}_1 - \bar{y}_3\rvert$ | $\lvert 1.75 - 4.75\rvert$ | $= 3$ | $>$ | 1.89 | // |
| $\lvert\bar{y}_1 - \bar{y}_4\rvert$ | $\lvert 1.75 - 5.75\rvert$ | $= 4$ | $>$ | 1.89 | // |
| $\lvert\bar{y}_1 - \bar{y}_5\rvert$ | $\lvert 1.75 - 3.50\rvert$ | $= 1.75$ | $<$ | 1.89 | Insignificant |
| $\lvert\bar{y}_1 - \bar{y}_6\rvert$ | $\lvert 1.75 - 1.75\rvert$ | $= 0$ | $<$ | 1.89 | // |
| $\lvert\bar{y}_2 - \bar{y}_3\rvert$ | | | | 1.89 | |
| $\lvert\bar{y}_2 - \bar{y}_4\rvert$ | | | | 1.89 | |
| $\lvert\bar{y}_2 - \bar{y}_5\rvert$ | | | | 1.89 | |
| $\lvert\bar{y}_2 - \bar{y}_6\rvert$ | | | | 1.89 | |
| $\lvert\bar{y}_3 - \bar{y}_4\rvert$ | | | | 1.89 | |
| $\lvert\bar{y}_3 - \bar{y}_5\rvert$ | | | | 1.89 | |
| $\lvert\bar{y}_3 - \bar{y}_6\rvert$ | | | | 1.89 | |
| $\lvert\bar{y}_4 - \bar{y}_5\rvert$ | | | | 1.89 | |
| $\lvert\bar{y}_4 - \bar{y}_6\rvert$ | | | | 1.89 | |
| $\lvert\bar{y}_5 - \bar{y}_6\rvert$ | | | | 1.89 | |

$^6C_2 = 15$ Pairs

# EXAMPLE

The effects of four types of graphite coaters on light box readings are to be studied. As these readings might differ from day to day, observations are to be taken on each of the four types every day for three days. The results are:

| | Graphite Coater Type | | | |
|---|---|---|---|---|
| **DAY** | ***M*** | ***A*** | ***K*** | ***L*** |
| **1** | 4.0 | 4.8 | 5.0 | 4.6 |
| **2** | 4.8 | 5.0 | 5.2 | 4.6 |
| **3** | 4.0 | 4.8 | 5.6 | 5.0 |

Construct an analysis of variance table. Determine at 0.05 level whether there is significant difference between Coater Types. Apply the *LSD* test to locate significant difference between the various pairs of coater types.

(1) Hypothesis

$H_0$ : There is no significant difference between coater types.

$H_1$ : There is significant " " " .

(2) $\alpha = 0.05$

(3) $F = \dfrac{MS \text{ for Coater types}}{MSE}$

(4) Calculation

| S.V | d.f | SS | MS | F |
|---|---|---|---|---|
| b/w coater types | | | 0.51 | |
| b/w Days | | | 0.10 | |
| Error | | | 0.09 | |
| Total | | | | |

$$LSD = 0.60$$

$$\bar{Y}_M = 4.97, \quad \bar{Y}_A = 4.87, \quad \bar{Y}_K = 5.27, \quad \bar{Y} = 4.73$$

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# THANK YOU

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# COURSE: STA 408
# DESIGN & ANALYSIS OF EXPERIMENT

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COURSE: STA 408

DESIGN & ANALYSIS OF EXPERIMENT

# FACTORIAL DESIGN

LECTURE NO.8

Experiments are often planned to investigate the effects of say, different rates of fertilizers , different dates of planting, different categories of education, different intensities of a stimulus, etc.

- The independent variables such as fertilizers, planting, education, stimulus, etc. are called ***factors***.

- While the values such as rates, dates, categories or intensities at which a factor is held fixed, are known as ***levels***.

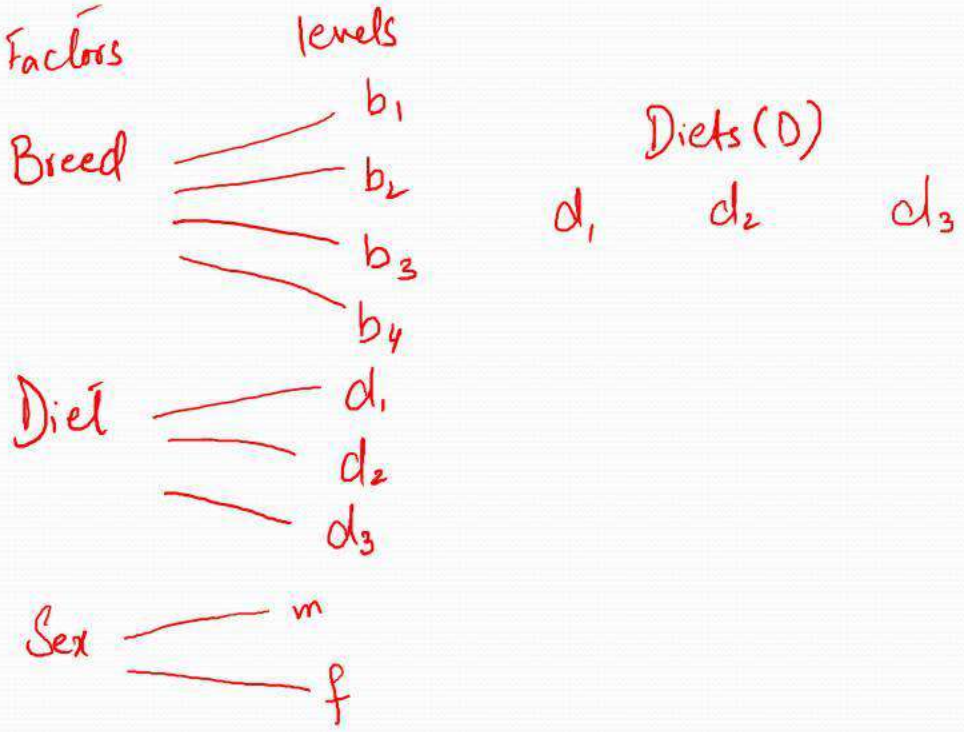
# WHAT IS FACTORIAL DESIGN?

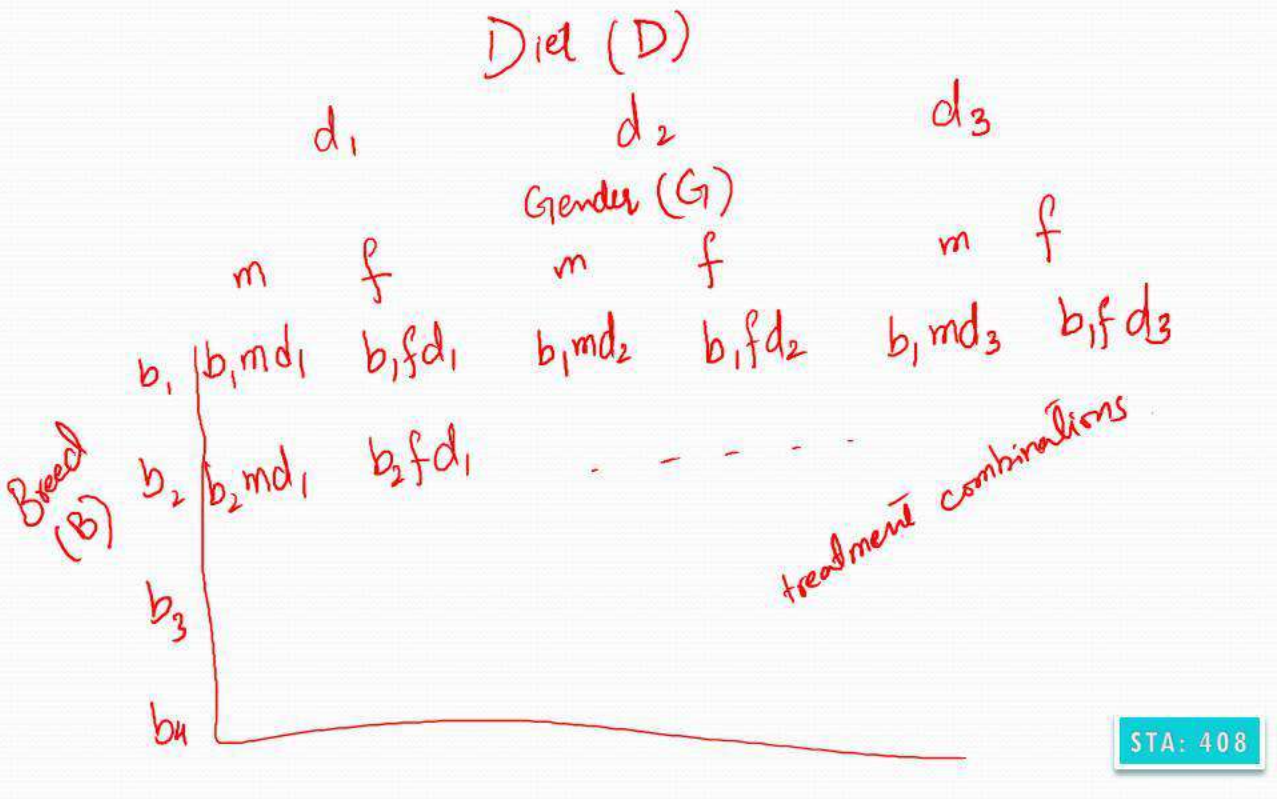
*Audio Lecture*

Level of watering — 4 times in a month
— 8 times in a month

Amount of Seeding — 60 kg
— 50 kg
— 80 kg

Cultural Practice — crop in rotation
— crop in gaps

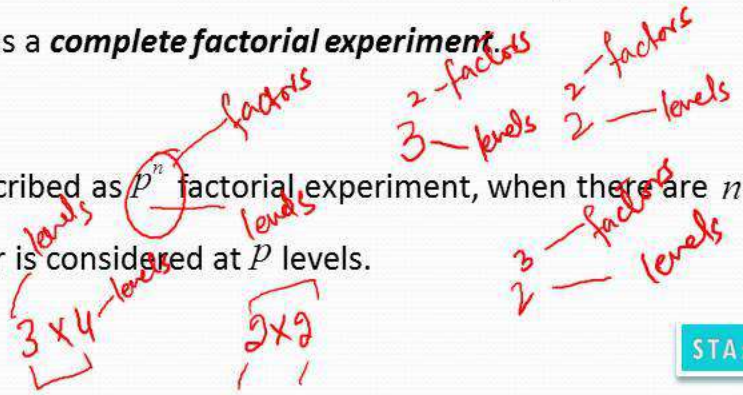

Factors
levels
Breed
$b_1$
$b_2$
$b_3$
$b_4$
Diet
$d_1$
$d_2$
$d_3$
Sex
m
f
Diets (D)
$d_1$
$d_2$
$d_3$


Diet (D)
$d_1$
$d_2$
$d_3$
Gender (G)
m
f
m
f
m
f
Breed (B)
$b_1$
$b_2$
$b_3$
$b_4$
$b_1md_1$
$b_1fd_1$
$b_1md_2$
$b_1fd_2$
$b_1md_3$
$b_1fd_3$
$b_2md_1$
$b_2fd_1$
treatment combinations

- It is customary to represent the factors by capital letters A, B, C etc. and levels in small letters a, b, c, etc.

- An experiment is called a ***factorial experiment*** if the treatments consist of all possible combinations of several levels of several factors. When each treatment combination is used the same number of times, the factorial experiment is known as a ***complete factorial experiment***.

- The experiment is described as $p^n$ factorial experiment, when there are $n$ factors and each factor is considered at $p$ levels.



# MAIN EFFECTS

A main effect of a factor is defined as a measure of the average change in effect produced by changing the level of the factor. It is measured independently of other factors and is the effect of the factor only. A main effect is sometimes regarded as an interaction of zero order.

# INTERACTION EFFECTS

Factors are said to be interact, when they are not independent. But interaction in a factorial experiment, is a measure of the extent to which the effect of changing the levels of one or more factors depends on the levels of the other factors. Interactions between two factors are referred to as first order interactions, those concerning three factors, as second order interactions and so on.

A, B, C AB 1st AC BC ABC 2nd A BC

# ADVANTAGES & DISADVANTAGES OF FACTORIAL EXPERIMENT

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- A factorial experiment is usually economical.
- All the experimental units are used in computing the main effects and interactions.
- The use of all treatment combinations makes the experiment more efficient and comprehensive.
- The interaction effects are easily estimated and tested through the usual analysis of variance.
- The experiment yields unbiased estimates of effects, which are of wider applicability.

- A factorial experiment requires an excessive amount of experimentation when there are several factors at several levels.

8 factors, each at 2 levels
256 combinations
7 factors, each at 3 levels
2187

fractional Replication

- A large number of combinations when used, cause decrease in the efficiency of the experiment. The experiment may be reduced to a manageable size by confounding some effects considered of little practical consequence.

- The experiment set up and the resulting statistical analysis are more complex.

# EXAMPLE

A soft drink company is studying the effect of percent carbonation (A), the speed (C) and operating pressure (B) on the filler on the volume of the carbonated drink pack. In each bottle, 3 levels of carbonation, 2 levels of pressure and 2 levels of speed are selected and a factorial experiment with 2 replicates is conducted. The order in which 24 observations are taken is determine randomly. The coded data are given in the table.

| | | Operating pressure (B) | | | |
|---|---|---|---|---|---|
| | | 25 PSI | | 30 PSI | |
| | | Line Speed (C) | | | |
| | | 200 | 250 | 200 | 250 |
| Percent Carbonation (A) | 10% | -3<br>-1 | -1<br>0 | -1<br>0 | 1<br>1 |
| | 12% | 0<br>1 | 2<br>1 | 2<br>3 | 6<br>5 |
| | 14% | 5<br>4 | 7<br>6 | 7<br>9 | 10<br>11 |

i → A → a
j → B → b
k → C → c
l → no of replicates
n

- HYPOTHESIS

$H_0$ : Effect of main effects is insignificant.

$H_0'$ : " 1st order interaction is ".

$H_0''$ : " 2nd order interaction is ".

or

$H_0$ : Effect of factor A is insignificant.

B

C

AB

⋮

ABC

| | | Operating pressure (B) | | | | |
|---|---|---|---|---|---|---|
| | | 25 PSI | | 30 PSI | | |
| | | Line Speed (C) | | | | |
| | | 200 | 250 | 200 | 250 | $\sum y_i \ldots$ |
| Percent Carbonation (A) | 10% | -3<br>-1 (-4) | -1<br>0 (-1) | -1<br>0 (-1) | 1<br>1 (2) | -4 |
| | 12% | 0<br>1 (1) | 2<br>1 (3) | 2<br>3 (5) | 6<br>5 (11) | 20 |
| | 14% | 5<br>4 (9) | 7<br>6 (13) | 7<br>9 (16) | 10<br>11 (21) | 54 |
| $\sum y_j \ldots$ | | 6 | 15 | 20 | 34 | 75 |

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$$C.F = (Y_{\cdots})^2 / abcn \qquad abcn = 3 \times 2 \times 2 \times 2 = 24$$

$Y_{\cdots}$ : Grand Total

$$C.F = (75)^2/24 = 234.375$$

$$TSS = \sum_i \sum_j \sum_k \sum_l Y_{ijkl}^2 - C.F$$

$$= (-4)^2 + (1)^2 + (9)^2 + (-1)^2 + (3)^2 + \cdots (21)^2 - 234.375$$

$$TSS = 1118 - 234.375 = 336.625$$

$$SSA = \frac{\sum_i y_{i\cdots}^2}{bcn} - C.F$$

$$SSA = \frac{(-4)^2 + (20)^2 + (59)^2}{3\times2\times2} - 234.375 = 252.75$$

$$SSB = \frac{\sum_j y_{\cdot j\cdot\cdot}^2}{acn} - C.F$$

$$SSB = \frac{(6+15)^2 + (20+34)^2}{3\times2\times2} - 234.375 = 45.375$$

$$SSC = \frac{\sum_k y_{\cdot\cdot k\cdot}^2}{abn} - C.F$$

$$= \frac{(6+20)^2 + (15+34)^2}{3\times2\times2} - C.F$$

$$SSC = 22.042$$

# Sub Table (A x B)

| | | Operating pressure (B) | |
|---|---|---|---|
| | | 25 PSI | 30 PSI |
| Percent Carbonation (A) | 10% | -5 | 1 |
| | 12% | 4 | 16 |
| | 14% | 22 | 37 |

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$$TSS(A\times B) = \frac{\sum\sum y_{ij}^2}{cn} - C.F$$

$$TSS(A\times B) = \frac{(-5)^2 + (1)^2 + \cdots + (37)^2}{2\times 2} - 234.375$$

$$TSS(A\times B) = 303.375$$

$$SSAB = TSSAB - SSA - SSB$$

$$= 303.375 - 252.75 - 45.375$$

$$SSAB = 5.25$$

# Sub Table (A x C)

| | | Line Speed (C) | |
|---|---|---|---|
| | | 200 | 250 |
| Percent Carbonation (A) | 10% | 5 | 1 |
| | 12% | 6 | 14 |
| | 14% | 25 | 34 |

$$TSSAC = \frac{\sum\sum y_{i.k.}^2}{bn} - C.F$$

$$TSSAC = 509.75 - 234.375$$

$$TSSAC = 275.375$$

$$SSAC = TSSAC - SSA - SSC$$

$$SSAC = 0.583$$

# Sub Table (B x C)

| | | Line Speed (C) | |
|---|---|---|---|
| | | 200 | 250 |
| Operating pressure (B) | 25 PSI | 6 | 15 |
| | 30 PSI | 20 | 34 |

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$a = 3$
$n = 2$

$$TSSBC = \frac{\sum\sum Y_{.jx.}^2}{jk} - C.F$$

$$TSSBC = \frac{(6)^2 + \cdots + (34)^2}{3x2} - 234.375$$

$$TSSBC = 68.46$$

$$SSBC = TSSBC - SSB - SSC$$

$$SSBC = 1.041$$

# Sub Table (A x B x C)

| | | Operating pressure (B) | | | |
|---|---|---|---|---|---|
| | | 25 PSI | | 30 PSI | |
| | | Line Speed (C) | | | |
| | | 200 | 250 | 200 | 250 |
| Percent Carbonation (A) | 10% | -4 | -1 | -1 | 2 |
| | 12% | 1 | 3 | 5 | 11 |
| | 14% | 9 | 13 | 16 | 21 |

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$$TSS(ABC) = \frac{\sum_i \sum_j \sum_k y_{ijk}^2.}{n} - C.F$$

$$= \frac{(-4)^2 + (11)^2 + \cdots + (21)^2}{2} - 234.375$$

$$TSS\ ABC = 328.125$$

$$SSABC = TSS\ ABC - SSA - SSB - SSC - SSAB - SSAC - SSBC$$

$$SSABC = 1.084$$

$$Error\ SS = TSS - TSS(ABC)$$

$$= 336.625 - 328.125 = 8.5$$

| Source of Variation | Degrees of Freedom |
|---|---|
| Factor A | $a-1$ |
| Factor B | $b-1$ |
| Factor C | $c-1$ |
| AB | $(a-1)(b-1)$ |
| AC | $(a-1)(c-1)$ |
| BC | $(b-1)(c-1)$ |
| ABC | $(a-1)(b-1)(c-1)$ |
| Error | $abc(n-1)$ |
| Total | $abcn-1$ |

ANOVA TABLE

| S.V | d.f | SS | MS | $F_{cal}$ | $F_{tab}$ | Decision |
|---|---|---|---|---|---|---|
| Main Effects | | | | | | |
| A | 2 | 252.75 | 126.375 | 178.412 | 3.89 | Sig |
| B | 1 | 45.375 | 45.375 | 64.059 | 4.75 | Sig |
| C | 1 | 22.042 | 22.042 | 31.118 | 4.75 | Sig |
| 1st Order Interaction | | | | | | |
| AB | 2 | 5.25 | 2.625 | 3.706 | 3.89 | Insig |
| AC | 2 | 0.583 | 0.292 | 0.412 | 3.89 | // |
| BC | 1 | 1.041 | 1.042 | 1.471 | 4.75 | // |
| 2nd Order Interaction | | | | | | |
| ABC | 2 | 1.084 | 0.542 | 0.765 | 3.89 | Insig |
| Error | 12 | 8.5 | 0.708 | | | |
| Total | 23 | 336.625 | | | | |

# PRACTICE QUESTION

The quality control department of a fabric finishing plant is studying the effect of several factors on the dying of cotton-synthetic cloth used to manufacture men's shirts. Two operators, three cycle times and two temperatures were selected, and a factorial experiment with 2 replicates is conducted. The results are as follows. Analyze the data and draw conclusions.

$i \to C \to c = 3$

$j \to R \to r = 2$

$K \to P \to p = 2$

$l \to n = 2$

$i = 1, 2, \ldots c$

$j = 1, 2, \ldots r$

$K = 1, 2, \ldots p$

$l = 1, 2, \ldots n$

| | | TEMPERATURE (R) | | | |
|---|---|---|---|---|---|
| | | 300°C | | 350°C | |
| | | OPERATOR (P) | | | |
| | | 1 | 2 | 1 | 2 |
| CYCLE TIME (C) | 40 | 23, 24 — 47 | 30, 28 — 58 | 24, 23 — 47 | 38, 36 — 74 |
| | 50 | 36, 35 — 71 | 34, 38 — 72 | 37, 39 — 76 | 34, 38 — 72 |
| | 60 | 28, 24 — 52 | 35, 35 — 70 | 26, 29 — 55 | 36, 37 — 73 |

C.F, TSS, SSR, SSC, SSP

SSCR, SSCP, SSRP

SSCRP, ESS

C, R, P } Main Effects

CR, CP, RP } 1st order interactions

CRP → 2nd order interaction

C  C-1 = 3-1 = 2

R  r-1 = 2-1 = 1

P  p-1 = 2-1 = 1

CR (C-1)(r-1)

CP - - -

# THANK YOU

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# COURSE: STA 408
# DESIGN & ANALYSIS OF EXPERIMENT

LECTURE DELIVERED BY

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COURSE: STA 408

DESIGN & ANALYSIS OF EXPERIMENT

# $2^K$ AND $2^2$ FACTORIAL DESIGN

LECTURE NO.9

# $2^K$ FACTORIAL DESIGN

The *K* factors each at only two levels. These levels may be quantitative, such as two values of temperature, pressure, or time; or they may be qualitative, such as two machines, two operators, the "high" and "low" levels of a factor, or perhaps the presence and absence of a factor. A complete replicate of such a design requires 2x2x2.....x2 = $2^k$ observations and is called a $2^k$ factorial design.

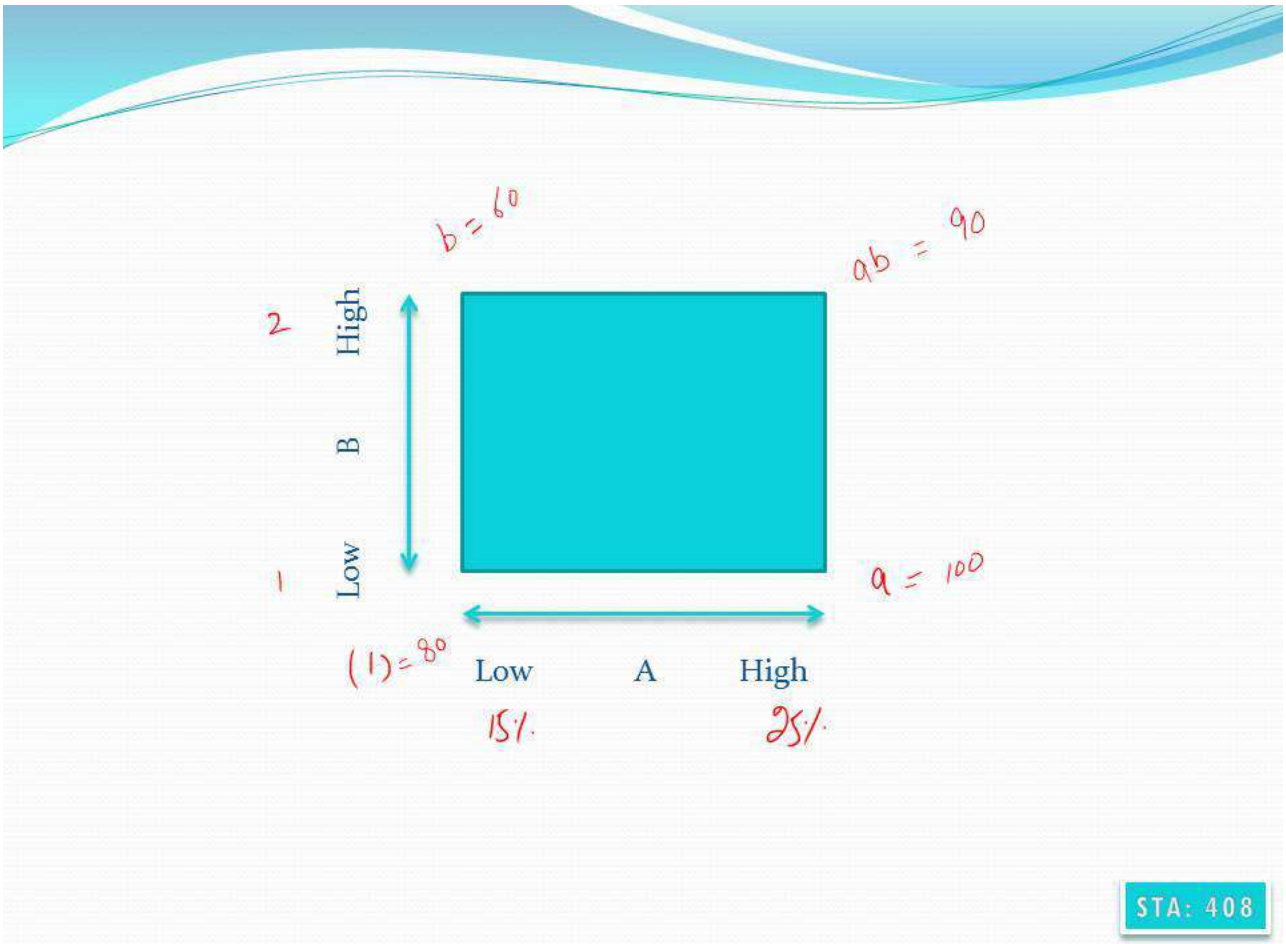
# $2^2$ FACTORIAL DESIGN

The simplest case is one in which two factors are of interest and each factor is said at only 2 levels. This design is called a $2^2$ factorial design. The levels of the factors may be arbitrarily called low and high or presence and absence.

# EXAMPLE

Consider an investigation into the effect of the concentration of the reactant and the amount of the catalyst on the conversion (yield) in a chemical process. The objective of the experiment was to determine if adjustments to either of these two factors would increase the yield. Let the reactant concentration be factor A and let the two levels of interest be 15 and 25 percent. The catalyst is factor B, with the high level denoting the use of 2 pounds of the catalyst and the low level denoting the use of only 1 pound. The experiment is replicated three times, so there are 12 runs. The order in which the runs are made is random so this is a completely randomized experiment. The data are obtained as follows:

| | | | REPLICATE | | | |
|---|---|---|---|---|---|---|
| | | | 1 | 2 | 3 | TOTAL |
| TREATMENT COMBINATIONS | A low, B low | (1) | 28 | 25 | 27 | 80 |
| | A high, B low | $a$ | 36 | 32 | 32 | 100 |
| | A low, B high | $b$ | 18 | 19 | 23 | 60 |
| | A high, B high | $ab$ | 31 | 30 | 29 | 90 |


b = 60
ab = 90
2
High
B
Low
1
a = 100
(1) = 80
Low
A
High
15%
25%
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# SIGN TABLE

| | | EFFECTS | | | |
|---|---|---|---|---|---|
| | | **M** | **A** | **B** | **AB** |
| TREATMENT COMBINATIONS | (1) | + | − | − | + |
| | *a* | + | + | − | − |
| | *b* | + | − | + | − |
| | *ab* | + | + | + | + |

$C.F = \frac{(330)^2}{12} = 9075$

# EFFECTS

Effect M

$C.F = \frac{[\text{contrast } M]^2}{2^k n}$

| | EFFECTS | | | |
|---|---|---|---|---|
| | M | A | B | AB |
| (1) | + | - | - | + |
| *a* | + | + | - | - |
| *b* | + | - | + | - |
| *ab* | + | + | + | + |

$M = \frac{1}{2^k n}$ [ Contrast of M ]

$2^k n \rightarrow$ no. of replicates

$M = \frac{1}{2^2(3)} [(1) + a + b + ab]$

$M = 27.5$

$M = \frac{1}{4 \times 3} [80 + 100 + 60 + 90] = \frac{1}{12} \times 330 = 27.5$

Effect A

$$A = \frac{1}{2^{k-1}n}[\text{contrast } A] = \frac{1}{2^{2-1}(3)}[-(1) + a - b + ab]$$

| | EFFECTS | | | |
|---|---|---|---|---|
| | M | A | B | AB |
| (1) | + | - | - | + |
| *a* | + | + | - | - |
| *b* | + | - | + | - |
| *ab* | + | + | + | + |

$$A = \frac{1}{2 \times 3}[-80 + 100 - 60 + 90]$$

$$A = \frac{1}{6}[50] = 8.33 \quad \text{——— effect A}$$

$$SSA = \frac{[\text{contrast } A]^2}{2^k n} = \frac{(50)^2}{2^2(3)} = \frac{(50)^2}{12} = 208.33$$

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Effect B

$$B = \frac{1}{2^{k-1} \cdot n} \left[ -(1) - a + b + ab \right]$$

$$B = \frac{1}{2(3)} \left[ -80 - 100 + 60 + 90 \right] = \frac{-30}{6}$$

$$\boxed{B = -5}$$

$$SSB = \frac{[\text{contrast } B]^2}{2^k n} = \frac{(-30)^2}{12} = 75$$

| | EFFECTS | | | |
|---|---|---|---|---|
| | M | A | B | AB |
| (1) | + | - | - | + |
| *a* | + | + | - | - |
| *b* | + | - | + | - |
| *ab* | + | + | + | + |

Effect AB

| | EFFECTS | | | |
|---|---|---|---|---|
| | M | A | B | AB |
| (1) | + | - | - | + |
| *a* | + | + | - | - |
| *b* | + | - | + | - |
| *ab* | + | + | + | + |

$$AB = \frac{1}{2^{k-1}n}\left[(1) - a - b + ab\right]$$

$$AB = \frac{1}{2(3)}\left[80 - 100 - 60 + 90\right]$$

$$AB = \frac{1}{6}(10) = 1.667 = \underline{1.67}$$

$$SSAB = \frac{[\text{contrast } AB]^2}{g^k \cdot n} = \frac{(10)^2}{12} = 8.33$$

$$TSS = \sum_{ijk} y_{ijk}^2 - C.F$$

$$TSS = (28)^2 + (25)^2 + \cdots\cdots + (29)^2 - 9075$$

$$TSS = 9398 - 9075 = 323$$

$$ESS = TSS - SSA - SSB - SSAB$$

$$ESS = 323 - 208.33 - 75 - 8.33$$

$$ESS = 31.34$$

$$\boxed{SSE = 31.34}$$

$F \gtrless F_{\alpha(v_1, v_E)}$

Cal  Tab

| ANOVA TABLE | | | | | | |
|---|---|---|---|---|---|---|
| S.O.V | d.f | SS | MS | $F_{cal}$ | $F_{tab}$ | Decision |
| Main Effects | | SSA = | | | | |
| A | $a-1 = 2-1 = 1$ | 208.33 | 208.33 | $F_A = 53.15$ | $> 5.32$ | Sig |
| B | $b-1 = 2-1 = 1$ | SSB = 75 | 75 | $F_B = 19.13$ | $> 5.32$ | Sig |
| 1st Order Interaction | | | | | | |
| AB | $(a-1)(b-1) = 1 \times 1 = 1$ | SSAB = 8.33 | 8.33 | $F_{AB} = 2.13$ | $< 5.32$ | Insig |
| Error | $ab(n-1) = 2 \times 2 (2) = 8$ | SSE = 31.34 | 3.92 | | | |
| Total | $abn - 1 = (2 \times 2 \times 3) - 1 = 11$ | | | | | |

① $H_0$: A is insig  $H_1$: A is sig ✓

② $H_0$: B " ✓  $H_1$: B " ✓

③ $H_0$: AB " ✓  $H_1$: AB "

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# EXAMPLE

A bacteriologist is interested in the effects of two different culture media and two different times on the growth of a particular virus. He or she performs six replicates of a $2^2$ design, making the runs in random order. Analyze the bacterial growth data that follow and draw appropriate conclusions.

$2^2 = 4$ Treatment combinations

| | Culture Medium | | | |
|---|---|---|---|---|
| Time | 1 | | 2 | |
| 12 | 21 | 22 | 25 | 26 |
| | 23 | 28 | 24 | 25 |
| | 20 | 26 | 29 | 27 |
| 18 | 37 | 39 | 31 | 34 |
| | 38 | 38 | 29 | 33 |
| | 35 | 36 | 30 | 35 |

| TREATMENT COMBINATIONS | REPLICATES | | | | | |
|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 |
| (1) | 21 | 22 | 23 | 28 | 20 | 26 |
| m | 37 | 39 | 38 | 38 | 35 | 36 |
| c | 25 | 26 | 24 | 25 | 29 | 27 |
| mc | 31 | 34 | 29 | 33 | 30 | 35 |

| M / Time | Culture Medium 1 (low) | | 2 (high) | |
|---|---|---|---|---|
| 12 (low) | 21 | 22 | 25 | 26 |
| | 23 | 28 | 24 | 25 |
| | 20 | 26 | 29 | 27 |
| 18 (high) | 37 | 39 | 31 | 34 |
| | 38 | 38 | 29 | 33 |
| | 35 | 36 | 30 | 35 |

C

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| TREATMENT COMBINATIONS | REPLICATES | | | | | | |
|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | TOTAL |
| (1) | 21 | 22 | 23 | 28 | 20 | 26 | 140 |
| m | 37 | 39 | 38 | 38 | 35 | 36 | 223 |
| c | 25 | 26 | 24 | 25 | 29 | 27 | 156 |
| mc | 31 | 34 | 29 | 33 | 30 | 35 | 192 |

(1) Hypothesis

$H_0$: The main effect M is insignificant.

$H_1$: " " significant

$H_0'$: The effect of C is insignificant.

$H_1'$: " " significant.

$H_0''$: The first order interaction effect MC is insignificant.

$H_1''$: " " significant.

(2) level of significance

(3) Test Statistic

$$F = \frac{MS\,Effect}{MSE}$$

(4) Calculations

Sign table

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n - no. of replicates
m - levels of factor M
c - levels of factor C

ANOVA TABLE

| Source of Variation | Degrees of Freedom | Sum of Squares | Mean Square | *F*-Ratio |
|---|---|---|---|---|
| M | $m-1 = 2-1 = 1$ | SSM | $MS(M) = SSM/m-1$ | $F_M = \frac{MS(M)}{MSE}$ |
| C | $c-1 = 2-1 = 1$ | SS C | $MS(C) = SSC/c-1$ | $F_C = \frac{MSC}{MSE}$ |
| MC | $(m-1)(c-1) = 1$ | SSMC | $MS(MC) = SSMC/(m-1)(c-1)$ | $F_{MC} = \frac{MS(MC)}{MSE}$ |
| Error | $mc(n-1)$ | SSE | $MSE = SSE/mc(n-1)$ | |
| Total | $mcn-1$ | SST | | |

# EXAMPLE

Find out the effect of *A*, *B* and *AB* by using the data given below. Also, make a column of signs *AB*.

| A | B | Treatment combination | REPLICATES | | | |
|---|---|---|---|---|---|---|
| | | | 1 | 2 | 3 | 4 |
| - | - | (1) | 18.2 | 18.9 | 12.9 | 14.4 |
| + | - | $a$ | 27.2 | 24.0 | 22.4 | 22.5 |
| - | + | $b$ | 15.7 | 14.5 | 15.1 | 14.2 |
| + | + | $ab$ | 41.0 | 43.9 | 36.3 | 39.9 |

| A | B | AB | Treatment combination | REPLICATES | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | 1 | 2 | 3 | 4 | Total |
| - | - | + | (1) | 18.2 | 18.9 | 12.9 | 14.4 | **64.4** |
| + | - | − | *a* | 27.2 | 24.0 | 22.4 | 22.5 | **96.1** |
| - | + | − | *b* | 15.7 | 14.5 | 15.1 | 14.2 | **59.5** |
| + | + | + | *ab* | 41.0 | 43.9 | 36.3 | 39.9 | **161.1** |

(1) = 64.4 b = 59.5

a = 96.1 ab = 161.1

**Effect of A**

$$A = \frac{1}{2^{k-1}(n)}\left[-(1) + a - b + ab\right]$$

$$A = \frac{1}{2^{2-1}(4)}\left\{-64.4 + 96.1 - 59.5 + 161.1\right\}$$

$$A = 133.3/8 = 16.6625$$

**Effect of B**

$$B = \frac{1}{2^{k-1}(n)}\left[-(1) - a + b + ab\right]$$

$$B = \frac{1}{2(4)}\left[-64.4 - 96.1 + 59.5 + 161.1\right]$$

$$B = 60.1/8 = 7.5125$$

**Effect of AB**

$$AB = \frac{1}{2^{k-1}(n)}\left[(1) - a - b + ab\right]$$

$$AB = \frac{1}{2(4)}\left\{64.4 - 96.1 - 59.5 + 161.1\right\}$$

$$AB = \frac{69.9}{8} = 8.7375$$

# THANK YOU

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DESIGN & ANALYSIS OF EXPERIMENT

# $2^3$ FACTORIAL DESIGN

LECTURE NO.10

# $2^3$ FACTORIAL DESIGN

In this case, three factors each at two levels, are of interest. The design is called $2^3$ factorial design, and the eight treatment combinations are possible in this design.

# SIGN TABLE

| TREATMENT COMBINATIONS | EFFECTS | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | **M** | **A** | **B** | **AB** | **C** | **AC** | **BC** | **ABC** |
| (1) | + | − | − | + | − | + | + | − |
| *a* | + | + | − | − | − | − | + | + |
| *b* | + | − | + | − | − | + | − | + |
| *ab* | + | + | + | + | − | − | − | − |
| *c* | + | − | − | + | + | − | − | + |
| *ac* | + | + | − | − | + | + | − | − |
| *bc* | + | − | + | − | + | − | + | − |
| *abc* | + | + | + | + | + | + | + | + |

# PROPERTIES OF SIGN TABLE

- Except for the first column, every column has an equal number of "+" and "-" signs.

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| A | − | + | − | + | − | + | − | + |
| B | − | − | + | + | − | − | + | + |

- The sum of products of signs is in any two columns is zero that is any two columns are orthogonal. $(-1)(-1)$ $(+1)(-1)$ $(-1)(+1)$ $(+1)(+1)$
- If we multiply any two columns of the sign table another column is achieved i.e. A and B are ,multiplied and get AB. A and B are
- Column 1 multiplied by another column leaves column unchanged that is column 1 is an identity column.

# MODE RULE

Let ABC x BC

So, it must be equal to A.

$ABC \times BC = A$

$AB^2C^2 = AB^0C^0 = A$

$\therefore B^0C^0 = 1$

$$2\overline{)2}\quad 1,\ 2,\ 0$$

# EXAMPLE

A $2^3$ factorial design was used to develop a nitride etch process on a single wafer plasma etching tool. The design factors are the gap between the electrodes, the gas flow ($C_2F_6$ is used as the reactant gas), and the RF power applied to the cathode. Each factor is run at two levels and the design is replicated twice. The response variable is the etch rate for silicon nitride. The etch rate data are shown as

| ETCH RATE | | | | |
|---|---|---|---|---|
| | | REPLICATES | | |
| | | 1 | 2 | Total |
| TREATMENT COMBINATIONS | (1) | 604 | 550 | 1154 |
| | *a* | 650 | 669 | 1319 |
| | *b* | 601 | 633 | 1234 |
| | *ab* | 635 | 642 | 1277 |
| | *c* | 1052 | 1037 | 2089 |
| | *ac* | 868 | 749 | 1617 |
| | *bc* | 1063 | 1075 | 2138 |
| | *abc* | 860 | 729 | 1589 |

# EFFECTS

Effect of A

$$A = \frac{1}{2^{k-1}n}[\text{contrast } A]$$

$$A = \frac{1}{2^{3-1}(2)}\left[-(1)+a-b+ab-c+ac-bc+abc\right]$$

| TREATMENT COMBINATIONS | EFFECTS | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | M | A | B | AB | C | AC | BC | ABC |
| (1) | + | - | - | + | - | + | + | - |
| *a* | + | + | - | - | - | - | + | + |
| *b* | + | - | + | - | - | + | - | + |
| *ab* | + | + | + | + | - | - | - | - |
| *c* | + | - | - | + | + | - | - | + |
| *ac* | + | + | - | - | + | + | - | - |
| *bc* | + | - | + | - | + | - | + | - |
| *abc* | + | + | + | + | + | + | + | + |

$$A = \frac{1}{4\times 2}[-1154+1319-1234+1277-2089+1617-2138+1589]$$

$$A = \frac{1}{8}[-813] = -101.625$$

$$SSA = \frac{(-813)^2}{16} = 41310.5625$$

# Effect of B

$$B = \frac{1}{2^2(2)}\left[b + ab + bc + abc - (1) - a - c - ac\right]$$

$$B = \frac{1}{8}\left[1234 + 1277 + 2138 + 1589 - 1154 - 1319 - 2089 - 1617\right]$$

$$B = \frac{1}{8}\left[59\right] = 7.375$$

$$SSB = \frac{(59)^2}{16} = 217.5625$$

| | | EFFECTS | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | M | A | B | AB | C | AC | BC | ABC |
| TREATMENT COMBINATIONS | (1) | + | - | - | + | - | + | + | - |
| | *a* | + | + | - | - | - | - | + | + |
| | *b* | + | - | + | - | - | + | - | + |
| | *ab* | + | + | + | + | - | - | - | - |
| | *c* | + | - | - | + | + | - | - | + |
| | *ac* | + | + | - | - | + | + | - | - |
| | *bc* | + | - | + | - | + | - | + | - |
| | *abc* | + | + | + | + | + | + | + | + |

effect of C

$$C = \frac{1}{8}[c + ac + bc + abc - (1) - a - b - ab]$$

$$C = \frac{1}{8}\{2089 + 1617 + 2138 + 1589 - 1134 - 1319 - 1234 - 1277\}$$

$$C = \frac{1}{8}[2449] = 306.125$$

$$SSC = \frac{(2449)^2}{16} = 374850.0625$$

| TREATMENT COMBINATIONS | EFFECTS | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | M | A | B | AB | C | AC | BC | ABC |
| (1) | + | - | - | + | - | + | + | - |
| *a* | + | + | - | - | - | - | + | + |
| *b* | + | - | + | - | - | + | - | + |
| *ab* | + | + | + | + | - | - | - | - |
| *c* | + | - | - | + | + | - | - | + |
| *ac* | + | + | - | - | + | + | - | - |
| *bc* | + | - | + | - | + | - | + | - |
| *abc* | + | + | + | + | + | + | + | + |

Effect of AB

$$AB = \frac{1}{8}[ab - a - b + (1) + abc - bc - ac + c]$$

$$AB = \frac{1}{8}[1277 - 1319 - 1234 + 1154 + 1589 - 2138 - 1617 + 2089]$$

$$AB = \frac{1}{8}(-199) = -24.875$$

$$SSAB = \frac{(-199)^2}{16} = 2475.0625$$

| TREATMENT COMBINATIONS | EFFECTS | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | M | A | B | AB | C | AC | BC | ABC |
| (1) | + | - | - | + | - | + | + | - |
| *a* | + | + | - | - | - | - | + | + |
| *b* | + | - | + | - | - | + | - | + |
| *ab* | + | + | + | + | - | - | - | - |
| *c* | + | - | - | + | + | - | - | + |
| *ac* | + | + | - | - | + | + | - | - |
| *bc* | + | - | + | - | + | - | + | - |
| *abc* | + | + | + | + | + | + | + | + |

Effect of AC

| TREATMENT COMBINATIONS | EFFECTS | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | M | A | B | AB | C | AC | BC | ABC |
| (1) | + | - | - | + | - | + | + | - |
| *a* | + | + | - | - | - | - | + | + |
| *b* | + | - | + | - | - | + | - | + |
| *ab* | + | + | + | + | - | - | - | - |
| *c* | + | - | - | + | + | - | - | + |
| *ac* | + | + | - | - | + | + | - | - |
| *bc* | + | - | + | - | + | - | + | - |
| *abc* | + | + | + | + | + | + | + | + |

$$AC = \frac{1}{8}[(1) - a + b - ab - c + ac - bc + abc]$$

$$AC = \frac{1}{8}[1184 - 1319 + 1234 - 1277 - 2089 + 1617 - 2138 + 1589]$$

$$AC = \frac{1}{8}[-1299] = -153.625$$

$$SSAC = \frac{(-1299)^2}{16} = 94402.5625$$

# Effect of BC

| TREATMENT COMBINATIONS | EFFECTS | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | M | A | B | AB | C | AC | BC | ABC |
| (1) | + | - | - | + | - | + | + | - |
| *a* | + | + | - | - | - | - | + | + |
| *b* | + | - | + | - | - | + | - | + |
| *ab* | + | + | + | + | - | - | - | - |
| *c* | + | - | - | + | + | - | - | + |
| *ac* | + | + | - | - | + | + | - | - |
| *bc* | + | - | + | - | + | - | + | - |
| *abc* | + | + | + | + | + | + | + | + |

$$BC = \frac{1}{8}[(1) + a - b - ab - c - ac + bc + abc]$$

$$BC = \frac{1}{8}[1154 + 1319 - 1234 - 1277 - 2089 - 1617 + 2438 + 1589]$$

$$BC = \frac{1}{8}(-17) = -2.125$$

$$SSBC = \frac{(-17)^2}{16} = 18.0625$$

Effect of ABC

$$ABC = \frac{1}{8}[abc - bc - ac + c - ab + b + a - (1)]$$

$$= \frac{1}{8}[1589 - 2138 - 1617 + 2089 - 1277 + 1234 + 1319 - 1154]$$

$$ABC = 5.625 = \frac{1}{8}(45)$$

$$SSABC = \frac{(45)^2}{16} = 126.5625$$

| TREATMENT COMBINATIONS | EFFECTS | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | M | A | B | AB | C | AC | BC | ABC |
| (1) | + | - | - | + | - | + | + | - |
| *a* | + | + | - | - | - | - | + | + |
| *b* | + | - | + | - | - | + | - | + |
| *ab* | + | + | + | + | - | - | - | - |
| *c* | + | - | - | + | + | - | - | + |
| *ac* | + | + | - | - | + | + | - | - |
| *bc* | + | - | + | - | + | - | + | - |
| *abc* | + | + | + | + | + | + | + | + |

TSS = 531420.9375

ESS = 18090.50

| | | EFFECTS | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | M | A | B | AB | C | AC | BC | ABC |
| TREATMENT COMBINATIONS | (1) | + | - | - | + | - | + | + | - |
| | *a* | + | + | - | - | - | - | + | + |
| | *b* | + | - | + | - | - | + | - | + |
| | *ab* | + | + | + | + | - | - | - | - |
| | *c* | + | - | - | + | + | - | - | + |
| | *ac* | + | + | - | - | + | + | - | - |
| | *bc* | + | - | + | - | + | - | + | - |
| | *abc* | + | + | + | + | + | + | + | + |

# ANOVA - Table

| S.O.V | d.f | SS | MS | $F_{cal}$ | $F_{tab}$ | Decision |
|---|---|---|---|---|---|---|
| Main Effects | | | | | | |
| A | a-1 = 1 | 41310.5625 | → | $F_A = 18.34$ | 5.32 | Sig |
| B | b-1 = 1 | 217.5625 | → | $F_B = 0.10$ | 5.32 | Insig |
| C | c-1 = 1 | 374850.0625 | → | $F_C = 166.41$ | 5.32 | Sig |
| 1st Order Interaction | | | | | | |
| AB | (a-1)(b-1) 1 | 2475.0625 | → | $F_{AB} = 1.10$ | 5.32 | Insig |
| AC | (a-1)(c-1) 1 | 94402.5625 | → | $F_{AC} = 41.91$ | 5.32 | Sig |
| BC | (b-1)(c-1) 1 | 18.0625 | → | $F_{BC} = 0.01$ | 5.32 | Insig |
| 2nd Order Interaction | | | | | | |
| ABC | (a-1)(b-1)(c-1) 1 | 126.5625 | → | $F_{ABC} = 0.06$ | 5.32 | Insig |
| Error | ab(n-1) = 8 | 18020.5000 | 2252.5625 | | | |
| Total | abcn - 1 | 531420.9375 | | | | |

(2×2×2×2) - 1 = 15

# EXAMPLE

An engineer is interested in the effects of cutting speed ($A$), tool geometry ($B$) and cutting angle ($C$) on the life (in hours) of a machine tool. Two levels of each factor are chosen, and three replicates of a $2^3$ factorial design are run. Compute the signs of $AC$ and $ABC$, and hence find the sum of squares of effects of $C$, $AC$ and $ABC$ by using the following results:

| TREATMENT COMBINATIONS | REPLICATES | | | Sign Table | | |
|---|---|---|---|---|---|---|
| | I | II | III | A | B | C |
| (1) | 22 | 31 | 25 | - | - | - |
| *a* | 32 | 43 | 29 | + | - | - |
| *b* | 35 | 34 | 50 | - | + | - |
| *ab* | 55 | 47 | 46 | + | + | - |
| *c* | 44 | 45 | 38 | - | - | + |
| *ac* | 40 | 37 | 36 | + | - | + |
| *bc* | 60 | 50 | 54 | - | + | + |
| *abc* | 39 | 41 | 47 | + | + | + |

| | | REPLICATES | | | | Sign Table | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | I | II | III | Total | A | B | C | AB | ABC |
| TREATMENT COMBINATIONS | (1) | 22 | 31 | 25 | 78 | - | - | - | + | − |
| | *a* | 32 | 43 | 29 | 104 | + | - | - | − | + |
| | *b* | 35 | 34 | 50 | 119 | - | + | - | − | + |
| | *ab* | 55 | 47 | 46 | 148 | + | + | - | + | − |
| | *c* | 44 | 45 | 38 | 127 | - | - | + | + | + |
| | *ac* | 40 | 37 | 36 | 113 | + | - | + | − | − |
| | *bc* | 60 | 50 | 54 | 164 | - | + | + | − | − |
| | *abc* | 39 | 41 | 47 | 197 | + | + | + | + | + |

STA: 408

# SUM OF SQUARES OF C

Effect of C

$$C = \frac{1}{2^{k-1}(n)} [c + ac + bc + abc - (1) - a - b - ab]$$

$$C = \frac{1}{2^{3-1}(3)} [127 + 113 + 164 + 127 - 78 - 104 - 119 - 148]$$

$$C = \frac{1}{4 \times 3} [82] = \frac{1}{12}(82) = 6.8333$$

$2^k(n)$

$$SSC = \frac{(82)^2}{24} = 280.1667$$

# SUM OF SQUARES OF *AC*

Effect of AC

$$AC = \frac{1}{2^{k-1}(n)} \left[ (1) - a + b - ab - c + ac - bc + abc \right]$$

$$AC = \frac{1}{2^{3-1}(3)} \left[ 78 - 104 + 119 - 148 - 127 + 113 - 164 + 127 \right]$$

$$AC = \frac{1}{4 \times 3} [-208] = \frac{-208}{12} = -17.3333$$

$$SSAC = \frac{(208)^2}{24} = 1802.6667$$

## SUM OF SQUARES OF *ABC*

Effect of ABC

$$ABC = \frac{1}{2^{k-1}(n)}\left[abc - bc - ac + c - ab + b + a - (1)\right]$$

$$ABC = \frac{1}{12}\left[127 - 164 - 113 + 127 - 148 + 119 + 104 - 78\right]$$

$$ABC = \frac{1}{12}(-26) = -2.1667$$

$$SSABC = \frac{(26)^2}{24} = 28.1667$$



# EXERCISE

The following data were obtained from a $2^3$ factorial experiment repeated 3 times

| | Treatment Combinations | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Replication | (1) | *a* | *b* | *ab* | *c* | *ac* | *bc* | *abc* |
| *1* | **12** | **15** | **24** | **23** | **17** | **16** | **24** | **28** |
| *2* | **19** | **20** | **16** | **17** | **25** | **19** | **23** | **25** |
| *3* | **10** | **16** | **17** | **27** | **21** | **19** | **29** | **20** |

Evaluate the sum of squares for all factorial effects.

# THANK YOU

# COURSE: STAT 408
# EXPERIMENTAL DESIGN ANALYSIS

LECTURE DELIVERED BY

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# COURSE: STAT 408

## EXPERIMENTAL DESIGN ANALYSIS

LECTURE NO. 11

# Comparing Means

# Anova

- ✓F-test can be used to determine whether the expected responses at the t levels of an experimental factor differ from each other
- ✓When the null hypothesis is rejected, it may be desirable to find which mean(s) is (are) different, and at what ranking order.
- ✓In practice, it is actually not primary interest to test the null hypothesis, instead the investigators want to make specific comparisons of the means.

# Means comparison

✓Three categories:

1. Pair-wise comparisons (Post-Hoc Comparison)
2. Comparison specified prior to performing the experiment (Planned comparison)
3. Comparison specified after observing the outcome of the experiment (Un-planned comparison)

✓Statistical inference procedures of pair-wise comparisons:

- Fisher's least significant difference (LSD) method
- Duncan's Multiple Range Test (DMRT)
- Student Newman Keul Test (SNK)
- Tukey's HSD ("Honestly Significantly Different") Procedure

# Pair Comparison

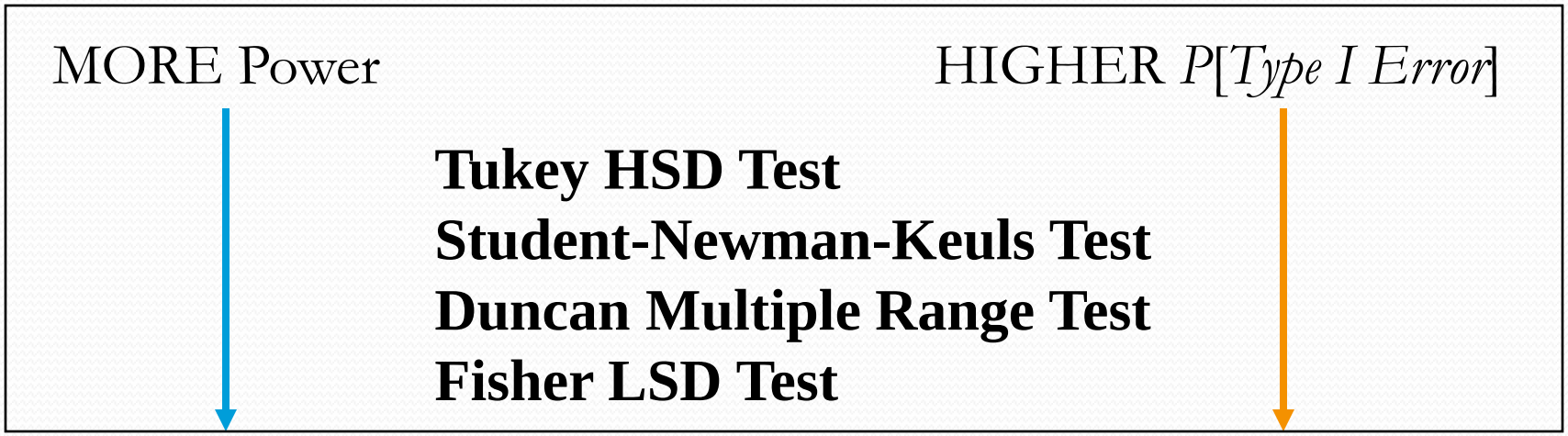
- ✓ An F-test has revealed that there are significant differences amongst the t means
- ✓ Performing an analysis to determine precisely where the differences exist.

# Pair Comparison

- ✓Two means are considered different if the difference between the corresponding sample means is larger than a critical number. Then, the larger sample mean is believed to be associated with a larger population mean.
- ✓Conditions common to all the methods:
  - The ANOVA model is the one way analysis of variance
  - The conditions required to perform the ANOVA are satisfied.
  - The experiment is fixed-effect.

# Comparing Pair-comparison methods

- ✓ With the exception of the F-*LSD* test, there is no good theoretical argument that favors one pair-comparison method over the others. Professional statisticians often disagree on which method is appropriate.
- ✓ In terms of Power and the probability of making a *Type I* error, the tests discussed can be ordered as follows:




- ✓ Pairwise comparisons are traditionally considered as "post hoc" and not "a priori", if one needs to categorize all comparisons into one of the two groups

# Duncan's Multiple Range Test

- D.B Duncan in 1955 introduced a multiple range test for comparing all pairs of sample means. This test, which has different significant ranges, is based on the Studentized-range statistic $q(p,v)$, where p is the number of steps between two means being compared, but the values of $q(p,v)$ for appropriate values of p and v are obtained from Duncan's table for significant ranges. The $\alpha$ percent critical ranges for the p means, called the *least significant range (lsr)*, and denoted by $R_p$ are given by

$$R_p = q_\alpha(\mathrm{p},\mathrm{v})\sqrt{s^2/r}, \quad p = 2,3,\ldots,k-1,k$$

- To perform the test, arrange the sample means in increasing order of magnitude and compute a set of k-1 *least significant ranges* by multiplying the standard error of mean, i.e. $\sqrt{MSE/r}$ By the tabulated values of $q_\alpha(p,v)$. Compare the observed difference between the largest mean and the smallest mean with the least significant range $R_k$; the difference between the largest mean and the second smallest mean with *lsr* $R_{k-1}$, and so on until all means have been compared with the largest mean. The process continues until the differences of all possible $k(k-1)/2$ pairs of means are considered.

A pair of means is declared to be significantly different when the observed difference exceeds the corresponding least significant range. Duncan's test is quite powerful and is very popular.

It is of importance to note that in case of unequal sample sizes $r_i$, not differing too greatly, the harmonic mean, $r_h$ of all $r_i's$, *i.e.* $r_h = \frac{k}{\sum\left(1/r_i\right)}$ is used in place of r, the number of observations in equal sample sizes.

# Example 1

- Perform the analysis of variance on the following data and analyze the treatment means using Duncan's multiple range test with 0.05 level of significance.

| | | **Treatments** | | | | | |
|---|---|---|---|---|---|---|---|
| | | **1** | **2** | **3** | **4** | **5** | **6** |
| **Blocks** | **1** | 1 | 3 | 6 | 4 | 3 | 2 |
| | **2** | 1 | 4 | 4 | 8 | 5 | 1 |
| | **3** | 3 | 6 | 7 | 8 | 4 | 3 |
| | **4** | 2 | 3 | 2 | 3 | 2 | 1 |

# Solution:

(i) Formulate the hypothesis

(ii) Level of significance

(iii) Test-Statistic:

(iv) Calculations:

| Source of Variation | d.f. | SS | MS | F |
|---|---|---|---|---|
| Between Treatments | | | | |
| Between Blocks | | | | |
| Error | | | | |
| Total | | | | |

(v) Critical Region:

(vi) Conclusion:

# Duncan's Multiple Range Test

Arrange the treatment means:

# *Critical Points for Duncan's Multiple Range Statistic -- ALPHA = 0.05*

| Degrees of freedom n | p | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 20 | 50 | 100 |
| 14 | 3.03 | 3.18 | 3.27 | 3.33 | 3.37 | 3.39 | 3.41 | 3.42 | 3.44 | 3.47 | 3.47 | 3.47 |
| 15 | 3.01 | 3.16 | 3.25 | 3.31 | 3.36 | 3.38 | 3.40 | 3.42 | 3.43 | 3.47 | 3.47 | 3.47 |
| 16 | 3.00 | 3.15 | 3.23 | 3.30 | 3.34 | 3.37 | 3.39 | 3.41 | 3.43 | 3.47 | 3.47 | 3.47 |
| 17 | 2.98 | 3.13 | 3.22 | 3.28 | 3.33 | 3.36 | 3.38 | 3.40 | 3.42 | 3.47 | 3.47 | 3.47 |
| 18 | 2.97 | 3.12 | 3.21 | 3.27 | 3.32 | 3.35 | 3.37 | 3.39 | 3.41 | 3.47 | 3.47 | 3.47 |
| 19 | 2.98 | 3.11 | 3.19 | 3.26 | 3.31 | 3.35 | 3.37 | 3.39 | 3.41 | 3.47 | 3.47 | 3.47 |
| 20 | 2.95 | 3.10 | 3.18 | 3.25 | 3.30 | 3.34 | 3.36 | 3.38 | 3.40 | 3.47 | 3.47 | 3.47 |
| 30 | 2.89 | 3.04 | 3.12 | 3.20 | 3.25 | 3.29 | 3.32 | 3.35 | 3.37 | 3.47 | 3.47 | 3.47 |
| 40 | 2.86 | 3.01 | 3.10 | 3.17 | 3.22 | 3.27 | 3.30 | 3.33 | 3.35 | 3.47 | 3.47 | 3.47 |
| 60 | 2.83 | 2.98 | 3.08 | 3.14 | 3.20 | 3.24 | 3.28 | 3.31 | 3.33 | 3.47 | 3.48 | 3.48 |

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# COURSE: STAT 408

# EXPERIMENTAL DESIGN ANALYSIS

**LECTURE NO. 12**

# COMPARING MEANS

## *THE STUDENT NEWMAN-KEUL'S MULTIPLE RANGE TEST*

# Neuman-Keuls Test

## Definition

The Newman–Keuls or Student–Newman–Keuls (SNK) method is a stepwise multiple comparisons procedure used to identify sample means that are significantly different from each other. It was named after Student (1927), D. Newman (1939), and M. Keuls (1952).

# Neuman-Keuls Test

## Procedure

The Newman–Keuls method employs a stepwise approach when comparing sample means.

### Steps

- Prior to any mean comparison, all sample means are rank-ordered in ascending or descending order, thereby producing an ordered range (p) of sample means.

# Neuman-Keuls Test

## Procedure...

- Critical value

$$q = \frac{\bar{y}_{\max} - \bar{y}_{\min}}{\sqrt{\frac{MS_E}{2}\left(\frac{1}{n_{\max}} + \frac{1}{n_{\min}}\right)}}$$

is the studentized range value, $\bar{y}_{\max}$ and $\bar{y}_{\min}$ are the largest and smallest sample means. For $n_{\max} = n_{\min} = n$

$$q = \frac{\bar{y}_{\max} - \bar{y}_{\min}}{\sqrt{\frac{MS_E}{n}}}$$

# Neuman-Keuls Test

## Procedure...

$$K_p = q_\alpha(p, df_E)s_{\bar{y}_{i.}}$$

for $p = 2, 3, \ldots, a$

where $q_\alpha(p, df_E)$

is the percentage point of studentized range statistic for a specific $\alpha$ and error degree of freedom.

# Neuman-Keuls Test

## Procedure

Arrange treatment means in ascending order.

•Test the difference between pairs of means starting from difference of largest vs smallest means, then largest vs second smallest and so on, by comparing these differences with $K_p$ (for $p$ in descending order).

# Example 1

An engineer is interested in determining if varying the cotton content in a synthetic fiber affects the tensile strength.

| | | Treatments | | | | |
|---|---|---|---|---|---|---|
| | | 1 | 2 | 3 | 4 | 5 |
| Blocks | 1 | 7 | 12 | 14 | 19 | 7 |
| | 2 | 7 | 17 | 18 | 25 | 10 |
| | 3 | 15 | 12 | 18 | 22 | 11 |
| | 4 | 11 | 18 | 19 | 19 | 15 |
| | 5 | 9 | 18 | 19 | 23 | 11 |

# Neuman-Keuls Test-Example

## Tensile strength experiment

An engineer is interested in determining if varying the cotton content in a synthetic fiber affects the tensile strength.

| %age of cotton | Observed tensile strength | | | | | Total: $y_{i.}$ | Average: $\bar{y}_{i.}$ |
|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | | |
| **15** | 7 | 7 | 15 | 11 | 9 | 49 | 9.8 |
| **20** | 12 | 17 | 12 | 18 | 18 | 77 | 15.4 |
| **25** | 14 | 18 | 18 | 19 | 19 | 88 | 17.6 |
| **30** | 19 | 25 | 22 | 19 | 23 | 108 | 21.6 |
| **35** | 7 | 10 | 11 | 15 | 11 | 54 | 10.8 |
| | | | | | | $y_{..}$ =376 | $\bar{y}_{..}$ =15.06 |

# Neuman-Keuls Test-Example

## ANOVA

Response: Tensile strength.

| | Sum of Squares | df | Mean Square | F | Sig. |
|---|---|---|---|---|---|
| Between Groups | 475.760 | 4 | 118.940 | 14.757 | .000 |
| Within Groups | 161.200 | 20 | 8.060 | | |
| Total | 636.960 | 24 | | | |

**Decision:** Null hypothesis of equal average all cotton percentages is rejected as p-value is quite low.

# Neuman-Keuls Test-Example

## Procedure

- Critical value

$$K_p = q_\alpha(p, df_E)s_{\bar{y}_{i.}} \quad \text{where} \quad p = 2, 3, \ldots, a$$

with $s_{\bar{y}_{i.}} = \sqrt{8.06/5} = 1.27$ and $\alpha$=0.01.

$$K_2 = q_{0.01}(2, 20) \times 1.27 = 4.02 \times 1.27 = 5.10$$

$$K_3 = q_{0.01}(3, 20) \times 1.27 = 4.64 \times 1.27 = 5.89$$

$$K_4 = q_{0.01}(4, 20) \times 1.27 = 5.02 \times 1.27 = 6.37$$

$$K_5 = q_{0.01}(5, 20) \times 1.27 = 5.29 \times 1.27 = 6.72$$

# Neuman-Keuls Test-Example

## Procedure...

- The treatment means

$$\bar{y}_{1.} = 9.8,\ \bar{y}_{2.} = 15.4,\ \bar{y}_{3.} = 16.6,\ \bar{y}_{4.} = 21.6,\ \bar{y}_{5.} = 10.8.$$

We see that

$$\bar{y}_{4.} > \bar{y}_{3.} > \bar{y}_{2.} > \bar{y}_{5.} > \bar{y}_{1.}$$

# Neuman-Keuls Test-Example

## Comparisons

$$4\ vs\ 1: 21.6 - 9.8 = 11.8 > 6.72(K_5)$$

$$4\ vs\ 5: 21.6 - 10.8 = 10.8 > 6.37(K_4)$$

$$4\ vs\ 2: 21.6 - 15.4 = 6.2 > 5.89(K_3)$$

$$4\ vs\ 3: 21.6 - 17.6 = 4.0 < 5.10(K_2)^*$$

$$3\ vs\ 1: 17.6 - 9.8 = 7.8 > 6.37(K_4)$$

$$3\ vs\ 5: 17.6 - 10.8 = 6.8 > 5.89(K_3)$$

$$3\ vs\ 2: 17.6 - 15.4 = 2.2 < 5.10(K_2)^*$$

$$2\ vs\ 1: 15.4 - 9.8 = 5.6 < 5.89(K_3)^*$$

$$2\ vs\ 5: 15.4 - 10.8 = 4.6 < 5.10(K_2)^*$$

$$5\ vs\ 1: 10.8 - 9.8 = 1.0 < 5.10(K_2)^*$$

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# COURSE: STAT 408

## EXPERIMENTAL DESIGN ANALYSIS

LECTURE NO. 13

# When to use Scheffe's Post hoc comparison…

- Whenever an ANOVA model is used to examine the differences among more than 2 groups, a post hoc procedure can be used to compare differences between all pairs of means.
- Post hoc comparisons are very similar to *t-test*.
- However, post hoc comparisons are more appropriate for multiple tests, because they help to control the type-I error.
- Type I-error is the chance of wrongly accepting differences between means as significant.
- In a *t-test*, this chance is controlled to be at most 5%.
- In other words, we only accept 2 means as significantly different if the p-value in the *t-test* is less than "0.05."
- In a study that has several groups, we could do several *t-tests* to compare all the differences between means.

# Cont…

- However, the overall significance level will be larger than the individual significance level in each one of the two sample *t-tests*. This is because the probability of incorrectly rejecting a null hypothesis increases as the number of hypothesis tests performed increases.
- For example, if we were to perform three two sample *t-tests* each with a 0.05 significance level, the overall significance level for all three taken together could be as high as 0.14.
- To understand why this occurs, consider the probability of observing a head when you flip a fair coin. If you flip the coin only once, you have a ½ probability of observing a head; however, if you flip the coin three times, the probability that you observe at least one head will be larger than ½, and if you flip the coin 10 times, it is almost us that if we perform enough hypothesis tests simultaneously each with a given significance level (say, 0.05), we can expect a very high probability of incorrectly rejecting at least one null hypothesis.

# Cont…

- Fortunately, post hoc comparisons have been con structured to adjust for this problem.
- Post hoc comparisons tests are more conservative than *t-tests* and control for Type-I error.
- There are many post hoc comparison methods available most of which are named are after dead statisticians (e.g. Tukey's HSD, Fishers LSD and Duncan's Multiple Range Test).
- We shall introduce only one method here, known as Scheffe's Method of pair wise multiple comparison of means.
- In statistics, Scheffe's method, named after the American statistician Henry Scheffe's, is a method for adjusting significance levels in a linear regression analysis to account for multiple comparisons.

# Scheffe's Pair Wise Comparison of Means

- Scheffe's is the very general, implying that it can be used for the widest variety of different types of comparisons.
- It controls experiment wide error rate to 5%, or in other words, there at most a 5% chance of making any type-I errors using the procedure.
- While this is desirable, it also makes the Scheffe's procedure less sensitive.
- There are three steps to applying Scheffe's pair wise multiple comparison method.
- The first step is to obtain the absolute values of pair wise differences between sample means.
- The second step is to obtain values against which the pair wise differences between sample means are compared and identify the statistically significant pair wise differences.
- The third and last step is to state the results.

# Cont…

- Scheffe's test is customarily used with unequal sample sizes, although it could be used with equal sample sizes.
- The critical value for the Scheffe's test is the degrees of freedom for the between variance times the critical value for the one way ANOVA. This simplifies to be:

$$C.V = (k-1) f_{k-1,n-k;\alpha}$$

- The test statistic is a little bit harder to compute. Pure mathematicians will argue that this should not be called F because it doesn't have an F distribution (it's the degree of freedom times an F).

# Scheffe's Method of Pair Wise Multiple Comparison of Means

- Obtain the absolute values of pair wise differences between sample

$$|\bar{x}_i - \bar{x}_j|.$$

- Identify statistically significant differences by finding those with an absolute value larger than the corresponding comparison value

$$\sqrt{(k-1)\, f_{k-1,n-k;\alpha}\, MSE(1/n_i + 1/n_j)}$$

- Reject Ho, if the test statistic is greater than the critical value. Note that, this is right tail test.
- State the significance level chosen and the direction of statistically significant differences between pairs of means.

# Example

Perform the analysis of variance on the following data and analyze the treatment means using the Scheffe's test with $\alpha = 0.05$ level of significance.

| | **Treatments** | | | | | |
|---|---|---|---|---|---|---|
| | **1** | **2** | **3** | **4** | **5** | **6** |
| **1** | 1 | 3 | 6 | 4 | 3 | 2 |
| **Blocks 2** | 1 | 4 | 4 | 8 | 5 | 1 |
| **3** | 3 | 6 | 7 | 8 | 4 | 3 |
| **4** | 2 | 3 | 2 | 3 | 2 | 1 |

1. Hypothesis:

2. Level of Significance:

3. Test Statistics:

# 4. Calculations:

| | Treatments | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | **1** | **2** | **3** | **4** | **5** | **6** | | |
| **1** | 1 | 3 | 6 | 4 | 3 | 2 | | |
| **Blocks 2** | 1 | 4 | 4 | 8 | 5 | 1 | | |
| **3** | 3 | 6 | 7 | 8 | 4 | 3 | | |
| **4** | 2 | 3 | 2 | 3 | 2 | 1 | | |
| | | | | | | | | |
| | | | | | | | | |

| | Treatments | | | | | |
|---|---|---|---|---|---|---|
| | **1** | **2** | **3** | **4** | **5** | **6** |
| **1** | 1 | 3 | 6 | 4 | 3 | 2 |
| **Blocks 2** | 1 | 4 | 4 | 8 | 5 | 1 |
| **3** | 3 | 6 | 7 | 8 | 4 | 3 |
| **4** | 2 | 3 | 2 | 3 | 2 | 1 |
| **Total** | | | | | | |
| **Means** | | | | | | |

| **Source of Variation** | d.f. | Sum of Squares | Mean Square | F |
|---|---|---|---|---|
| Between Treatments | | | | |
| Between Blocks | | | | |
| Error | | | | |
| Total | | | | |

## 5. Critical Region:

## 6. Conclusion:

- The ANOVA table provides us with what is called the "omnibus F-test" or likewise "model significance."
- In the Two way ANOVA, this test shows that there is at least one significant difference between a pair of means.
- Of course, there are 15 unique pairs of means in our example.
- The omnibus F-test does not indicate which pairs are significantly different.
- Therefore, we need a post-hoc comparison to examine the difference between all pairs.

# Scheffe's Test

The first step to using the Scheffe's procedure involves calculating the differences in all pairs of means:

# Cont…

Next, we need to compare these differences to Scheffe's critical “S” value:

# Cont…

# Cont…

# Example

The effects of four types of graphite coaters on light box readings are to be studied. As these readings might differ from day to day., observations are to be taken on each of the four types every day for three days. The results are:

| | Graphite Coater Type | | | |
|---|---|---|---|---|
| Day | *M* | *A* | *K* | *L* |
| 1 | 4.0 | 4.8 | 5.0 | 4.6 |
| 2 | 4.8 | 5.0 | 5.2 | 4.6 |
| 3 | 4.0 | 4.8 | 5.6 | 5.0 |

Construct an analysis of variance table. Determine at 0.05 level whether there is significant difference between the various pairs of Coater types. Apply Scheffe's test to locate significant difference between the various pairs of Coater types.

1. Hypothesis:

2. Level of Significance:

3. Test Statistics:

# 4. Calculations:

| | Graphite Coater Type | | | | | |
|---|---|---|---|---|---|---|
| **Day** | ***M*** | ***A*** | ***K*** | ***L*** | | |
| **1** | 4.0 | 4.8 | 5.0 | 4.6 | | |
| **2** | 4.8 | 5.0 | 5.2 | 4.6 | | |
| **3** | 4.0 | 4.8 | 5.6 | 5.0 | | |
| | | | | | | |
| | | | | | | |

| | Graphite Coater Type | | | |
|---|---|---|---|---|
| **Day** | ***M*** | ***A*** | ***K*** | ***L*** |
| **1** | 4.0 | 4.8 | 5.0 | 4.6 |
| **2** | 4.8 | 5.0 | 5.2 | 4.6 |
| **3** | 4.0 | 4.8 | 5.6 | 5.0 |
| **Total** | | | | |
| **Means** | | | | |

| **Source of Variation** | d.f. | Sum of Squares | Mean Square | F |
|---|---|---|---|---|
| Between Treatments | | | | |
| Between Blocks | | | | |
| Error | | | | |
| Total | | | | |

## 5. Critical Region:

## 6. Conclusion:

- The ANOVA table provides us with what is called the "omnibus F-test" or likewise "model significance."
- In the Two way ANOVA, this test shows that there is at least one significant difference between a pair of means.
- Of course, there are 6 unique pairs of means in our example.
- The omnibus F-test does not indicate which pairs are significantly different.
- Therefore, we need a post-hoc comparison to examine the difference between all pairs.

# Scheffe's Test

The first step to using the Scheffe's procedure involves calculating the differences in all pairs of means:

# Cont…

Next, we need to compare these differences to Scheffe's critical “S” value:

# Cont…

# Example

Each of five varieties of corn is planted in three plots in a large field. The respective yields, in bushels per acre, are indicated below:

| Var 1 | Var 2 | Var 3 | Var 4 | Var 5 |
|---|---|---|---|---|
| 46.2 | 49.2 | 60.3 | 48.9 | 52.5 |
| 51.9 | 58.6 | 58.7 | 51.4 | 54.0 |
| 48.7 | 57.4 | 60.4 | 44.6 | 49.3 |

Construct an analysis of variance table. Determine at 0.05 level whether there is significant difference between the various pairs of varieties of corn. Apply Scheffe's test to locate significant difference between the various pairs of varieties of corn.

1. Hypothesis:

2. Level of Significance:

3. Test Statistics:

# 4. Calculations:

| **Varieties of Corn** | | | | |
|---|---|---|---|---|
| ***1*** | ***2*** | ***3*** | ***4*** | ***5*** |
| 46.2 | 49.2 | 60.3 | 48.9 | 52.5 |
| 51.9 | 58.6 | 58.7 | 51.4 | 54.0 |
| 48.7 | 57.4 | 60.4 | 44.6 | 49.3 |
| | | | | |
| | | | | |

# 4. Calculations:

| | **Varieties of Corn** | | | | |
|---|---|---|---|---|---|
| | ***1*** | ***2*** | ***3*** | ***4*** | ***5*** |
| | 46.2 | 49.2 | 60.3 | 48.9 | 52.5 |
| | 51.9 | 58.6 | 58.7 | 51.4 | 54.0 |
| | 48.7 | 57.4 | 60.4 | 44.6 | 49.3 |
| Total | | | | | |
| Mean | | | | | |

| **Source of Variation** | d.f. | Sum of Squares | Mean Square | F |
|---|---|---|---|---|
| Between Treatments | | | | |
| Between Blocks | | | | |
| Error | | | | |
| Total | | | | |

## 5. Critical Region:

## 6. Conclusion:

- The ANOVA table provides us with what is called the "omnibus F-test" or likewise "model significance."
- In the One way ANOVA, this test shows that there is at least one significant difference between a pair of means.
- Of course, there are 10 unique pairs of means in our example.
- The omnibus F-test does not indicate which pairs are significantly different.
- Therefore, we need a post-hoc comparison to examine the difference between all pairs.

# Scheffe's Test

The first step to using the Scheffe's procedure involves calculating the differences in all pairs of means:

# Cont…

Next, we need to compare these differences to Scheffe's critical “S” value:

# Cont…

# Advantages of Scheffe's Method

- This procedure is extremely conservative since it controls the overall significance for any possible contrast or set of contrasts, even when suggested by the data.

- It therefore should not be used if only pairwise comparisons are wanted.

# Disadvantages of Scheffe's Method

- If the number of contrasts being considered is not considerably greater than the number of groups (i.e. I) and the contrasts were not suggested by the data, a Bonferroni correction probably will be more powerful than Scheffe's.


Thank You

# COURSE: STAT 408
# EXPERIMENTAL DESIGN ANALYSIS

LECTURE DELIVERED BY

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# COURSE: STAT 408

## EXPERIMENTAL DESIGN ANALYSIS

LECTURE NO. 14

# When to use Tukey's HSD Post- hoc test…

- A post-hoc test is needed after we complete an ANOVA in order to determine which groups differ from each other.
- Don't need to conduct a post-hoc test unless you find an affect in the ANOVA problem. If you fail to reject the null, then there are no differences to find.
- T-test is not applicable here because the level of significance become higher and higher after applying the t-test on the groups of mean.
- The Type I error rate can be controlled by using a test called the "Tukey Honestly Significant Difference test or Tukey HSD test".

# Cont…

- The Tukey HSD test is based on a variation of the t-distribution that takes into the account the number of means being compared.
- This distribution is called the studentized range distribution.
- The main idea of HSD is used to compute the honestly significant difference between the two means using a statistical distribution defined by Student and called the *q* distribution.
- This distribution gives the exact sampling distribution of the largest difference between a set of means originating from the same population.
- The Tukey's test is only useable when the sample sizes are the same.

# Tukey' Test - How many pairs?

- This test allows us to compare every pair of population means with a single level of significance.
- So, the first step is to determine how many pairs of means there are. You can either write all of them down, or use the combination formula:

$$\binom{r}{2} = \frac{r!}{2!(r-2)!} = \frac{r(r-1)}{2}$$

- How many pairs are there if there are 3 means? 4 means? 5 means?

# Tukey' Test – Hypothesis

- We have a pair of hypotheses for each pair of means in the following form:

$$H_0 : \mu_i = \mu_j \ \ for\ each\ pair\ of\ means \qquad i \neq j$$

$$H_A : \mu_i \neq \mu_j \ \ for\ at\ least\ one\ pair\ of\ means$$

- where i and j are the appropriate subscripts for the means that you are comparing.

- This means we will have to go through as many hypothesis testing procedures, as there are pairs. Fortunately, the critical value is the same for all the pairs.

# Tukey Test - Critical Value

- The critical value for the Tukey Test is calculated as:

$$T = \left(q_{\alpha,(r,N-r)}\right) \times \left(\sqrt{\frac{MSE}{n_i}}\right)$$

where:

| | |
|---|---|
| MSE | mean square error from ANOVA |
| N | number of all observations |
| $n_i$ | the smallest sample size |
| $q_{\alpha,(r,\,N-r)}$ | the studentized range distribution with r and (N-r) degrees of freedom |

# Tukey Test – Test Statistic

- For each pair of hypothesis in the Tukey test;

$$H_0 : \mu_i = \mu_j \quad for\ each\ pair\ of\ means \qquad i \neq j$$

$$H_A : \mu_i \neq \mu_j \quad for\ at\ least\ one\ pair\ of\ means$$

- The calculated test statistic is equal to the absolute difference between the respective sample means

$$\left|\bar{x}_i - \bar{x}_j\right|$$

- We reject the null hypothesis, if for that particular pair, the test statistic is greater than the critical value, T:

$$\text{Re}\, ject\, H_0 : \mu_i = \mu_j \qquad if \quad \left|\bar{x}_i - \bar{x}_j\right| > T$$

$$Accept\, H_0 : \mu_i = \mu_j \qquad if \quad \left|\bar{x}_i - \bar{x}_j\right| < T$$

- The decision is made separately for each pair of hypotheses.

# Example

Four breeds of cattle $B_1, B_2, B_3, B_4$ were fed on three different rations $R_1, R_2, R_3$. Gains in weight in pounds over a given period were recorded as given below:

| | **Breads** | | | |
|---|---|---|---|---|
| | **1** | **2** | **3** | **4** |
| **1** | 46.5 | 62 | 41 | 45 |
| **Rations 2** | 47.5 | 41.5 | 22 | 31.5 |
| **3** | 50 | 40 | 25.5 | 28.5 |

Perform the analysis of variance on the following data and analyze the treatment means using the Tukey test with $\alpha = 0.05$ level of significance

1. Hypothesis:

2. Level of Significance:

3. Test Statistics:

# 4. Calculations:

| | Breads | | | | | |
|---|---|---|---|---|---|---|
| | **1** | **2** | **3** | **4** | | |
| **1** | 46.5 | 62 | 41 | 45 | | |
| **Rations 2** | 47.5 | 41.5 | 22 | 31.5 | | |
| **3** | 50 | 40 | 25.5 | 28.5 | | |
| | | | | | | |
| | | | | | | |

| | Breads | | | |
|---|---|---|---|---|
| | **1** | **2** | **3** | **4** |
| **1** | 46.5 | 62 | 41 | 45 |
| **Rations 2** | 47.5 | 41.5 | 22 | 31.5 |
| **3** | 50 | 40 | 25.5 | 28.5 |
| **Total** | | | | |
| **Mean** | | | | |

| **Source of Variation** | d.f. | Sum of Squares | Mean Square | F |
|---|---|---|---|---|
| Between Treatments | | | | |
| Between Blocks | | | | |
| Error | | | | |
| Total | | | | |

## 5. Critical Region:

## 6. Conclusion:

- The ANOVA table provides us with what is called the "omnibus F-test" or likewise "model significance."
- In the Two way ANOVA, this test shows that there is at least one significant difference between a pair of means.
- Of course, there are 6 unique pairs of means in our example.
- The omnibus F-test does not indicate which pairs are significantly different.
- Therefore, we need a post-hoc comparison to examine the difference between all pairs.

# Tukey Test

The first step to using the Tukey procedure involves calculating the differences in all pairs of means:

# Cont…

Next, we need to compare these differences to Tukey critical “T” value:

# Cont…

# Example

In a feeding experiment of some animals, the following results were obtained, the numbers in the table being the gains in weight in pounds. The animals were in groups of 3 each.

| | **Groups** | | | |
|---|---|---|---|---|
| | **1** | **2** | **3** | **4** |
| **A** | 7.0 | 16.0 | 10.5 | 13.5 |
| **Rations B** | 14.0 | 15.5 | 15.0 | 21.0 |
| **C** | 8.5 | 16.5 | 9.5 | 13.5 |

Perform the analysis of variance on the following data and analyze the treatment means using the Tukey test with $\alpha = 0.05$ level of significance.

1. Hypothesis:

2. Level of Significance:

3. Test Statistics:

# 4. Calculations:

| | Groups | | | | | |
|---|---|---|---|---|---|---|
| | **1** | **2** | **3** | **4** | | |
| **1** | 7.0 | 16.0 | 10.5 | 13.5 | | |
| **Rations 2** | 14.0 | 15.5 | 15.0 | 21.0 | | |
| **3** | 8.5 | 16.5 | 9.5 | 13.5 | | |
| | | | | | | |
| | | | | | | |

| | Groups | | | |
|---|---|---|---|---|
| | **1** | **2** | **3** | **4** |
| **1** | 7.0 | 16.0 | 10.5 | 13.5 |
| **Rations 2** | 14.0 | 15.5 | 15.0 | 21.0 |
| **3** | 8.5 | 16.5 | 9.5 | 13.5 |
| **Total** | | | | |
| **Mean** | | | | |

| **Source of Variation** | d.f. | Sum of Squares | Mean Square | F |
|---|---|---|---|---|
| Between Treatments | | | | |
| Between Blocks | | | | |
| Error | | | | |
| Total | | | | |

## 5. Critical Region:

## 6. Conclusion:

- The ANOVA table provides us with what is called the "omnibus F-test" or likewise "model significance."
- In the Two way ANOVA, this test shows that there is at least one significant difference between a pair of means.
- Of course, there are 6 unique pairs of means in our example.
- The omnibus F-test does not indicate which pairs are significantly different.
- Therefore, we need a post-hoc comparison to examine the difference between all pairs.

# Tukey Test

The first step to using the Tukey procedure involves calculating the differences in all pairs of means:

# Cont…

Next, we need to compare these differences to Tukey critical “T” value:

# Cont…

# Example

Five fertilizers A, B, C, D and E were tested by arranging plants in Latin square design in a field. The rows and columns in the table are rows and columns in the field. The yields in bushels per plot are as shown.

| | | **Columns** | | | | |
|---|---|---|---|---|---|---|
| | | **1** | **2** | **3** | **4** | **5** |
| | **1** | B 4.9 | D 6.4 | E 3.3 | A 9.5 | C 11.8 |
| **Rows** | **2** | C 9.3 | A 4.0 | B 6.2 | E 5.1 | D 5.4 |
| | **3** | D 7.6 | C 15.4 | A 6.5 | B 6.0 | E 4.6 |
| | **4** | E 5.3 | B 7.6 | C 13.2 | D 8.6 | A 4.9 |
| | **5** | A 9.3 | E 6.3 | D 11.8 | C 15.9 | B 7.6 |

Perform the analysis of variance on the following data and analyze the treatment means using the Tukey test with $\alpha = 0.05$ level of significance

1. Hypothesis:

2. Level of Significance:

3. Test Statistics:

# 4. Calculations:

| | Columns | | | | | | |
|---|---|---|---|---|---|---|---|
| | **1** | **2** | **3** | **4** | **5** | | |
| **1** | 4.9 | 6.4 | 3.3 | 9.5 | 11.8 | | |
| **Rows 2** | 9.3 | 4.0 | 6.2 | 5.1 | 5.4 | | |
| **3** | 7.6 | 15.4 | 6.5 | 6.0 | 4.6 | | |
| **4** | 5.3 | 7.6 | 13.2 | 8.6 | 4.9 | | |
| **5** | 9.3 | 6.3 | 11.8 | 15.9 | 7.6 | | |
| | | | | | | | |
| | | | | | | | |

# 4. Calculations:

| | | Columns | | | | |
|---|---|---|---|---|---|---|
| | | **1** | **2** | **3** | **4** | **5** |
| | **1** | 4.9 | 6.4 | 3.3 | 9.5 | 11.8 |
| **Rows** | **2** | 9.3 | 4.0 | 6.2 | 5.1 | 5.4 |
| | **3** | 7.6 | 15.4 | 6.5 | 6.0 | 4.6 |
| | **4** | 5.3 | 7.6 | 13.2 | 8.6 | 4.9 |
| | **5** | 9.3 | 6.3 | 11.8 | 15.9 | 7.6 |
| | | | | | | |
| | | | | | | |

| **Source of Variation** | d.f. | Sum of Squares | Mean Square | F |
|---|---|---|---|---|
| Rows | | | | |
| Columns | | | | |
| Fertilizers | | | | |
| Error | | | | |
| Total | | | | |

## 5. Critical Region:

## 6. Conclusion:

- The ANOVA table provides us with what is called the "omnibus F-test" or likewise "model significance."
- In the Two way ANOVA, this test shows that there is at least one significant difference between a pair of means.
- Of course, there are 10 unique pairs of means in our example.
- The omnibus F-test does not indicate which pairs are significantly different.
- Therefore, we need a post-hoc comparison to examine the difference between all pairs.

# Tukey Test

The first step to using the Tukey procedure involves calculating the differences in all pairs of means:

# Cont…

Next, we need to compare these differences to Tukey critical “T” value:

# Cont…

# Advantages of Tukey's Test

- Useful in confirmatory research.
- Combination of groups is not meaningful.
- Available in many statistical packages.
- Reduced risk of type I error.
- May be used when group sizes are unequal.

# Disadvantages of Tukey's Test

- Does not test complex contrasts.
- Subject to type II errors and not as powerful as other tests.
- Not idea for exploratory studies.
- Not as available as Scheffe's test.


Thank You

# COURSE: STAT 408
# EXPERIMENTAL DESIGN ANALYSIS

LECTURE DELIVERED BY

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# COURSE: STAT 408

## EXPERIMENTAL DESIGN ANALYSIS

$2^4$ *Factorial Experiment*

LECTURE NO. 15

# Overview of Factorial Experiment

An experiment is called factorial experiment if the treatment consists of all possible combinations of several levels of factors.

The simplest factorial experiment contains two levels for each of two factors. Suppose an engineer wishes to study the total power used by each of two different speeds, 2000 or 3000 RPM. The factorial experiment would consist of four experimental units, motor A at 2000 RPM and motor B at 2000 RPM, motor A at 3000 RPM and motor B at 3000 RPM.

## $2^4$ *Factorial Experiment*

In this case, four factors each at two levels, are of interest. The design is called $2^4$ factorial design and 16 treatment combinations are possible in this design.

# SIGN TABLE

# Properties of Sign Table

- Except for the first column, every column has an equal number of “+” and “-” signs.
- The sum of products of sign is in any two columns is zero that is ay two columns are orthogonal.
- If we multiply any two columns of the sign table another column is achieved i.e. A and B are multiplied and get AB.
- Column 1 multiplied by another column leaves column unchanged that is 1 column is an identity column.

# MODE RULE

Let ABCD × BCD

So it must be equal to A.

Effect A $= \frac{1}{16}(-62) = -3.875$

Effect B $= \frac{1}{16}(182) = 11.375$

AB $= \frac{1}{16}(26) = 1.625$

Effect C $= \frac{1}{16}(0) = 0$,

Effect AC $= \frac{1}{16}(48) = 3$,

BC $= \frac{1}{16}(-36) = -2.25$


Thank You